



Professional Developer

Activity Guide

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Revision History

Publication Date	Summary
March 2020	Initial version for March 2020 Packaged Components release
May 2020	Updated error on page 20
Sept 2020	Quarterly Review
Dec 2020	Quarterly Review
May 2021	Quarterly Review
March 2022	Quarterly Review
June 2022	Updated iTunes Search link and removed logos
May 2023	Updated to use Materials section instead of Resources tab (LMS)

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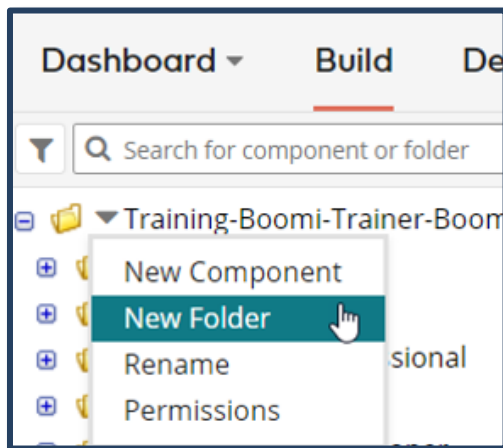
Extensions Activity

Extensions define configuration settings in a single process but specify different settings for each location when deploying to multiple environments.

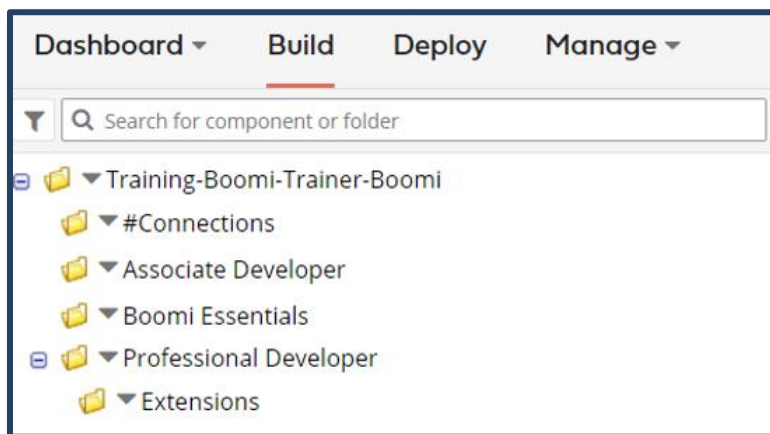
In this hands-on activity, you will change a company's login credentials from Test to Production by changing the parameter values using an extension.

Create a subfolder and install a process

1. Navigate to the **Component Explorer**, select the drop-down arrow next to the Training folder, and select **New Folder**.



2. Name the new folder **Professional Developer** and select **Save**.
3. Under the Professional Developer folder, create a new folder called **Extensions**.



Load the process from the Process Library

4. Open the Process Library by selecting **Browse Process Library** at the bottom of the Component Explorer.



The Boomi Process Library lets you browse, view, and install published processes into your account.

5. Locate **Extensions Activity** by filtering by Education Services Publisher and entering **Extensions** in the filter.
6. Select **Install** next to Extensions Activity.
7. Select **Choose...** next to Select Installation Location and navigate to the Extensions folder.
8. Select **Install** in the lower-right corner.
9. The process is now installed in your account. Select **View Process**.

Extensions Activity Scenario

Your process writes data to a database. When the development team tests the process, use the DEV database. When the process is running in Production, use the PROD database. Your goal is to have a single integration process to write to the appropriate database depending on your AtomSphere environment.

Goal

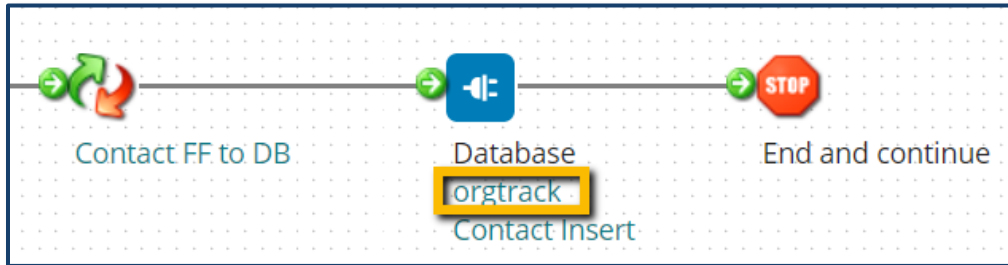
Modify a process to utilize different extension values.



Configure extensions

A recommended practice is to test the extension functionality in the Build page before moving the process to Production and setting the extensions via Atom Management.

1. First, examine the **orgtrack** database connection component by opening it on the Build page.



2. Verify the Username and Database Name says *****SET AS EXTENSION***** and select **Close**.

Database Connection

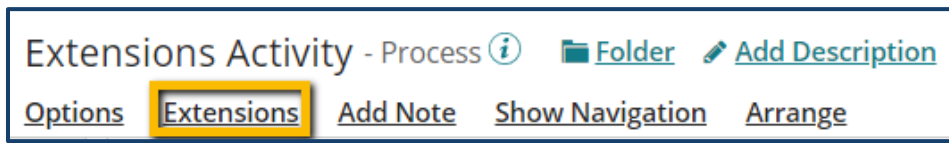
The authentication details and location of the hosted Database.

Database URL ⓘ	jdbc:mysql://traindb1.boomi.com:3306/*** SET AS EXTENSION ***
Driver Type ⓘ	MySQL ▼
Class Name ⓘ	com.mysql.jdbc.Driver
User Name ⓘ	*** SET AS EXTENSION ***
Password ⓘ	<Encrypted>
Host ⓘ	traindb1.boomi.com
Port ⓘ	3306
Database Name ⓘ	*** SET AS EXTENSION ***
Additional Options ⓘ	

Save Save and Close Close Previous Save on 20 Oct 2020 at 11:19:15 AM UTC-4 [Revert](#)

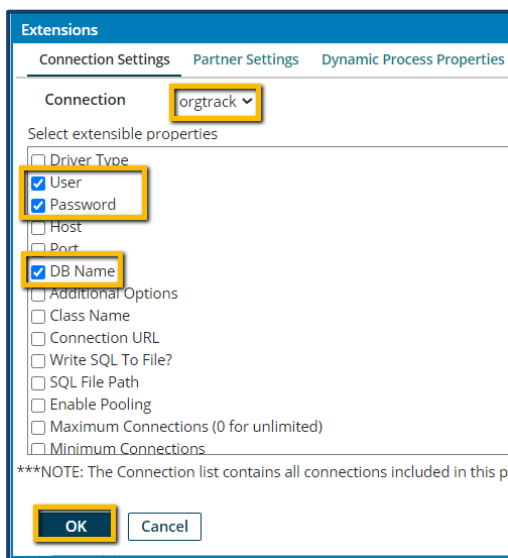
3. To set extensions in the Build page, select the **Extensions** link above the Process Canvas.

Extensions Activity



 The process contains one connection component, so it is the only connection available via the drop-down.

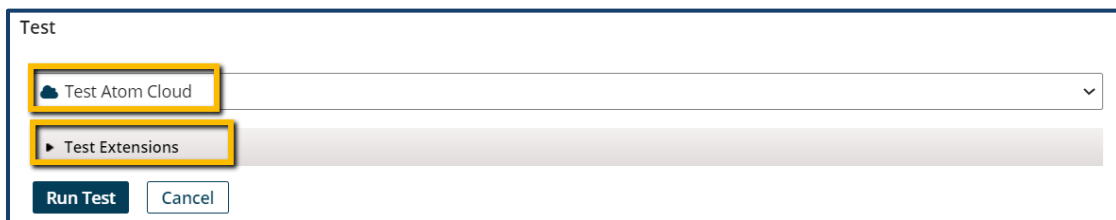
4. Confirm the **User**, **Password**, and **DB Name** are checked and select **OK**.



5. **Save** the Process.

Test the Extensions

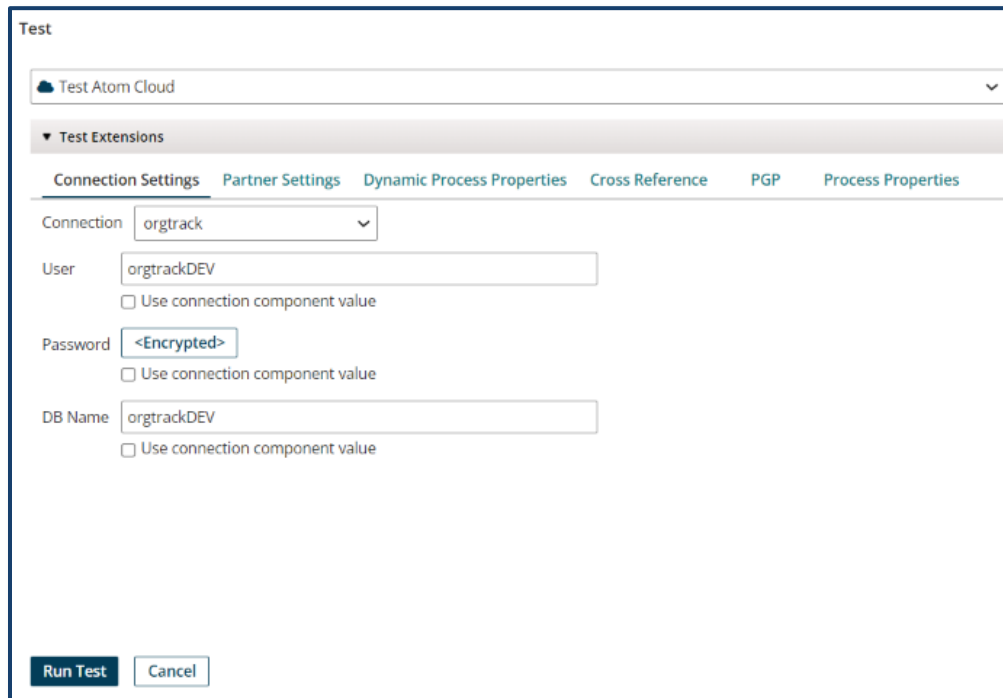
1. Execute the process in the Build page by selecting **Run Test**. Select the **Test Atom Cloud** as the Atom and expand **Test Extensions**.



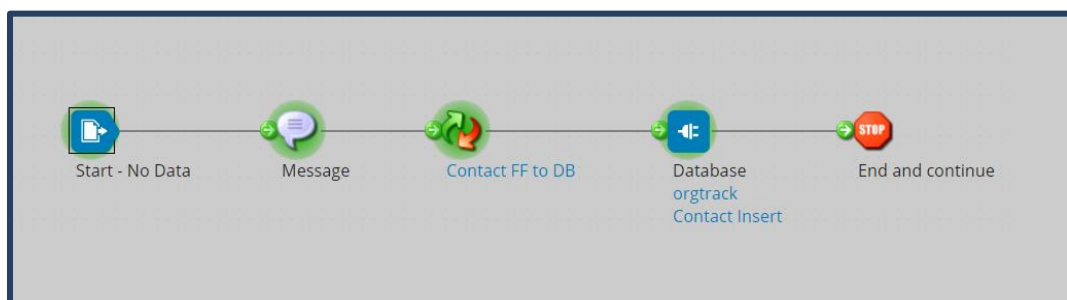
Extensions Activity

2. Select the **orgtrack** from the Connection drop-down. Uncheck **Use connection component value**, enter the following:

DB Name:	orgtrackDEV
User:	orgtrackDEV
Password:	orgtrackDEV



3. Select **Run Test**. Remember the purpose of the hands-on activity is to connect to the database.



✓ Remember the User and Database Names are set to values in the Connection component. If you configured the extensions, the process would have failed at the Database Connector.

4. Test changing the test extension values to reflect those assigned to the Production database by the task **Execute the process in the Build page to test extensions**. The values for the Production database are:

Extensions Activity

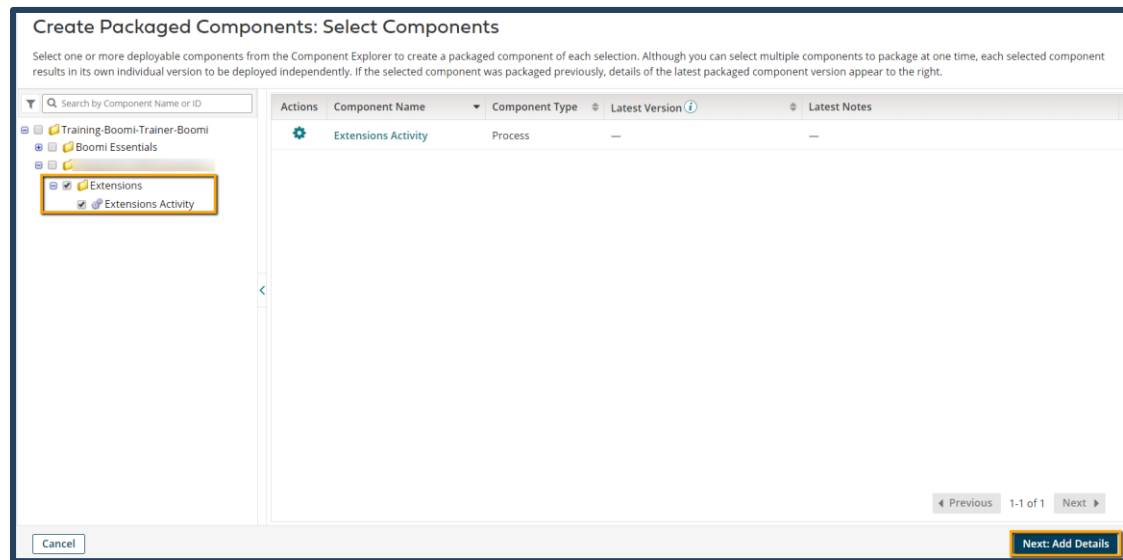
DB Name:	orgtrackPROD
User:	orgtrackPROD
Password:	orgtrackPROD

Package the Component and Deploy to Test and Production

1. In the Build page, select **Create Package Component**.



2. Select the **Extensions Activity** process from the Component Explorer and select **Add Details**.

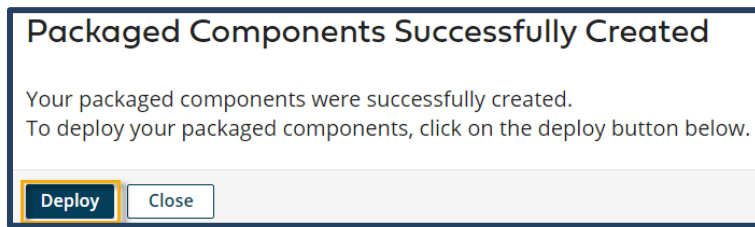


3. Complete the following, and select **Create Packaged Component**.

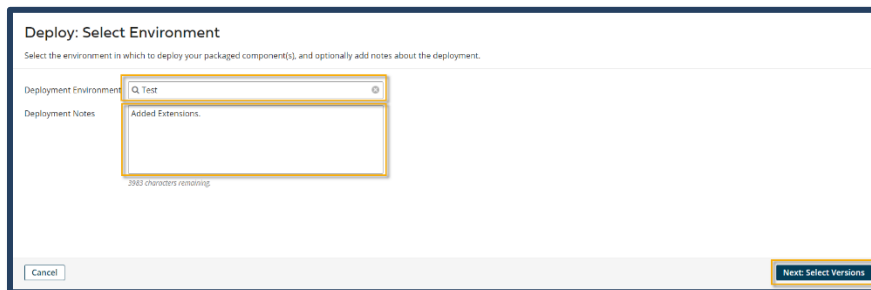
Field	Information to update
Version for All	PDEV-[Mon][Year]-01
Package Notes for All	Added Extensions
Sharing	Unchecked

4. Once you successfully create the packaged component, select **Deploy**.

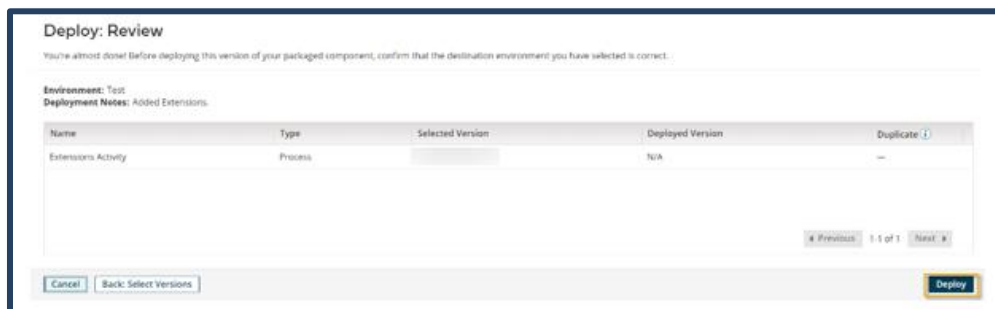
Extensions Activity



5. Select **Test** as the Deployment Environment, enter **Added Extensions** as the Deployment Notes and select **Next**.

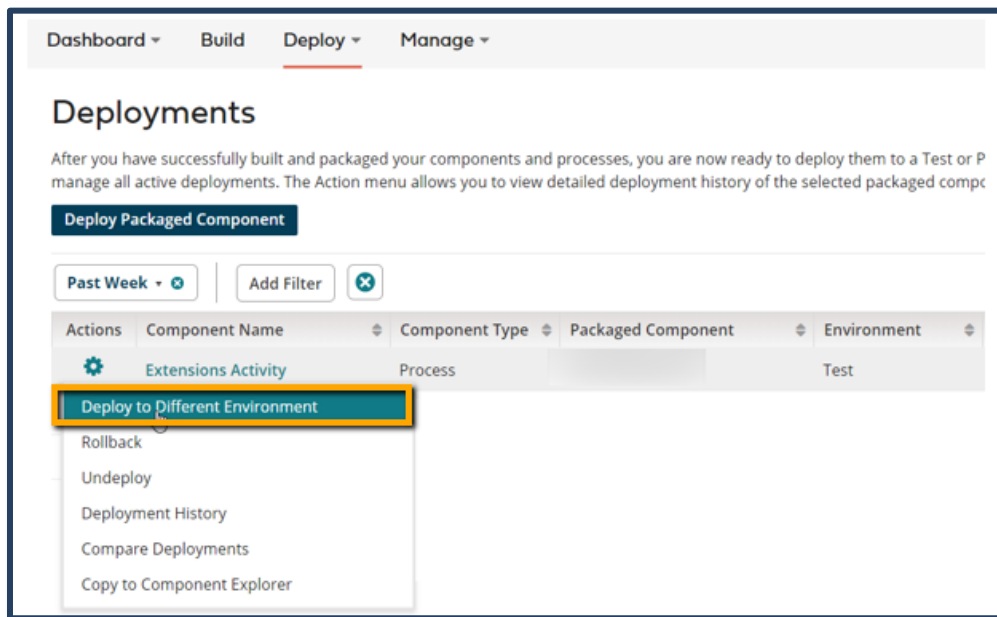


6. Select **Deploy**.

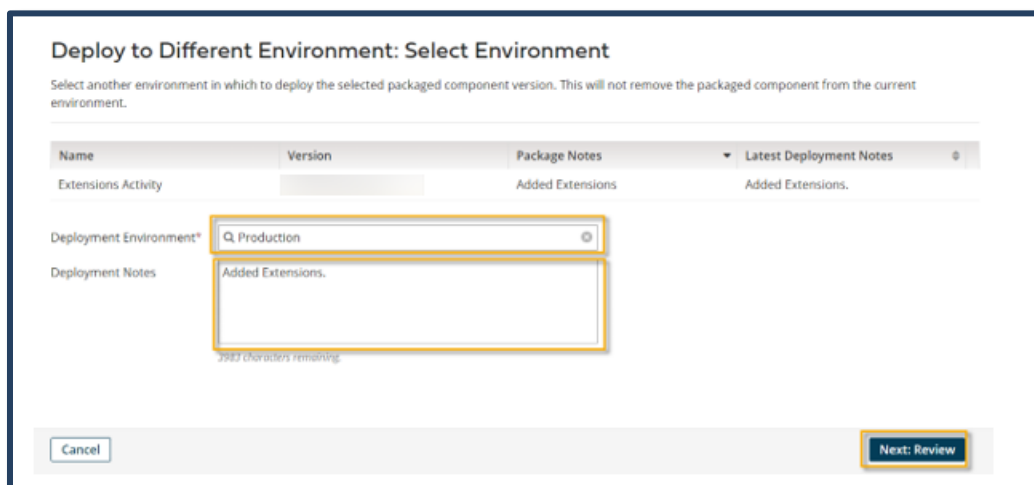


Extensions Activity

7. To deploy the Extensions Activity to Production, navigate to **Deploy > Deployments**. Under the Action icon for the Extensions Activity, choose to **Deploy to Different Environment**.

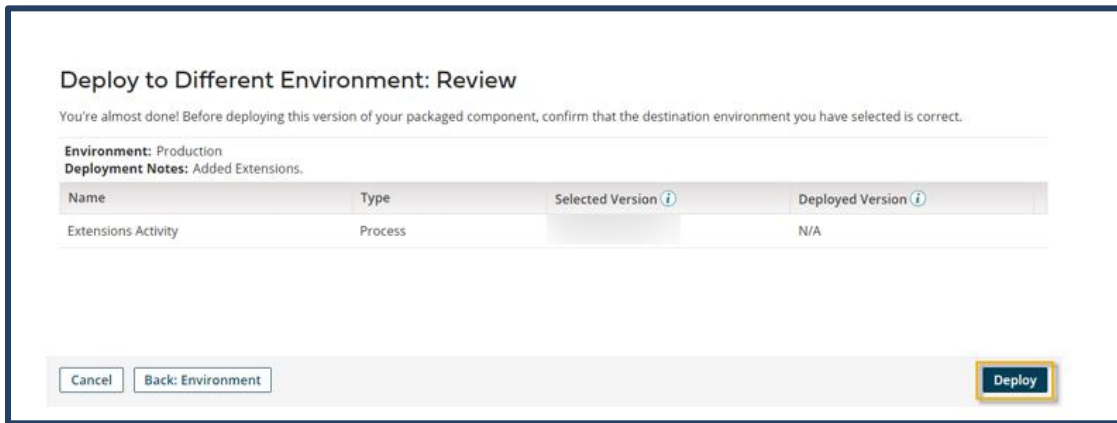


8. Select **Production** as the Deployment Environment, enter **Added Extensions** for the Deployment Notes, and select **Next**.



Extensions Activity

9. After reviewing, select **Deploy**.



Deploy to Different Environment: Review

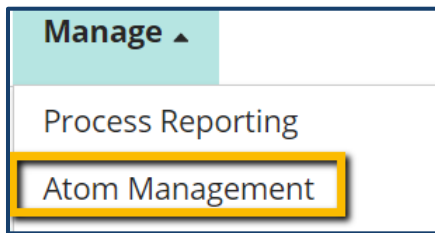
You're almost done! Before deploying this version of your packaged component, confirm that the destination environment you have selected is correct.

Environment: Production
Deployment Notes: Added Extensions.

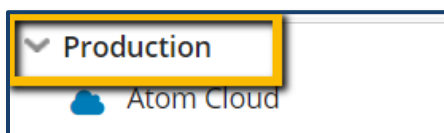
Name	Type	Selected Version 	Deployed Version 
Extensions Activity	Process		N/A

Set extension values at the Environment level

1. Navigate to **Atom Management** to change the extensions.



2. Select the **Production** environment.



Extensions Activity

3. Select **Environment Extensions**.

The screenshot shows the 'Environments » Production' page in Boomi. The page is divided into several sections: 'Production' (a tab), 'Information' (showing Environment ID and Classification), 'Configuration' (showing Roles with Access and Attachments), and 'Administration'. In the 'Administration' section, the 'Environment Extensions' link is highlighted with a yellow box, and the 'Delete Environment' link is also visible.

Environments » Production

Production

Information

Environment ID: 8f07ad56-ae5d-449c-bbc9-fc683c883d63

Classification: Production

Configuration

Roles with Access: 0 items selected.

Choose additional roles

- ☐ Administrator
- ☐ MDM Administrator
- ☐ MDM Data Steward
- ☐ MDM Developer
- ☐ MDM Standard User
- ☐ MDM View Only
- ☐ Production Support
- ☐ Restricted Docs & Data
- ☐ Standard User
- ☐ Support

Any role with Environment Management privilege has access by default.

Attachments: 1 items selected.

Choose Atoms to attach...

- ☒ Atom Cloud

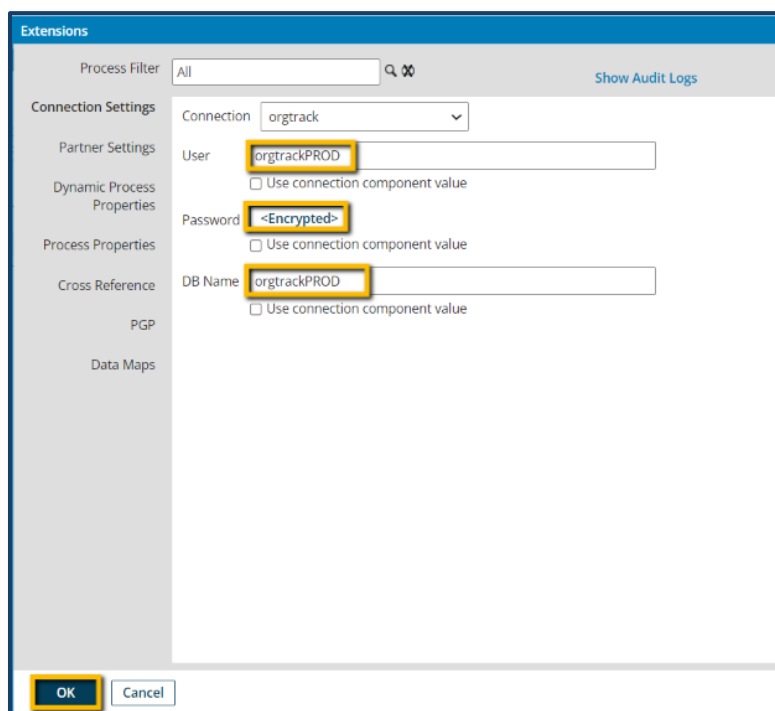
Administration

[+ Environment Extensions](#)

[x Delete Environment](#)

4. Select the drop-down to select **orgtrack** for Connection Settings to view the extendable values, uncheck **Use connection component value**, and enter the following:

Extensions Activity



5. Enter the values for the Production instance:

DB Name:	orgtrackPROD
User:	orgtrackPROD
Password:	orgtrackPROD

6. Repeat steps 1 to 4 in this section to set these values for the Test instance:

DB Name:	orgtrackDEV
User:	orgtrackDEV
Password:	orgtrackDEV

7. Navigate to **Process Reporting**.
8. Manually execute the Extensions Activity by selecting **Execute Process**. You can execute the process in the Production and Test Environments.

Using extensions, you can connect to two different databases without altering the process in the Build page.



If you have difficulties understanding extensions in either Test or Production, incorrectly enter a value. An error message like the one below is displayed:

Error Message

Access denied for user 'orgtrackDEV1'@'23.20.142.114' (using password: YES)

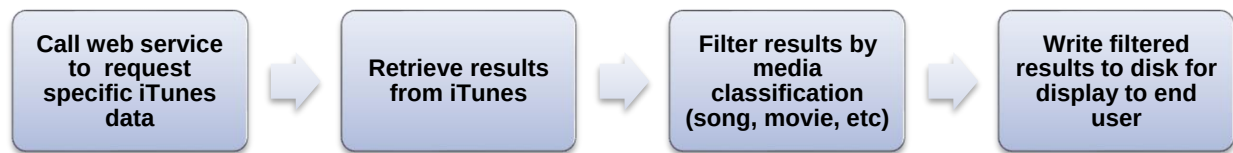
REST Activity

Scenario

A mobile application developer is implementing an interactive application for movie listings and searches. It is now in the development stage and is testing the process with an iTunes integration. When a user searches for information about a film, actor, or director, a search results page is displayed, and the user can rent or buy the selected title. The integration team will create, and unit test the back-end interface logic for the application.

Goal

Improve a partly built process, using the Web Services HTTP Client connector to tap into the iTunes Search API.



Configure the HTTP Connector

Create a subfolder within your account

1. Select the blue drop-down arrow next to the **Professional Developer** folder.
2. Choose New Folder and name it **REST**.

Load the process from the Process Library

1. Open the Process Library by selecting on **Browse Process Library** at the bottom of the Component Explorer.
2. In the right column, on the search bar, enter **REST** and select **Enter**.
3. Select **Install** next to REST Activity.
4. Select browse to Select Installation Location and navigate to the REST folder.
5. Select **Install** in the lower-right corner.
6. The process is now installed in your account. Select **View Process**.

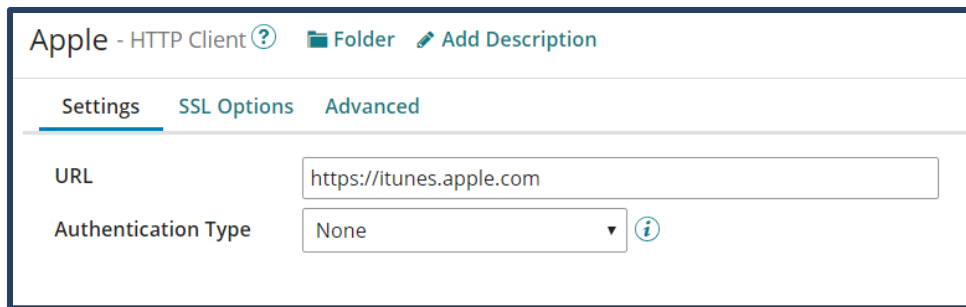
Configure the connection

1. Familiarize yourself with the iTunes Search API, at:

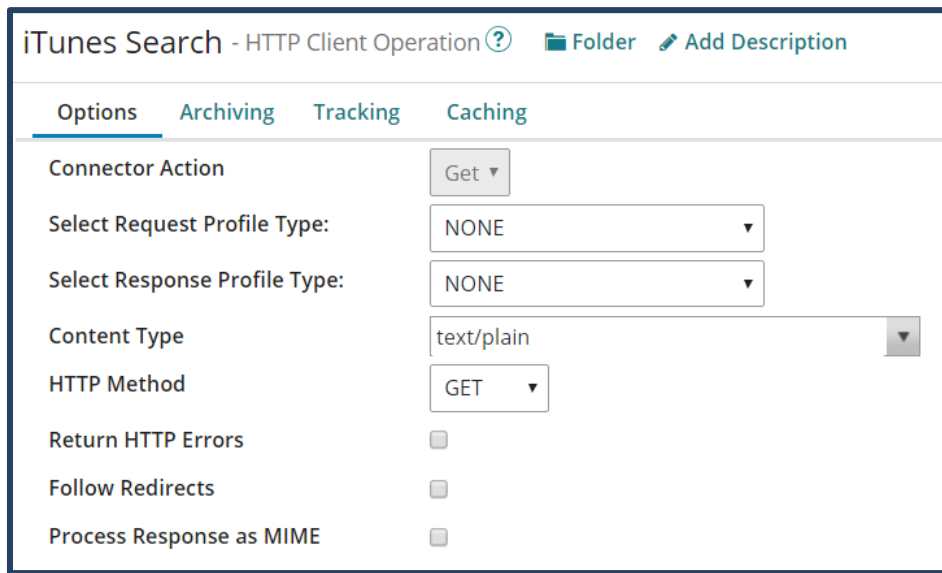
<https://developer.apple.com/library/archive/documentation/AudioVideo/Conceptual/iTuneSearchAPI/index.html>

A response document is needed to interact with a RESTful web service. The HTTP Client Connector does not support a dynamic response profile import. You must complete the following to create a Profile:

- Configure the connector.
 - Run the process in the Build page.
 - Save the response document to disk for later use to configure and create a profile
- ✓ *Remember, each connector is a two-part container with a connection and operation. The connection component contains the base URI in a RESTful web service integration. Building remaining URI using the resource path in the operation component.*
2. Open and view the **Apple** Connection, which contains the base URI.



3. Select **Close** when done viewing.
4. Open and configure the **iTunes Search** Operation.



iTunes Search - HTTP Client Operation ? Folder Add Description

Options Archiving Tracking Caching

Connector Action Get ▾

Select Request Profile Type: NONE ▾

Select Response Profile Type: NONE ▾

Content Type text/plain ▾

HTTP Method GET ▾

Return HTTP Errors ☐

Follow Redirects ☐

Process Response as MIME ☐

✓ You must scroll down to access the **Resource Path** section.

5. Replace the Resource Path elements with the associated substring in the Request URI required by the Apple iTunes Search API. Remember to use special characters, like '?', '+', and '&'. The following example will help you create a properly formatted URL:

✓ To search for **Francis Ford Coppola** and return **1** item, the URL would look like:

BASE URI	URI SEARCH CRITERIA	URI PARAMETERS
https://itunes.apple.com/	search?term=francis+ford+coppola	&country=us&limit=1

For this activity, you will search for **Francis Ford Coppola**, using the Country of **us** and a Limit of **1**. The example above will help you figure out the URI information in the Resource Path.

REST Activity

Resource Path

Replace with search criteria URI ☐

NAME_OF_PERSON ☒

Replace with country ☐

Replace with limit URI ☐

Name: NAME_OF_PERSON

☒ Is replacement variable?

Resource Path

search?term= ☐

NAME_OF_PERSON ☒

&country=us ☐

&limit=1 ☐

Name: NAME_OF_PERSON

☒ Is replacement variable?

- ✓ Enter 1 to return a single response document for the limit URI element to build the JSON profile needed for the next section.
- ✓ If 'Is replacement variable?' is checked next to a resource path element, this will replace the element name with a dynamic value. Note the element NAME_OF_PERSON is a replacement variable. You will enter the name (Francis Ford Coppola) as a Parameter in the HTTP Client Connector Parameters tab.

6. Select **Save and Close**.

7. Open the HTTP Client Connector and select the **Parameters** tab.

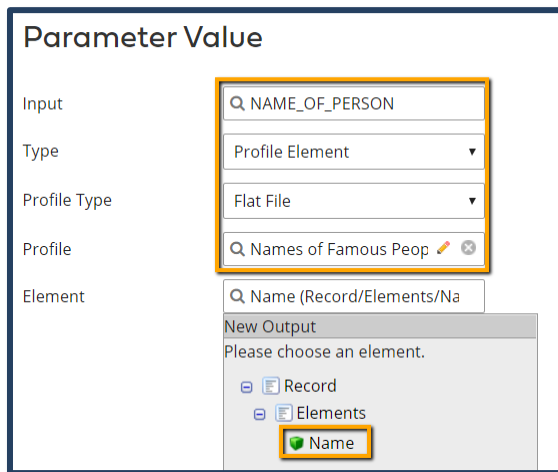
General Parameters

Add or edit the values of your parameters.

Name	Value	Source
------	-------	--------

8. Select **Add Parameter**, and configure the value according to the image shown below:

REST Activity



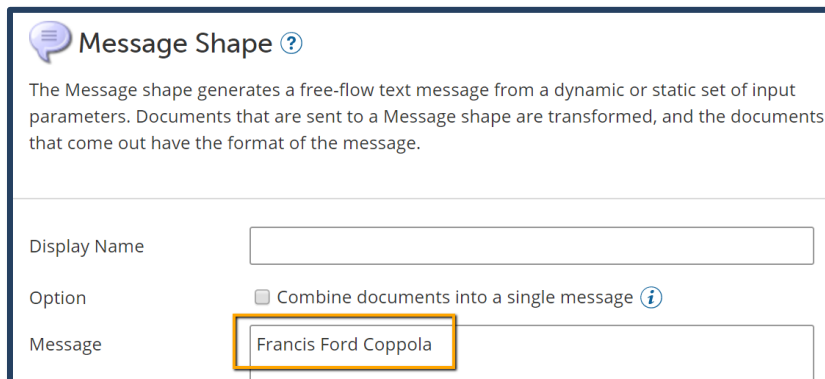
The 'Parameter Value' configuration window shows the following settings:

- Input: Q NAME_OF_PERSON
- Type: Profile Element
- Profile Type: Flat File
- Profile: Q Names of Famous Peop
- Element: Q Name (Record/Elements/Na)

The 'New Output' section is expanded, showing a list of elements: Record, Elements, and Name. The 'Name' element is selected and highlighted with an orange box.

 The process contains a No Data Start shape, Message, Data Process, and Map (which completes a URL encode) before the HTTP Client Connector, allowing you to enter the names of famous people, simulating multiple values entered the mobile app.

9. Enter **Francis Ford Coppola** into the Message.



The 'Message Shape' configuration window shows the following settings:

- Display Name: (empty)
- Option: ☐ Combine documents into a single message
- Message: Francis Ford Coppola

10. Run the Process in the Build page using the Test Atom Cloud. If successful, open and view the document in the Document Viewer.
11. Select the **HTTP Client Connector** and the **Connection Data** tab.

Documents

#

1

✓

Logs

Shape Source Data

Connection Data

View

Url

Size (kB)

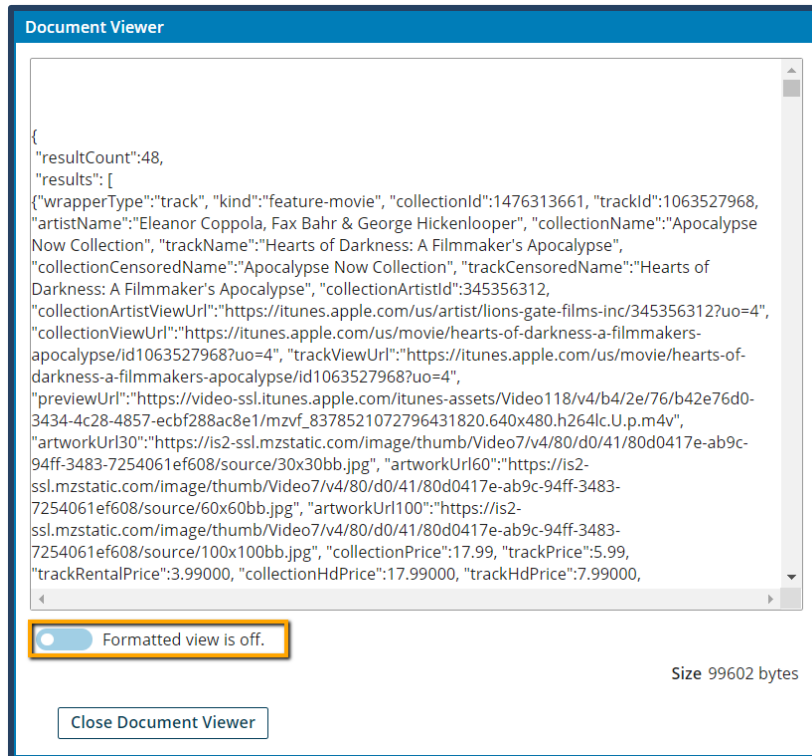
https://itunes.apple.com/search?term=Francis+Ford+Coppola&country=us&limit=1

2.07

REST Activity

Open the **Document Viewer**, copy the contents and paste the contents into a text editor. Save the contents locally to reference to later create the JSON Response Profile.

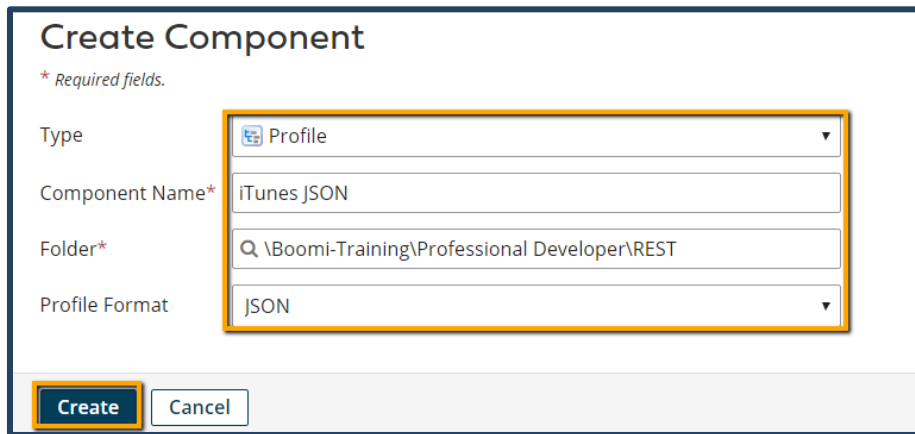
- ✓ *Make sure the Formatted view is off before you copy. If you copy with the formatting view on you can not import the JSON file.*



Configure and Test the Rest Activity

Create a JSON profile

1. Create a new JSON Profile with the following information:



The 'Create Component' dialog box is shown with the following fields and values:

- Type: Profile
- Component Name*: iTunes JSON
- Folder*: \Boomi-Training\Professional Developer\REST
- Profile Format: JSON

Buttons: Create, Cancel

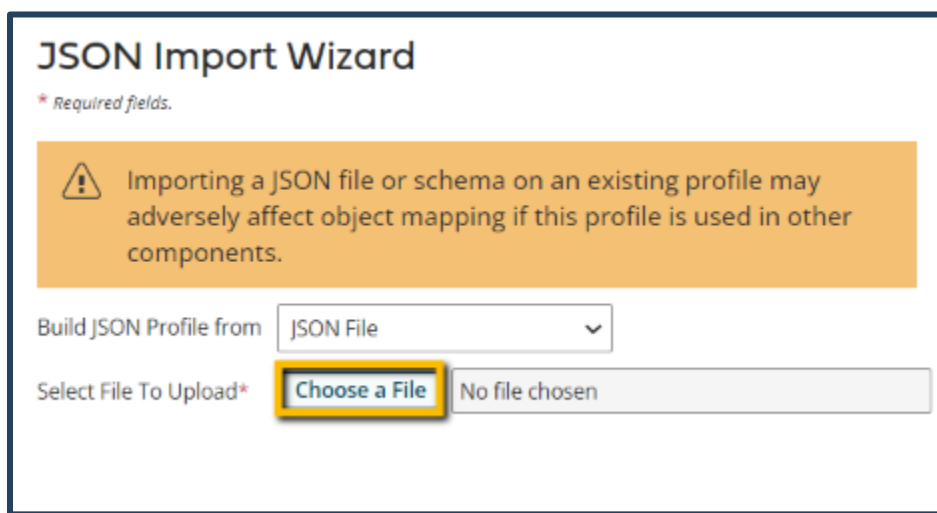
2. Select **Create**.
3. In the JSON Profile window, select **Import**.



The 'iTunes JSON - JSON Profile' window is shown with the following elements:

- Header: iTunes JSON - JSON Profile ?
- Buttons: Folder, Add Description
- Import button

4. In the JSON Import Wizard, select **Browse** and locate your JSON file.



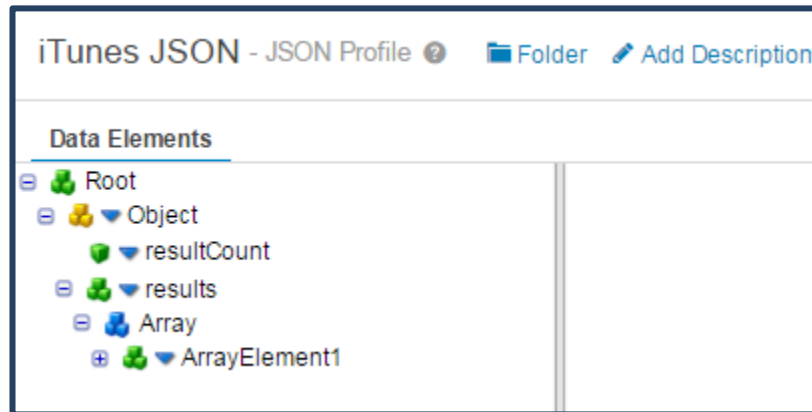
The 'JSON Import Wizard' dialog box is shown with the following elements:

- Warning message: Importing a JSON file or schema on an existing profile may adversely affect object mapping if this profile is used in other components.
- Build JSON Profile from: JSON File
- Select File To Upload*: Choose a File (highlighted), No file chosen

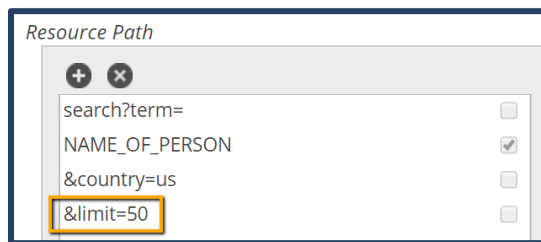
5. Select **Next** and **Finish**.

Configure and Test the Rest Activity

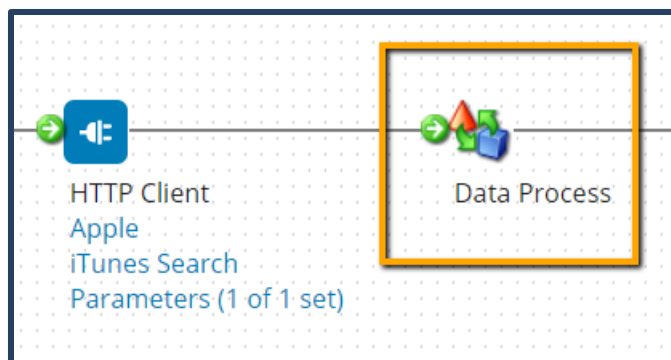
6. Open the profile to make sure it looks like the profile below:



7. Select **Save and Close**.
8. Open and configure the **iTunes Search** operation.
9. Scroll to the resource path and change the limit to 50.
✓ *Increasing the limit provides the ability to retrieve multiple records into the process.*



10. Select **Save and Close**.
11. Drag a **Data Process** shape onto the Process Canvas and place it after the HTTP Client Connector.



Configure and Test the Rest Activity

12. Configure the properties to split the incoming JSON data by the 'ArrayElement1' element to create a separate document for each iTunes entry.

Process Type:	Split Documents
Profile Type:	JSON
Profile:	iTunes JSON
Split Element:	Array Element 1

Data Process Shape ?

The Data Process shape provides a number of options for manipulating document data within a process, from splitting and combining documents to zipping and unzipping data. You can define multiple processing steps to perform more than one action on the document data. The processing steps will be executed in the order defined in the Data Process shape. Each processing step will operate on the data output from the previous processing step.

Display Name:

Processing Steps

- Split Documents**

Process Type:

Profile Type:

Split Options

Batch Count:

Profile:

Split Element:

13. Select **OK**.

The documents are split and flow through the process individually, which provides the ability to drill down to document-level details about a particular iTunes entry.

Configure the Decision

Configure the Decision, so if the kind field equals feature-movie, it continues the True path.

1. In the Decision, select the profile element 'kind' and compare it to a Static value.

Type:	Profile Element
Profile Type:	JSON
Profile:	iTunes JSON
Element:	kind

Compare Type: **Equal To**

Second Value: Static value of **feature-movie**

Configure and Test the Rest Activity



Decision Shape 

The Decision shape routes documents based on a true/false comparison of two values. Those values can be anything from field values in the current document (profile elements), static values, results of a database, an application query, or more.

Display Name

Media Type?

First Value

Q JSON Profile - iTunes JSON - kind (Root/Object/results/Array/1) 

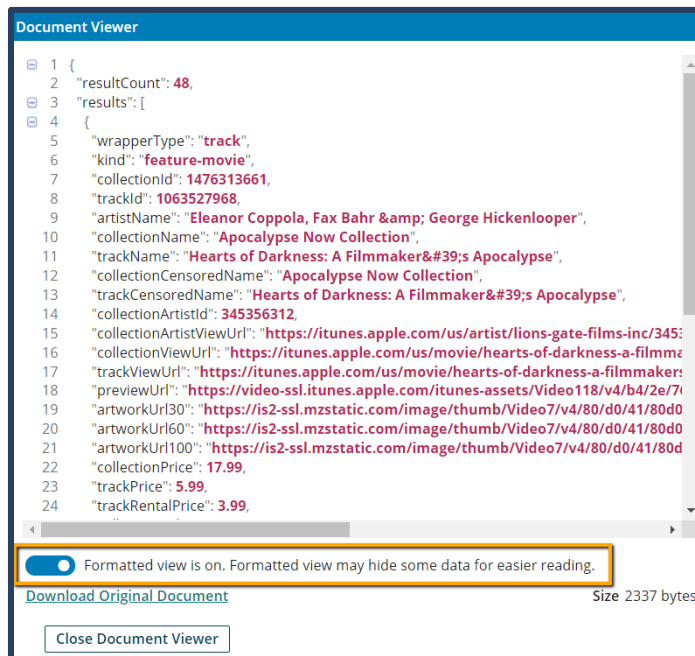
Comparison

Equal To ▼

Second Value

Q Static value of 'feature-movie' 

2. Select **OK**.
3. Select **Save and Close** and **OK**. Select **Save** to save the process.
4. Run the process in the Build page.
5. Select the **Stop** at the end of the True path and select the **Shape Source Data** tab.
6. Open the output to view the movies for Francis Ford Coppola and turn on Formatted View for the Document view below:



- ✓ Also, select the False path Stop to view the outputs.

SOAP Activity

Scenario

A web developer is implementing a weather app where a user enters a temperature to convert from Celsius to Fahrenheit or Fahrenheit to Celsius. You will create the back-end interface logic between the web service to perform the conversion and the UI to display the temperature conversion.

 You will need to return to the beginning of the activity and reinstall the process to change the temperature conversion.

Goal

Modify a pre-built process to use the Web Services SOAP Client connector to tap into the AtomSphere API Service performing the temperature conversion.



Create a subfolder within your account

1. Select the blue drop-down arrow next to the **Professional Developer** folder.
2. Choose New Folder and name it **SOAP**.

Load the process from the Process Library

1. Open the Process Library by selecting on **Browse Process Library** at the bottom of the Component Explorer.
2. In the right column, in the search bar, enter the word **SOAP**.
3. Select Install next to **SOAP Activity**.
4. Select **Choose...** next to Select Installation Location and navigate to the SOAP folder.
5. Select **Install** in the lower-right corner.
6. The process is now installed in your account. Select **View Process**.

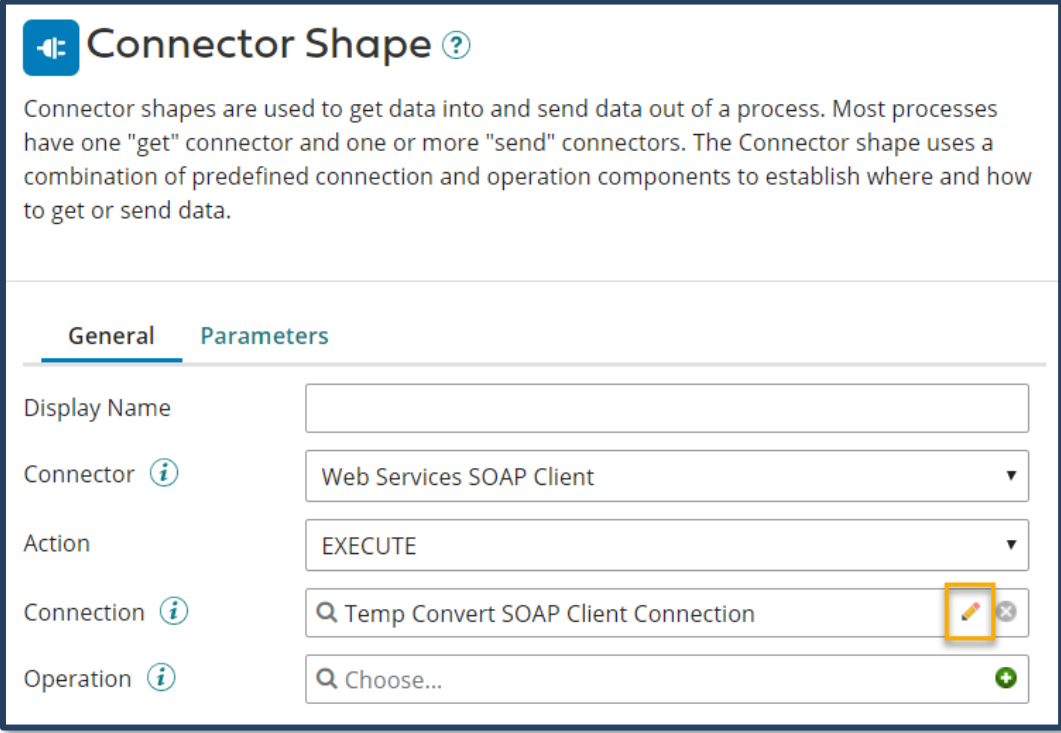
The Web Service

 For this exercise, you will use a SOAP Web Service published by Boomi, containing a WSDL file and two methods: *executeCelsiusToFahrenheit* and *executeFahrenheitToCelsius* you may use either.

Configure the Connection

Remember, connectors have a connection and an operation. You will configure the connection component to import the Request and Response profiles.

1. Open the Web Services Soap Client Connector.
2. In the **Connector General** tab, select the **Edit** icon next to Temp Convert SOAP Client Connection.



Connector Shape ?

Connector shapes are used to get data into and send data out of a process. Most processes have one "get" connector and one or more "send" connectors. The Connector shape uses a combination of predefined connection and operation components to establish where and how to get or send data.

General Parameters

Display Name

Connector  Web Services SOAP Client ▼

Action EXECUTE ▼

Connection  Temp Convert SOAP Client Connection  

Operation  Choose... 

 The Web Services SOAP Client Connection settings vary depending on the requirements and documentation provided by the SOAP web service.

3. In the Web Services Soap Client window, view the following URLs:
 - WSDL Url: <https://c01-usa-west.integrate.boomi.com/ws/soap/tempconverter?wsdl>
 - SOAP Endpoint Url: <https://c01-usa-west.integrate.boomi.com/ws/soap/tempconverter?>
 - User and Password: This web service uses basic authentication.

 Do not change the Connection.

The screenshot shows the 'Temp Convert SOAP Client Connection' configuration window with the 'WS Security' tab selected. The window has a title bar with a question mark icon, a 'Folder' icon, and an 'Add Description' link. Below the title bar are two tabs: 'Connection' and 'WS Security'. The 'WS Security' tab contains the following fields:

WSDL Url ⓘ	https://c01-usa-west.integrate.boomi.com/ws/soap/tempconvert
SOAP Endpoint Url ⓘ	https://c01-usa-west.integrate.boomi.com/ws/soap/tempconvert
Security Type ⓘ	Basic
User ⓘ	SOAPtest@boomitrainingparent-ALQN3P.R1T58M
Password ⓘ	<Encrypted>
Client SSL Certificate ⓘ	Q Choose...
Trust SSL Server Certificate ⓘ	Q Choose...

4. Select **Close**.

With the Connection component configured, use the Connection to help configure the Operation and import the profiles.

5. Add a new Operation to the Connector.

The screenshot shows the 'General' tab of the configuration window. It contains the following fields:

Display Name	
Connector	Web Services SOAP Client
Action	EXECUTE
Connection	Q Temp Convert SOAP Client Connection
Operation	Q Choose...

The 'Operation' field has a green plus icon in a box next to it, indicating where to click to add a new operation.

SOAP Activity

6. Name the Operation **Convert Temperature**.

Convert Temperature Web Services SOAP Client Operation ⓘ Folder Add Description

Options Archiving Tracking Caching

Connector Action EXECUTE ▼

Object

Tracking Direction ⓘ ☒ Input Documents ☐ Output Documents

Error Behavior ☐ Return Application Error Responses ⓘ

Option ☒ Expose Request Envelope ⓘ

Option ☐ Expose Response Envelope ⓘ

Attachment Cache ⓘ +

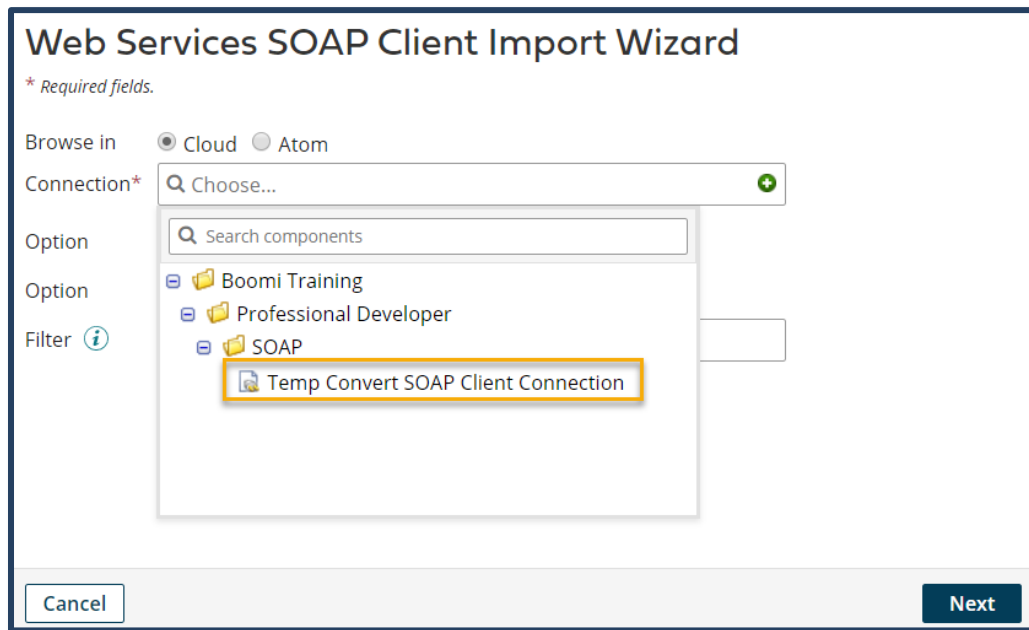
7. Select **Import**.

Convert Temperature - Web Services SOAP Client Operation ⓘ Folder Add Description

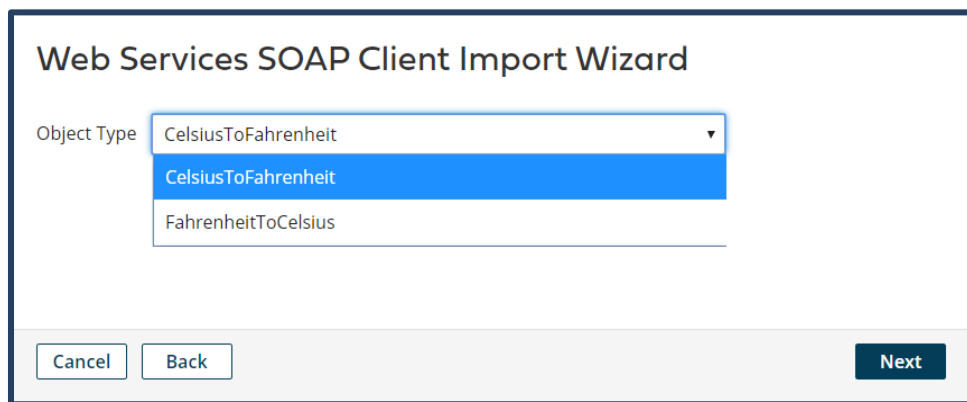
Options Archiving Tracking Caching Import

Connector Action EXECUTE ▼

8. Select the **Magnifier** icon to browse and load the **Temp Convert SOAP Client Connection** component.



9. After selecting the Connection component, select **Next**.
10. For Object Type (method): select **executeCelsiusToFahrenheit** or **executeFahrenheitToCelsius** from the drop-down menu and select **Next** to load the Operation's Profiles.



11. Select **Finish**.
The imported Request and Response profiles are displayed in the Operation window.

Convert Temperature - Web Services SOAP Client Operation ? Folder Add Description

Options Archiving Tracking Caching

Connector Action EXECUTE

Object executeCelsiusToFahrenheit

Request Profile Web Services SOAP Client executeCelsiusToFahrenheit E

Response Profile Web Services SOAP Client executeCelsiusToFahrenheit E

12. Select **Save and Close**, and **OK** to return to the Process Canvas.

Configure the Message and Map

 You are using a *No Data Start* shape because the source data comes from the *Message*. A common approach for complete initial testing is to use a *No Data Start* shape with a *Message*. Later you can exchange this for a *Connector* to retrieve data.

1. Open the *Message* and enter the temperature you want to convert in the *Message* field.

Message Shape ?

The Message shape generates a free-flow text message from a dynamic or static set of input parameters. Documents that are sent to a Message shape are transformed, and the documents that come out have the format of the message.

Display Name Temperature

Option ☒ Combine documents into a single message i

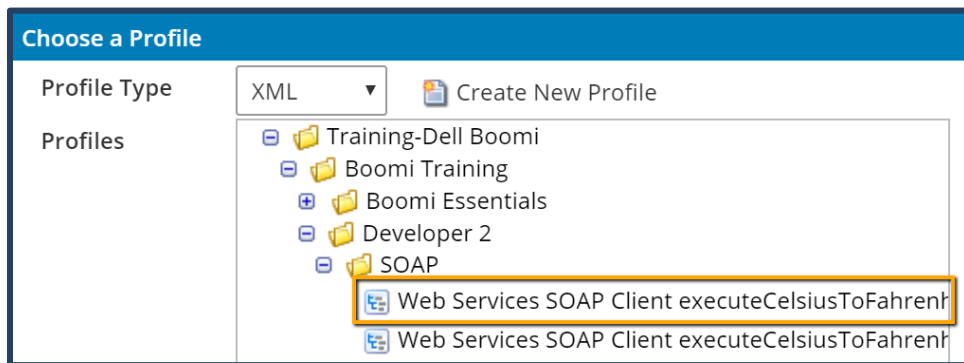
Message 22

2. Select **OK**.
3. Open the **Temperature (FF to XML)** map and choose the destination (XML) profile.
4. In the right pane, select **Choose** the Destination profile.

SOAP Activity



5. Browse to select the appropriate XML profile.



6. Select **OK**.



The document moves from the map to the Web Services SOAP Client connector, so you need to use the Request Profile as the destination profile. Your data is transformed into the proper format to send the request to the Web Services SOAP Client connector. The Response Profile format for the document returned from the called web service after the Web Services SOAP Client connector executes.

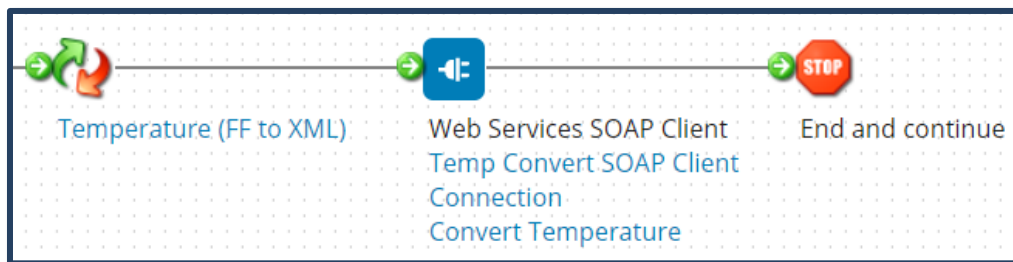
SOAP Activity

- Map the fields in the source and destination profiles as displayed below:



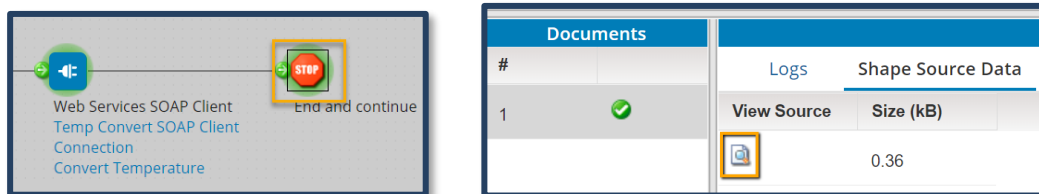
- Select **Save and Close** and **OK** to save the Mapconverted.
- Place the previously configured Web Service SOAP Client Connector after the Map and before the Stop, connect them and select **Save**.

✓ *The goal is to fully configure the client connector to obtain the appropriate Request and Response profiles to be used throughout the process.*



Run a test

- Run the process in the Build page, select the Stop, and use the Document Viewer (under Shape Source Data) to view results.
- ✓ *Remember, the temperature is converted by the web service called the Web Services SOAP Client connector. The conversion result is available under Shape Source Data only for the Stop since it is the only shape in the process after the Web Services SOAP Client connector.*



The output below displays the converted temperature entered in the Message.

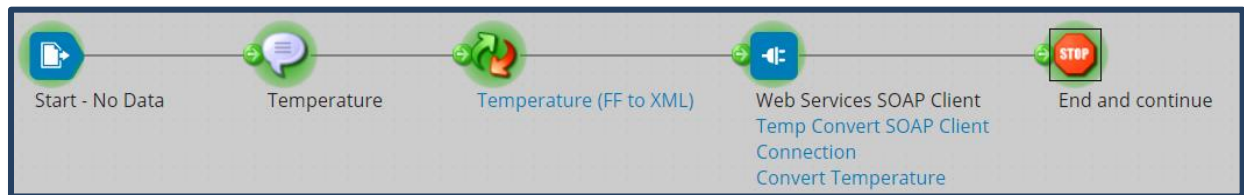
Document Viewer

```

1 <executeCelsiusToFahrenheitResponse>
2   <CelsiusToFahrenheitResponse>
3     <CelsiusToFahrenheitResult>71.6</CelsiusToFahrenheitResult>
4   </CelsiusToFahrenheitResponse>
5 </executeCelsiusToFahrenheitResponse>
  
```

Review the process flow and the document state at each step:

- Since the process is still in testing, begin with a No Data Start shape.
- Enter the temperature in the Message (for testing purposes)
- In the Map, translate the Flat File temperature value from the Message to the XML Request format needed by the web service called by the Web Services SOAP Client connector. (You will create the destination XML profile when you configure your Convert Temperature Operation.)
- Pass the value to the Web Services SOAP Client connector, which calls the web service.
- The web service returns the XML Response with temperature conversion is viewable under Shape Source Data for the Stop.



Process Call Activity

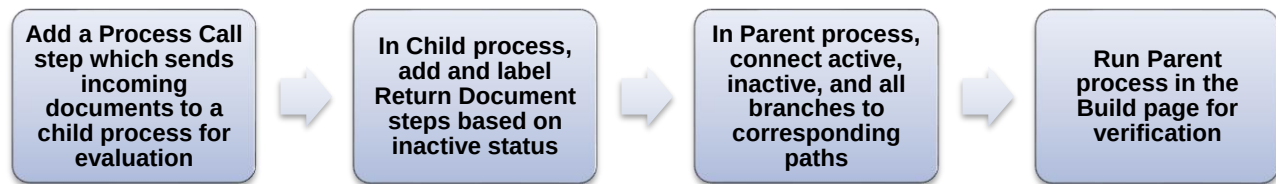
Process Call executes another process in a Process. When calling a sub-process, use the sub-process Start shape to get data. The sub-process call is a separate process execution, and the process execution results display in the Manage > Process Reporting page like a regularly executed process.

Scenario

A Parent process retrieves incoming CSV customer records to send to its Child process for further evaluation. The Child process determines if the customer record is active or inactive. The Child process returns all records and separates and returns all active and in-active customer records to the Parent process. The Parent process converts active customer records to XML and writes them to disk. It also will write the inactive records and archive the customer records to separate disk directories.

Goal

Modify a pre-built process using Process Call and Return Documents.



Create a subfolder within your account

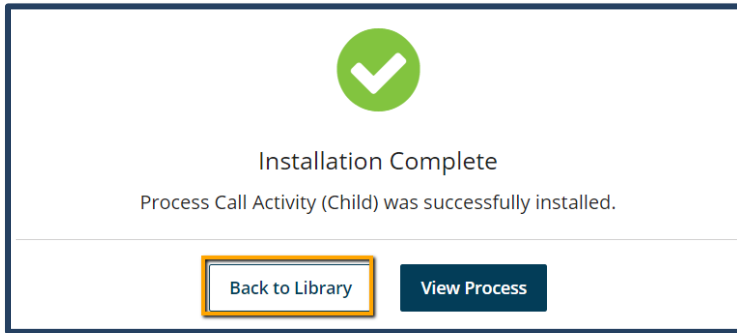
1. Select the drop-down arrow next to the **Professional Developer** folder.
2. Choose New Folder and name it **Process Call**.

Load the parent and child processes from the Process Library

1. Open the Process Library by selecting Browse Process Library at the bottom of the Component Explorer.
2. In the right column on the search bar, enter **Process Call** and select **Enter**.
3. Select Install next to **Process Call (Parent)**.
4. Select the browse symbol next to Select Installation Location and navigate to the **Process Call** folder.
5. Select **Install** in the lower-right corner.
6. The process is now installed in your account. Select **Back to Library** to return to the Process

Process Call Activity

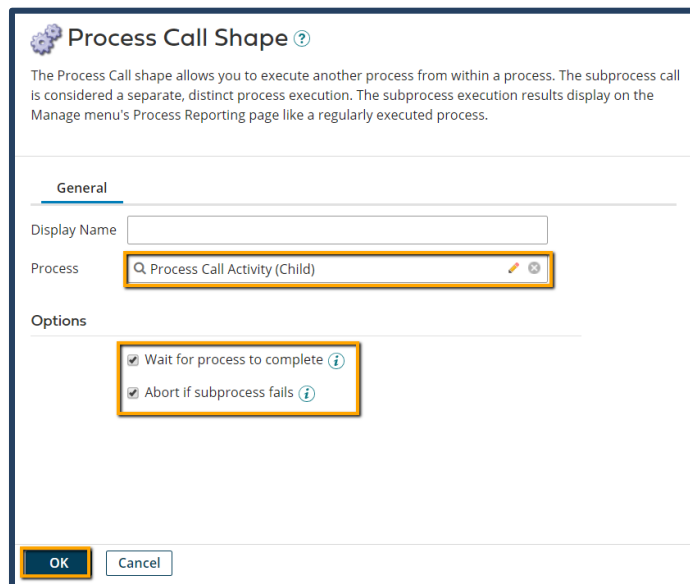
Library.



7. Select **Install** next to Process Call Activity (Child).
8. The Installation Location is already set to the Process Call folder.
9. Select **Install** in the lower-right corner.
10. The process is now installed in your account. Select **Close**.

Configure the Process Call

1. Open the Process Call folder and the **Process Call Activity (Parent)**. From the palette, Execute Tab, drag a **Process Call** to the Process Canvas.
2. In the Process Call, select **Choose...** and select the **Process Call Activity (Child)** process.



3. Select **OK**.
✓ *Wait for the process to complete option forces the parent process to wait until the*

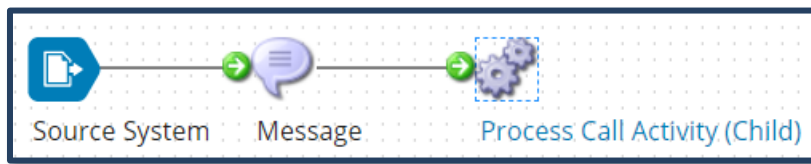
child process finishes. The default is checked.

- ✓ *Abort if the process fails option halts the parent process if the child process fails. The default is checked.*

4. Connect the Process Call to the Message.

The Message contains the incoming document containing the customer records.

No return paths extend from the Process Call. The return path appears when the child process is configured.



5. Switch from the Process Call Activity (Parent) process to the **Process Call Activity (Child)** process.
6. Open the Start shape.

 **Start Shape ?**

The Start shape is the main shape that begins the process flow. It is automatically added to each new process and it cannot be removed.

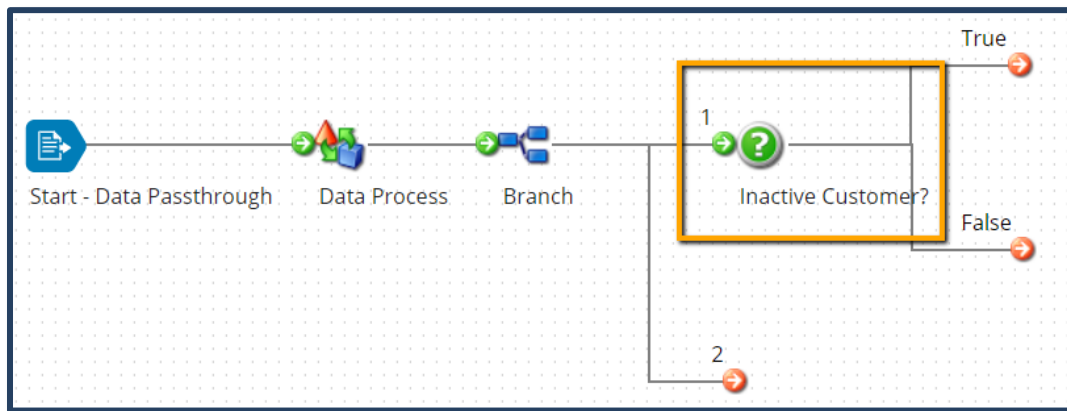
Process Mode	General
Type	☐ Connector ☐ Trading Partner ☐ Data Passthrough ☒ No Data

Select this option if this process should not receive or retrieve data from any source. In other words, the process will be scheduled and run similar to CRON or any other sort of job scheduler. The process will generate a single empty document.

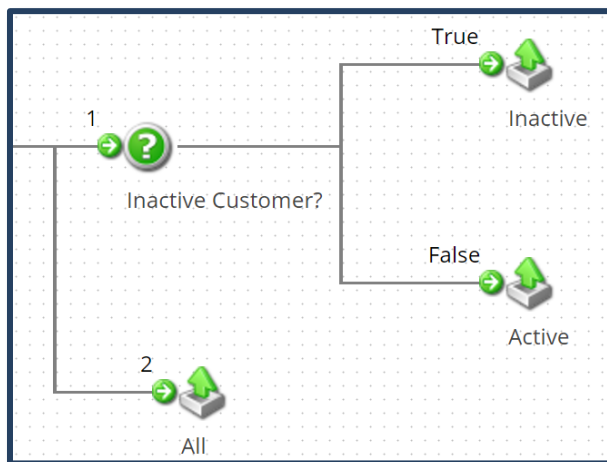
- ✓ *Should the Start shape type equal No Data? Think about if you want to send data to the Child Process.*

7. A Decision is already preconfigured to evaluate whether the customer record is Active or Inactive.

Process Call Activity

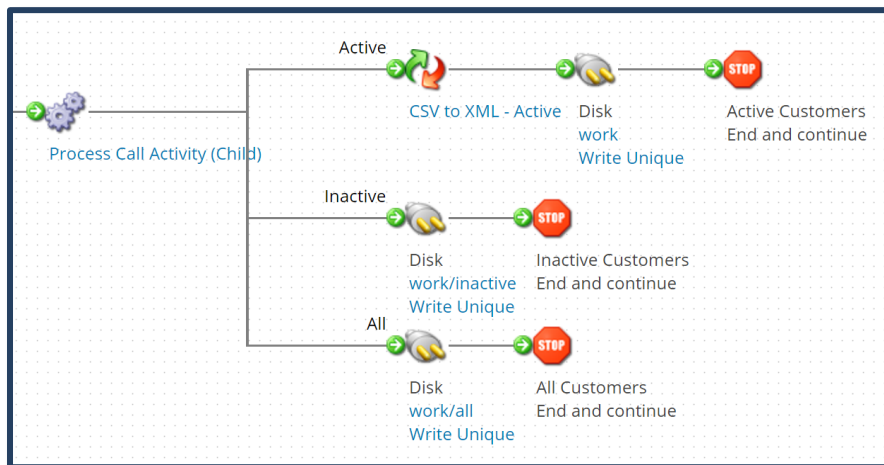


8. From the Shapes palette, Logic tab, add a **Return Document** for each branch.
9. Using the image below, enter a Display Name for all Return Documents. The Display Names offer contexts to the scenario and appear in the Parent Process.



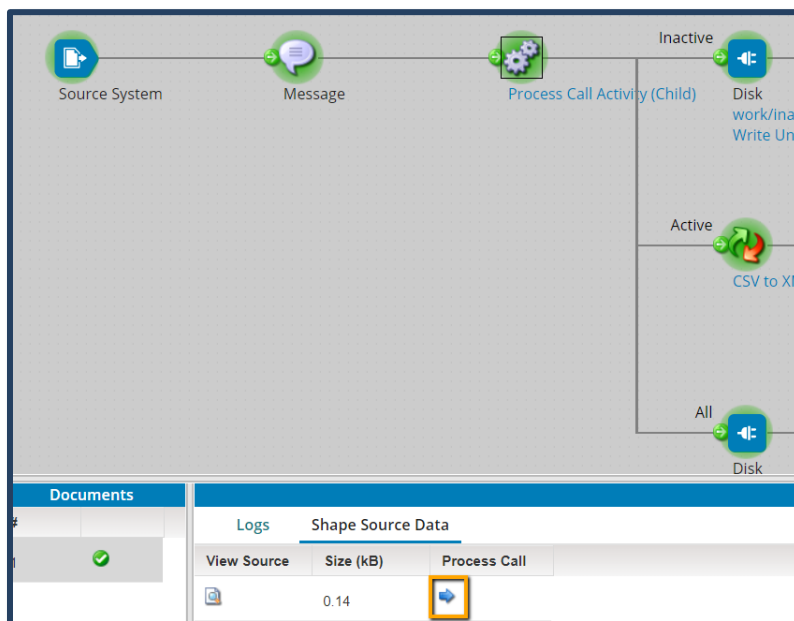
10. **Save** the Child Process.
11. Return to the **Process Call Activity (Parent)** process.
12. Reposition any branches off the canvas and connect each branch to the corresponding map or disk connector.

Process Call Activity



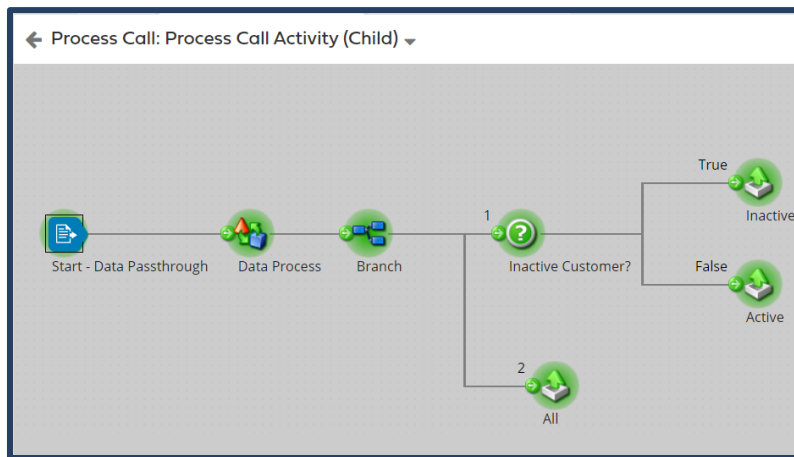
Test the process and view the results

1. Select **Save** and run a **Test**.
 - ✓ *Only the Parent process is executed. However, you will also observe the Child's process.*
2. Once the test is complete, select the **Process Call**.
3. In the **Shape Source Data** tab, under Process Call, select the **blue arrow**.

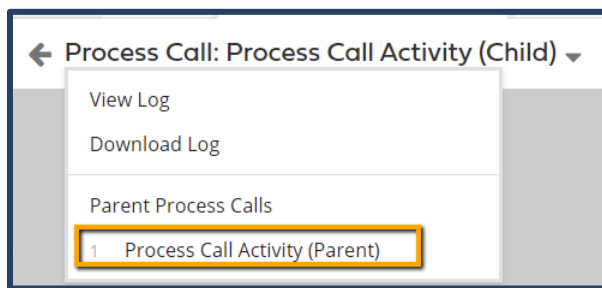


Notice the process view changes to the Child Process.

Process Call Activity



4. Select the **Process Call: Process Call Activity (Child)** and select the **Process Call Activity (Parent)**.



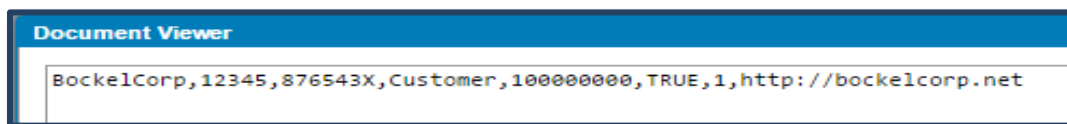
5. Select each **connector** in the Active, Inactive, and All paths to view output under the **Shape Source Data** tab.

Active

```
<Account>
  <Name>PaskoComm</Name>
  <Type>Customer</Type>
  <AccountNumber>976543X</AccountNumber>
  <Website>http://paskocomm.net</Website>
  <AnnualRevenue>100000000</AnnualRevenue>
  <NumberOfEmployees>1</NumberOfEmployees>
  <ExternalID__c>12346</ExternalID__c>
</Account>
```

Process Call Activity

Inactive



All

Logs	Shape Source Data	Connection Data
View Source	Size (kB)	
	0.07	
	0.07	

Business Rules Activity

One Business Rule can store multiple rules and conditions in a single catch-all container. A single Business Rule can define conditions in a Profile to replace multiple Decisions, using AND/OR operators to allow users to build compound logic.

Scenario

An online bookseller, Athens Books, purchases used books from other booksellers. The sellers upload their inventory lists, and Athens Books chooses which books to buy based on several criteria.

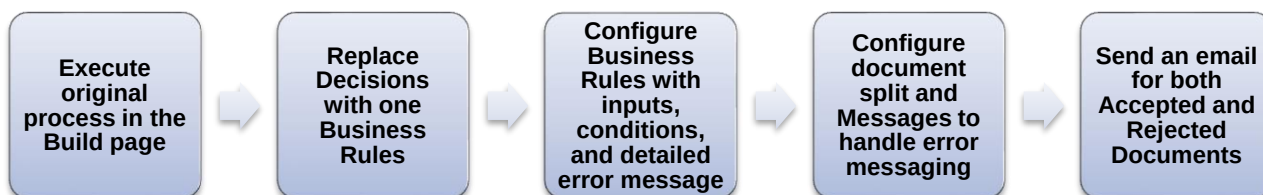
Athens Books specializes in Technology Books and only accepts items in specific genres, costing less than \$50. Assuming the book meets these criteria, Athens Books only buys copies when its book inventory is less than five.

Input field	Value(s)
Price	Less than \$50
Genre	Computer or Cloud-Based Integration or Web Programming
*Cross Reference Lookup	This function checks inventory levels which must be less than five books.

Goal

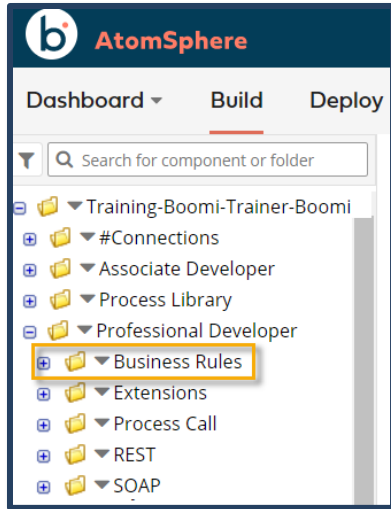
This activity changes a prebuilt process containing multiple Decisions to take advantage of the Business Rules and understand how the Business Rules Results Base Document Property is used for custom messaging.

The following diagram shows the high-level integration strategy and business workflow:



Create a subfolder within your account

1. Select the drop-down arrow next to the **Professional Developer** folder.
2. Choose New Folder and name it **Business Rules**.

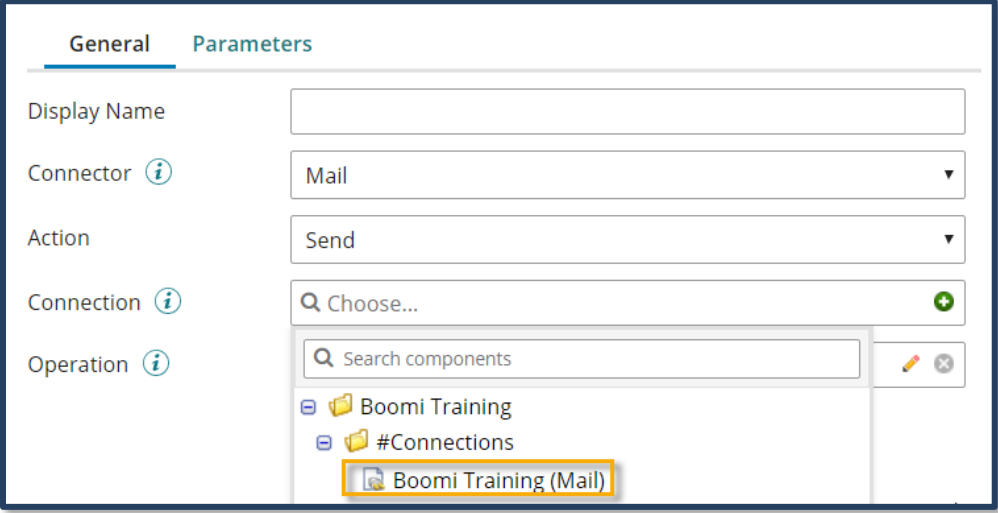


Load the process from the Process Library

1. Open the Process Library by selecting **Browse Process Library** at the bottom of the Component Explorer.
2. In the right column search bar, enter the phrase Business Rules.
3. Select Install next to **Business Rules Activity - Athens Books**.
4. Select Browse Select Installation Location and navigate to the Business Rules folder.
5. Select **Install** in the lower-right corner.
6. The process is now installed in your account. Select **View Process**.

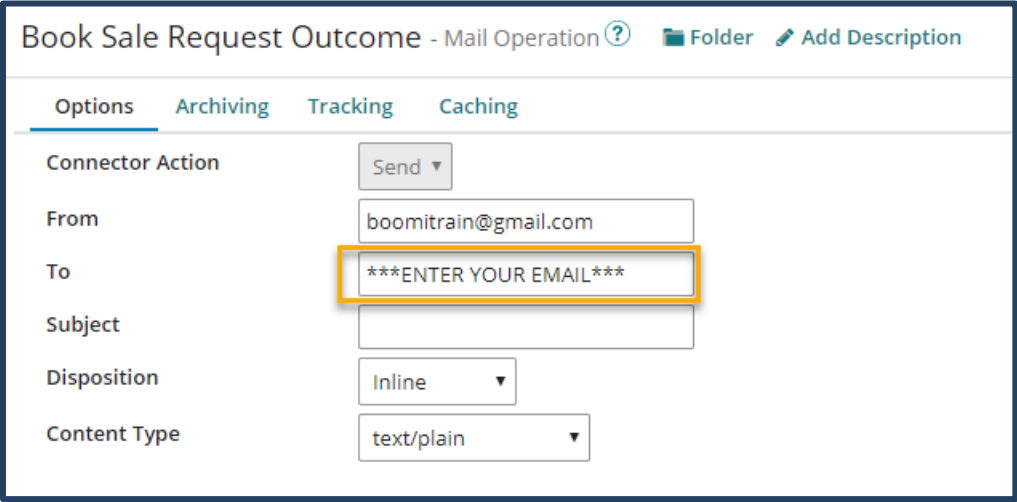
Configure the Mail Connector's Connection and Operation

1. The Business Rules Activity uses a reusable Connection already in your training account (from Associate Developer). Open the Mail **Connector**, select Browse next to Connection, and select **Boomi Training (Mail)**.



The screenshot shows the 'Mail Connector' configuration window with the 'Parameters' tab selected. The 'Connector' is set to 'Mail' and the 'Action' is 'Send'. The 'Connection' field is set to 'Boomi Training (Mail)', which is highlighted with a yellow box. The 'Operation' field is also highlighted with a yellow box and shows a search bar with 'Search components' and a list of components including 'Boomi Training', '#Connections', and 'Boomi Training (Mail)'.

2. Open the **Book Sale Request Outcome** Mail Operation, and in the **To** field, enter your email address.

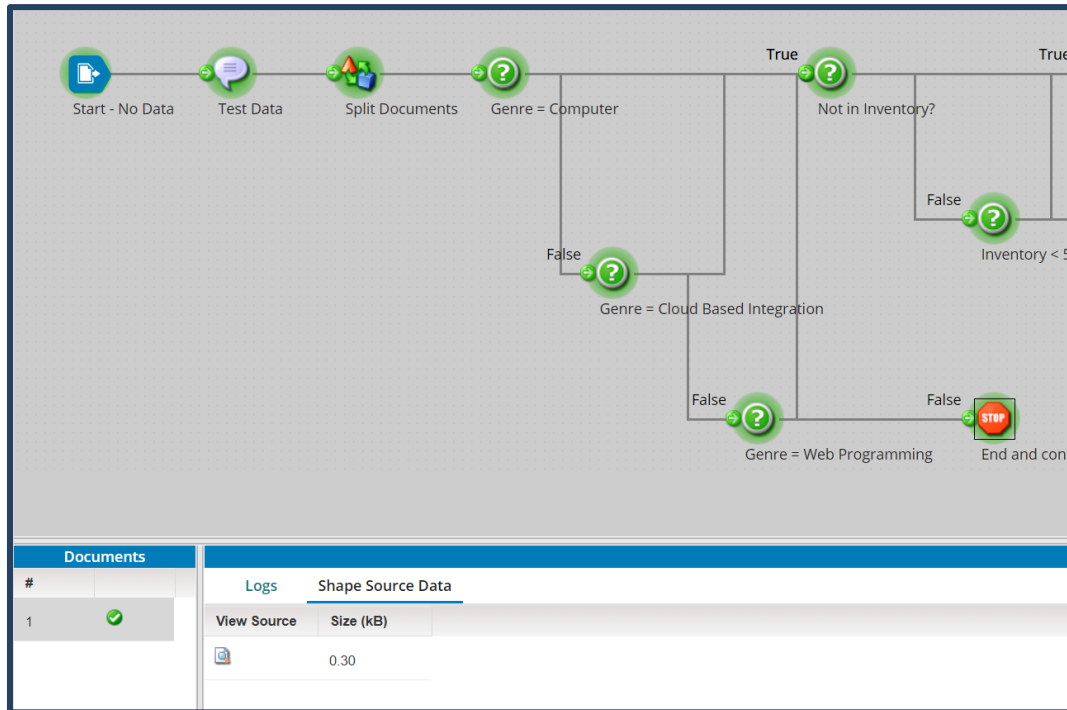


The screenshot shows the 'Book Sale Request Outcome - Mail Operation' configuration window. The 'Options' tab is selected. The 'Connector Action' is 'Send'. The 'From' field is 'boomitrain@gmail.com'. The 'To' field is highlighted with a yellow box and contains the text '***ENTER YOUR EMAIL***'. The 'Subject' field is empty. The 'Disposition' is 'Inline' and the 'Content Type' is 'text/plain'.

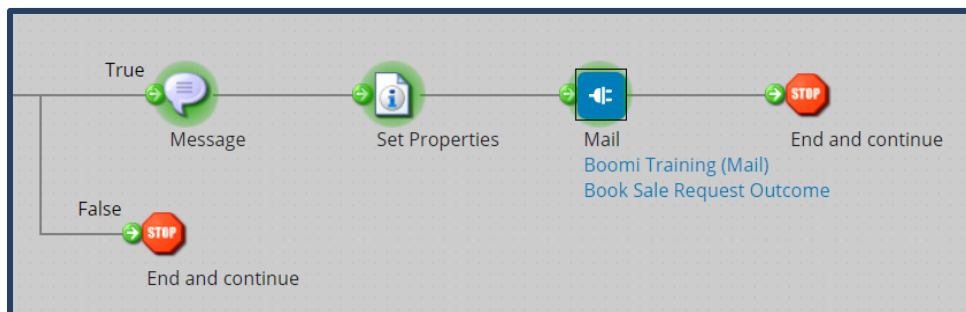
3. Select **Save and Close** and then **OK**.

Test the process and view the results

1. Run the process on the Build page to examine the functionality and output documents. Documents will flow down several different paths. Use the document viewer to examine them.



✓ Notice only one document meets all conditions.



Documents		Logs	Shape Source Data	Connection Data
#		View Source	Size (kB)	
1	✓		0.07	

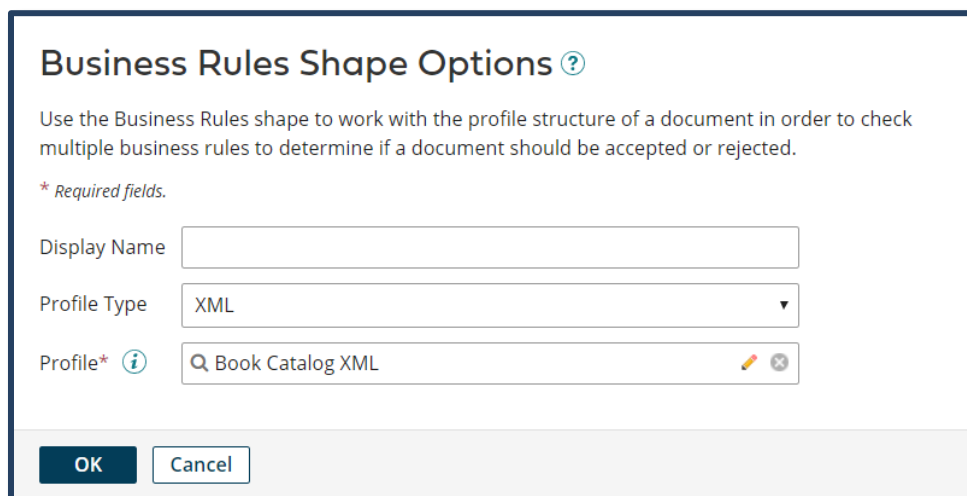
Business Rules Activity

2. Select **Return to Edit Mode** located above the process.
3. Detach the six Decisions and move them to the canvas side or bottom.
 - ✓ *Do not delete the Decisions; instead, replicate the same logic in the Business Rules.*

 You can move multiple shapes as a group. Select the shapes for the group and select any shape in the group to move them.

Configure the Business Rules

1. To configure the Business Rules from the Shapes palette, locate a Business Rules and drag it to the canvas. Select **XML** for the Profile type and load the provided Profile, **Book Catalog XML** for the Profile.



Business Rules Shape Options ?

Use the Business Rules shape to work with the profile structure of a document in order to check multiple business rules to determine if a document should be accepted or rejected.

* Required fields.

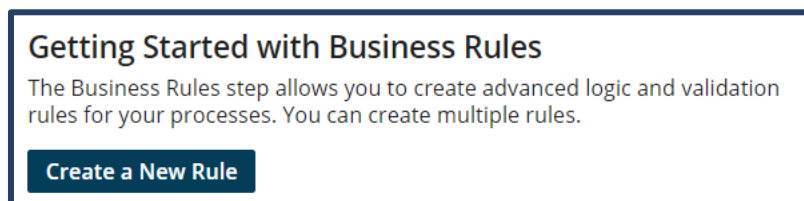
Display Name

Profile Type

Profile*   

OK **Cancel**

2. Select **OK**.
- The Business Rules Properties pane opens.



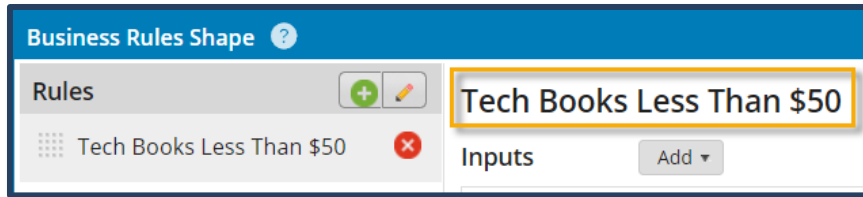
Getting Started with Business Rules

The Business Rules step allows you to create advanced logic and validation rules for your processes. You can create multiple rules.

Create a New Rule

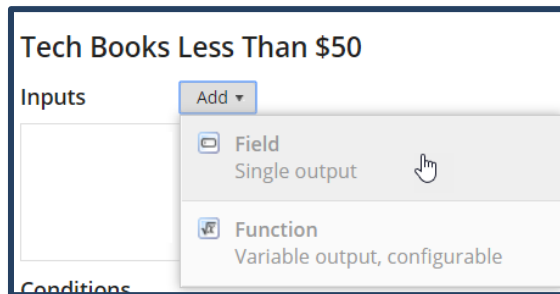
3. Select **Create a New Rule**.
4. Name the Business Rule **Tech Books Less than \$50**.

Business Rules Activity

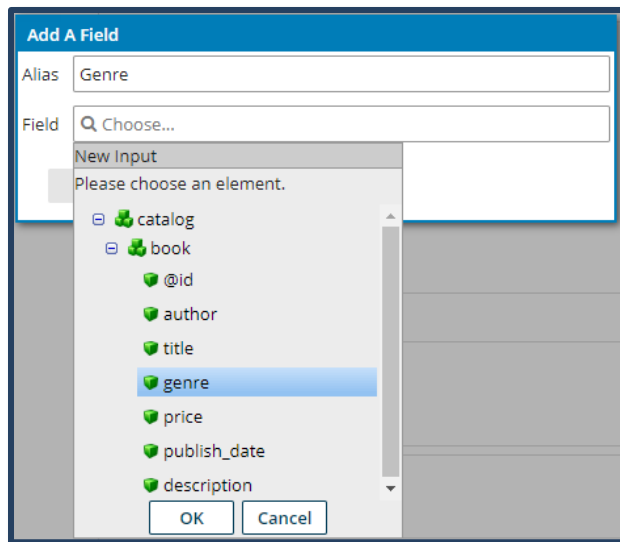


Add Inputs to the Business Rule

1. Select **Add** and select **Field** from the drop-down list.



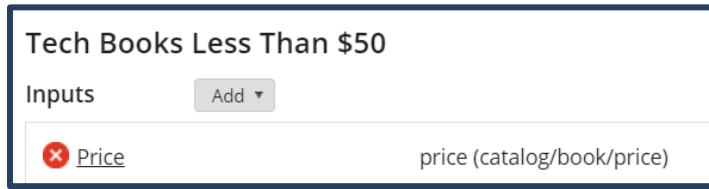
2. Enter **Genre** as the Alias, select **Browse** and select the **Genre** field.



3. Select **OK**.

Business Rules Activity

When adding Inputs, they appear in the Business Rules Properties 'Inputs' pane.



Tech Books Less Than \$50

Inputs Add ▾

✖ Price price (catalog/book/price)

4. Add Input for the Field listed below by repeating the previous 3 steps in the Add Inputs section.

- Alias = Price (Field = Price)

Add Conditions to the Business Rule

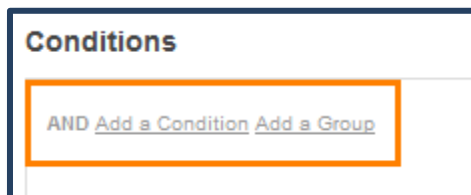
Follow the instructions below to set the following conditions:

- Price < 50
- Genre = Computer OR Cloud-Based Integration OR Web Programming



The instructions to set the Genre condition begins in the next section - Add a Group.

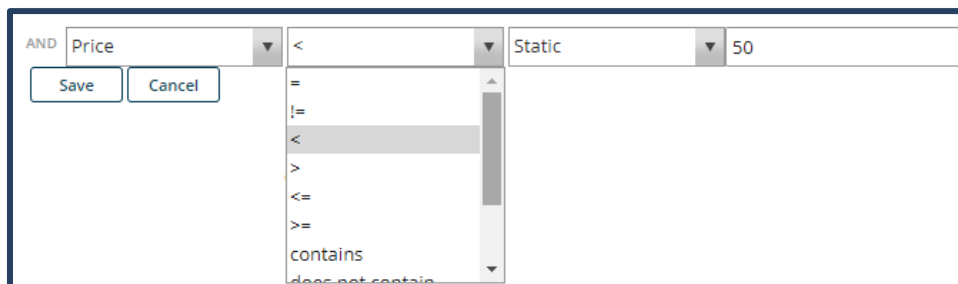
1. First, add your condition of Price < 50. Select **Add a Condition**.



Conditions

AND Add a Condition Add a Group

2. Use the **Less Than (<)** option and enter a static value of **50**.



AND Price < Static 50

Save Cancel

=
<=
>=
contains
does not contain

Business Rules Activity

3. Select **Save**.

 Notice the overall condition is based on conjunction, so the Top-Level Operator is **AND**.



Conditions

Top Level Operator **AND** OR

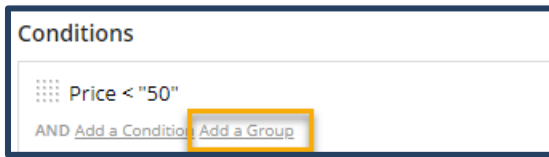
Price < "50"

 Added conditions appear in the Business Rules Properties pane conditions section.

Add a Group

The Genre can be either Computer or Cloud-Based Integration or Web Programming, meaning you must add a Group.

1. Select **Add a Group**.

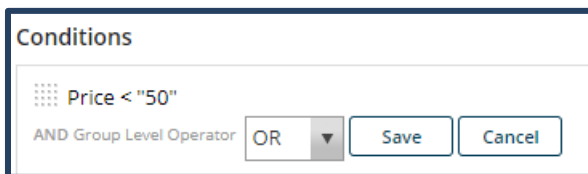


Conditions

Price < "50"

AND Add a Condition **Add a Group**

2. Select **OR** as the Group Level Operator and select **Save**.



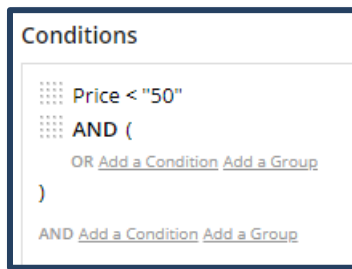
Conditions

Price < "50"

AND Group Level Operator **OR** Save Cancel

3. Select **Add a Condition** for each 'Genre' Condition:
(Genre = Computer or Genre = Cloud Based Integration or Genre = Web Programming)

Business Rules Activity

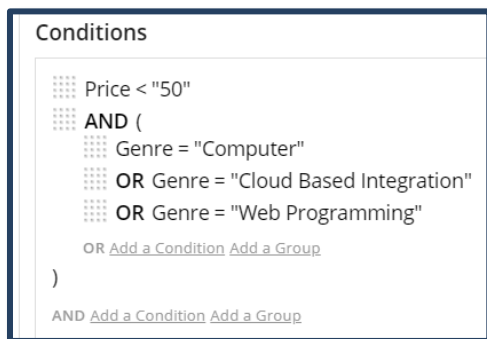


Conditions

```
Price < "50"
AND (
  OR Add a Condition Add a Group
)
AND Add a Condition Add a Group
```

Experiment with the logic options in the Business Rules Properties pane conditions section.

- ✓ Remember, there are multiple ways to set up your Conditions. Below is an example of one possible way to set up your conditions.



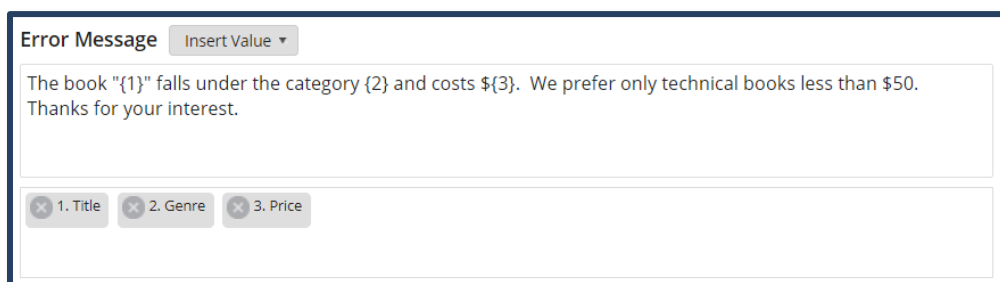
Conditions

```
Price < "50"
AND (
  Genre = "Computer"
  OR Genre = "Cloud Based Integration"
  OR Genre = "Web Programming"
  OR Add a Condition Add a Group
)
AND Add a Condition Add a Group
```

Configure the error message

It is time to configure a dynamic error message for each book NOT meeting the Business Rule.

1. Use the Error Message template below to assist you in configuring your error message:
- ✓ The Error Message references dynamic parameters unique to each document analyzed by the Business Rules. **Before configuring the Error Message**, define the **Title** as an input at the top of the Business Rules window.



Error Message Insert Value ▾

The book "{1}" falls under the category {2} and costs \${3}. We prefer only technical books less than \$50. Thanks for your interest.

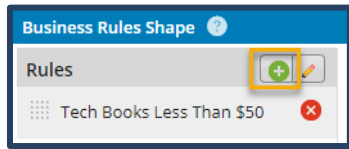
✕ 1. Title ✕ 2. Genre ✕ 3. Price

Business Rules Activity

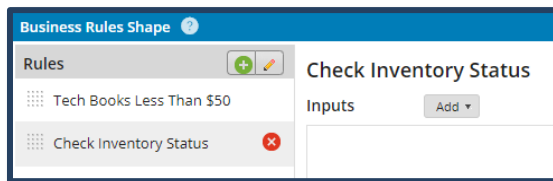
- ✓ *An error occurs if the error message contains a hard-coded return.*

Add Business Rule

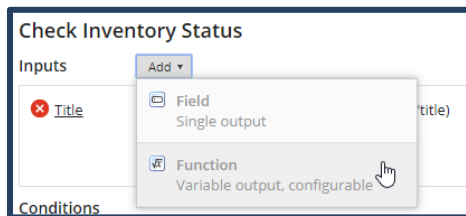
1. In the same Business Rules, add a second rule by selecting the plus icon.



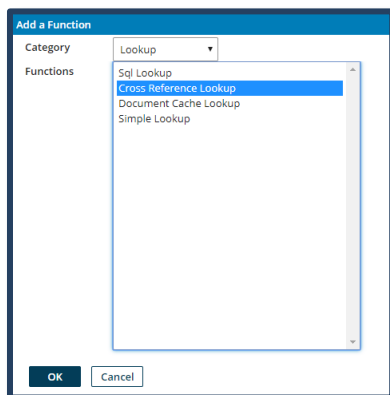
2. Call the rule **Check Inventory Status**.



3. Add the **Title** as an input.
4. Select **Add** and select **Function** from the drop-down list.

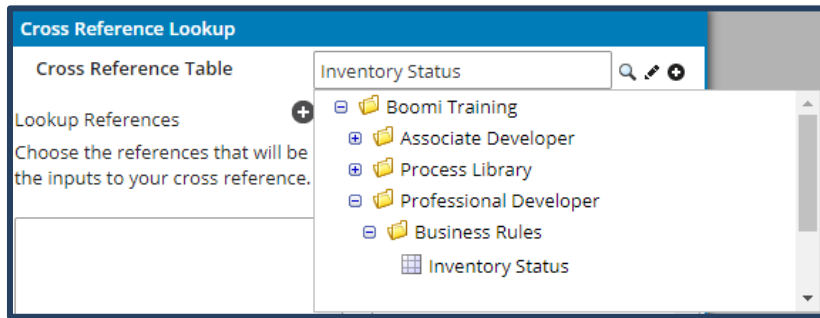


5. Choose **Lookup** as the **Category**, select **Cross Reference Lookup** from the Functions list, and select **OK**.

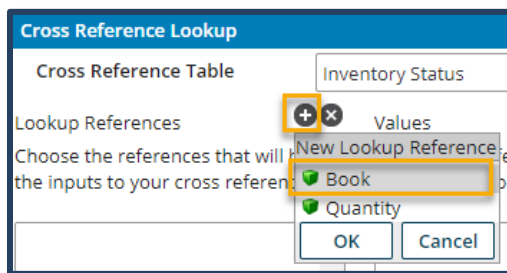


Business Rules Activity

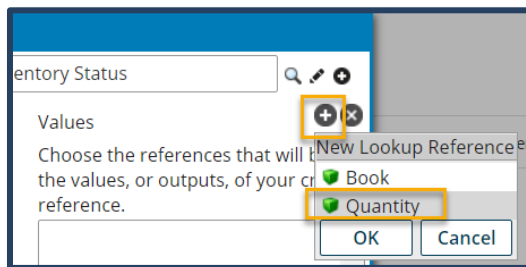
6. Select **Browse** and select the **Inventory Status** cross-reference table.



7. Select the plus icon to add a Lookup Reference of **Book**.

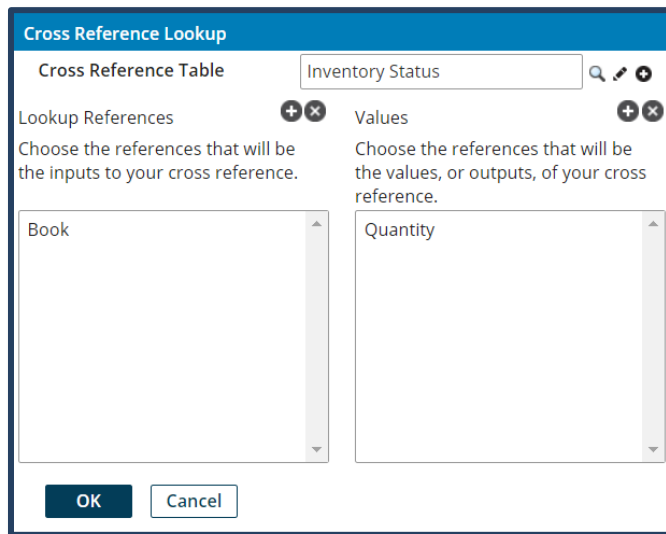


8. Select the plus icon to add an output Value of **Quantity**.



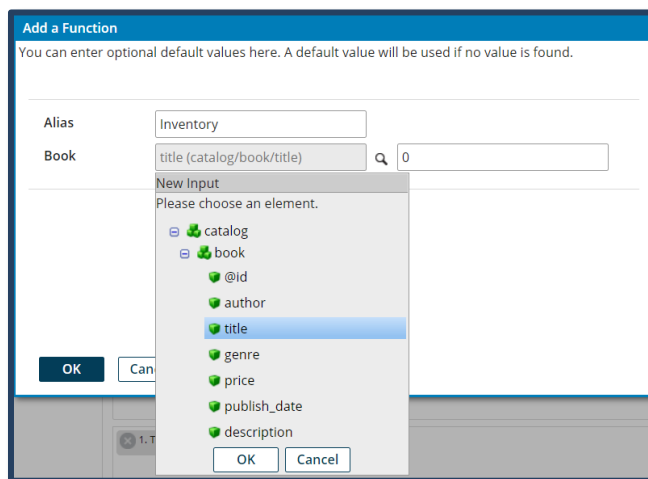
9. Select **OK**.

Business Rules Activity



The **Cross Reference Lookup** dialog box is shown. The **Cross Reference Table** is set to **Inventory Status**. Under **Lookup References**, the instruction is "Choose the references that will be the inputs to your cross reference." The list contains **Book**. Under **Values**, the instruction is "Choose the references that will be the values, or outputs, of your cross reference." The list contains **Quantity**. **OK** and **Cancel** buttons are at the bottom.

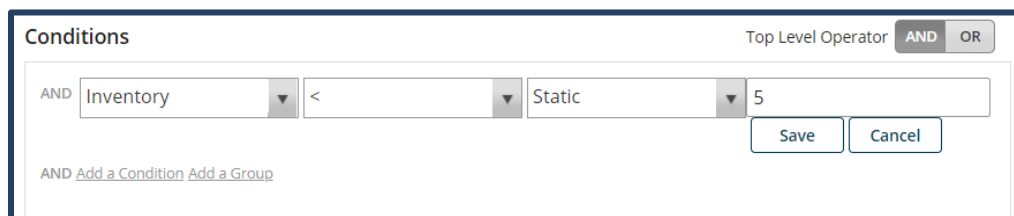
10. Enter **Inventory** as the Alias. Choose **Title** as the element, and enter a default value of **0**.



The **Add a Function** dialog box is shown. The **Alias** is **Inventory**. The **Book** is **title (catalog/book/title)**. A **New Input** dropdown menu is open, showing a list of elements: **catalog**, **book**, **@Id**, **author**, **title** (highlighted), **genre**, **price**, **publish_date**, and **description**. The **Default Value** is **0**. **OK** and **Cancel** buttons are at the bottom.

- ✓ If a **Title** does not exist in your Cross-Reference Table, then assign a default **Quantity** of **0**.

11. Add a Condition to check for titles with less than five items in the inventory.

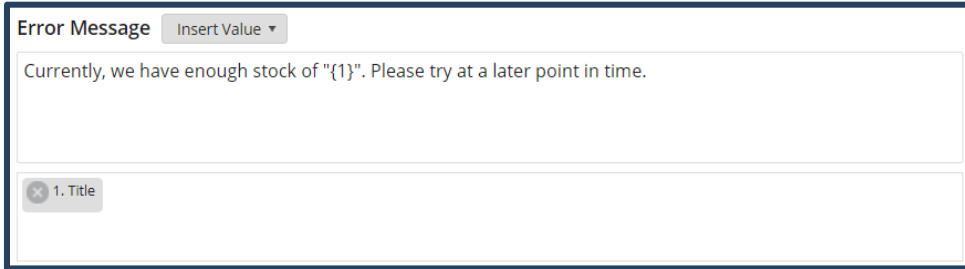


The **Conditions** dialog box is shown. The **Top Level Operator** is **AND**. The condition is: **Inventory** **<** **Static** **5**. **Save** and **Cancel** buttons are at the bottom. Below the condition, there are links: **AND** [Add a Condition](#) [Add a Group](#).

Business Rules Activity

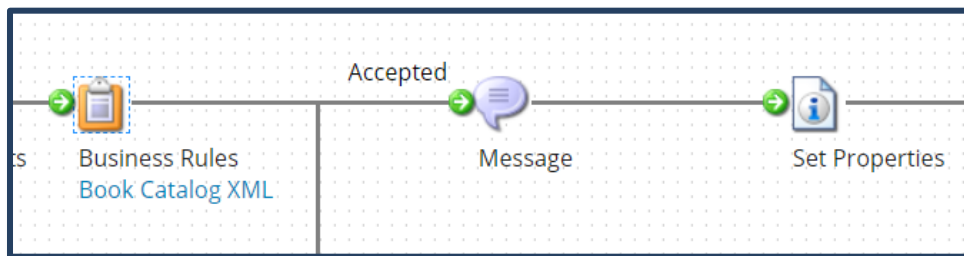
12. Configure the Error Message as shown below.

 *Add the Title first as input.*



13. Select **OK** to close the Business Rules.

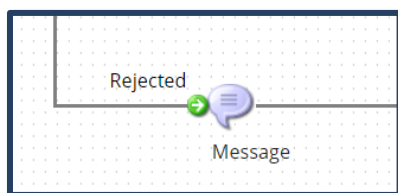
14. Connect the Data Process to the Business Rules and the Accepted path to the Message as shown below.



Format and Capture the Error Message

Next, set up the rejected path to capture the error message configured in the Business Rules Properties pane.

1. Add a Message to the Rejected path to store the “Base - Business Rules Result Message” Document Property.



2. Configure the Message Properties as shown below.

- ✓ *The “Base - Business Rules Result Message” is stored as an XML Document Property and references the Error Message configured earlier in the Business Rules.*

Business Rules Activity

Display Name: Business Rules Result XML

Option: ☒ Combine documents into a single message ⓘ

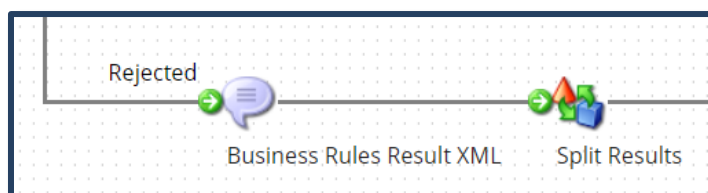
Message: {1}

Variables: {1} Document Property - Base - Business Rules Result M
essage

3. Select **OK** when finished.

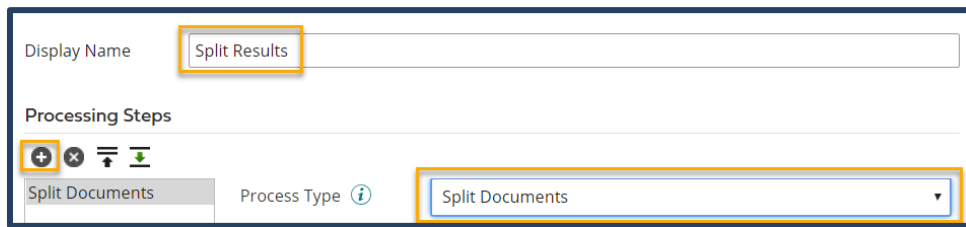
The Message is configured to capture the error message in the Business Rules, with the errors stored as XML. Next, you will format the error message to make the error message easier to read.

4. Within the Learning Management tool, the Materials Section of the course, download Materials for Activities – Pro Dev.zip. Unzip the file to use the BRR.xml file.
5. Add a Data Process to the canvas to split the Business Rules Result.



6. Configure the Data Process with a display name of **Split Results** and a **Split Documents** step.

Business Rules Activity



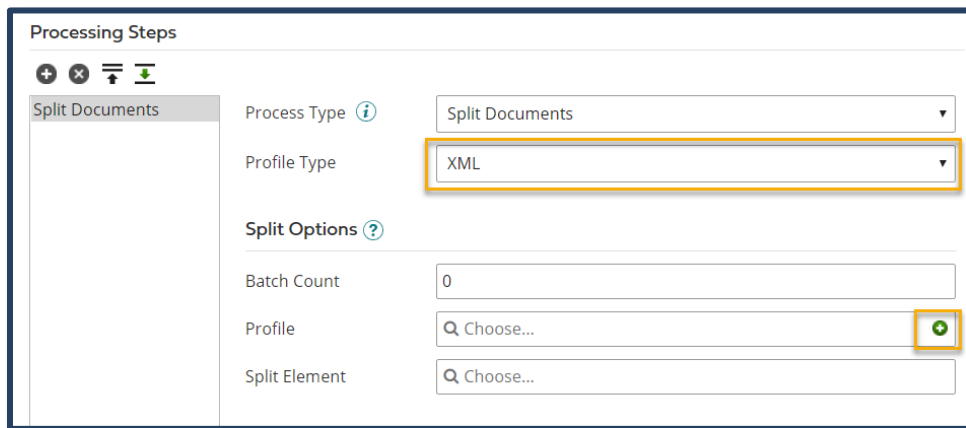
Display Name: Split Results

Processing Steps

Split Documents

Process Type: Split Documents

7. The Profile Type is **XML**. For the Profile, select the plus icon to create a new profile.



Processing Steps

Split Documents

Process Type: Split Documents

Profile Type: XML

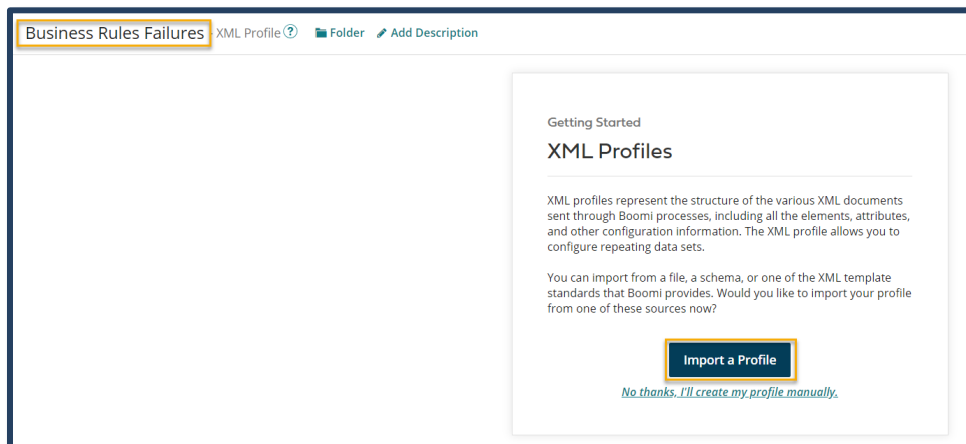
Split Options

Batch Count: 0

Profile: Choose... (+)

Split Element: Choose...

8. Name your profile Business Rules Failures and select **Import a Profile**.



Business Rules Failures XML Profile Folder Add Description

Getting Started

XML Profiles

XML profiles represent the structure of the various XML documents sent through Boomi processes, including all the elements, attributes, and other configuration information. The XML profile allows you to configure repeating data sets.

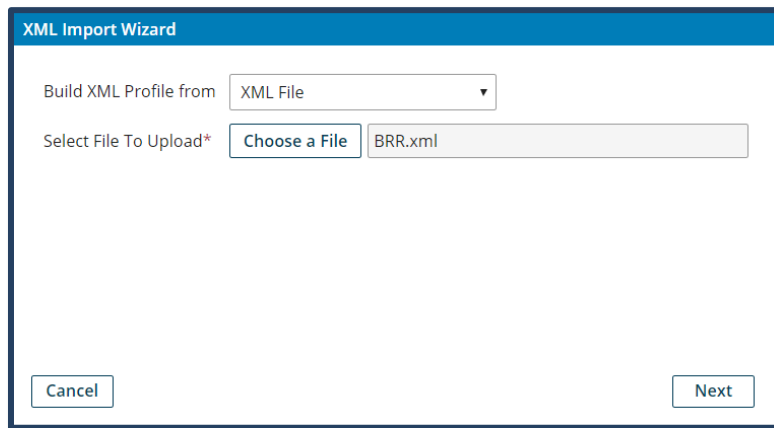
You can import from a file, a schema, or one of the XML template standards that Boomi provides. Would you like to import your profile from one of these sources now?

Import a Profile

No thanks, I'll create my profile manually.

9. Choose **XML File** from the Build XML Profile from the drop-down, choose your locally saved **BRR.xml** file, and select **Next**.

Business Rules Activity

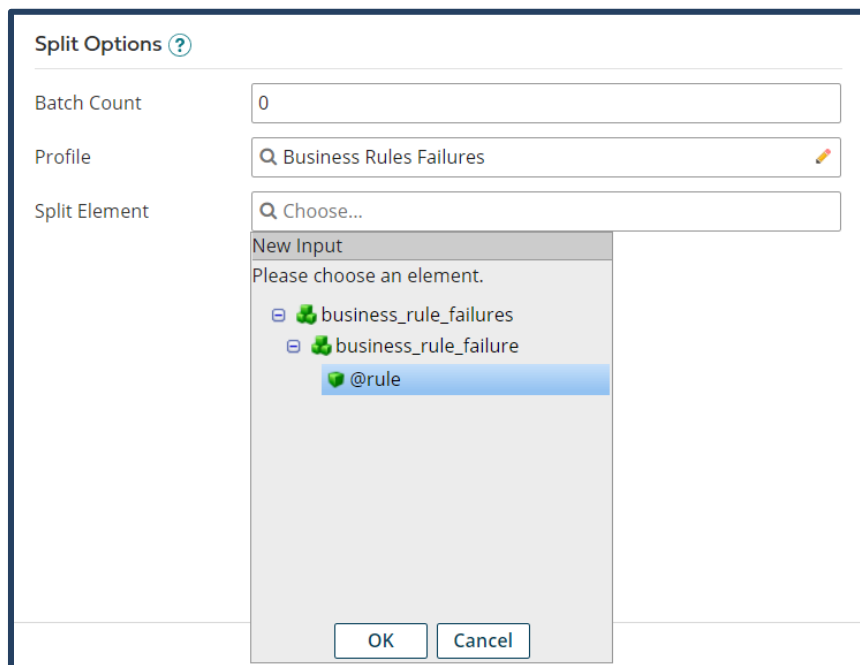


The XML Import Wizard dialog box has a title bar "XML Import Wizard". It contains a label "Build XML Profile from" with a dropdown menu showing "XML File". Below it is a label "Select File To Upload*" with a "Choose a File" button and a text field containing "BRR.xml". At the bottom left is a "Cancel" button and at the bottom right is a "Next" button.

10. On the next screen, select **Finish** and **Save, and Close**.

 *In this exercise, you created the BRR.xml sample file for yourself as a time saver. If you needed to create your own, you would run the process with the Message in place, copy and paste the XML output to a text editor, and save the file.*

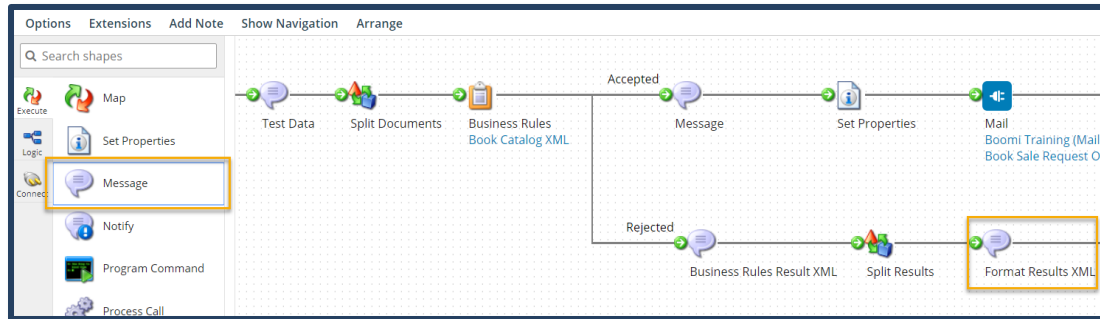
11. Leave the Batch Count at **0**, and set the Split Element to the **@rule** element.



The Split Options dialog box has a title bar "Split Options ?". It contains three fields: "Batch Count" with a value of "0", "Profile" with a value of "Business Rules Failures", and "Split Element" with a value of "Choose...". Below the "Split Element" field is a dropdown menu with the following options: "New Input", "Please choose an element.", "business_rule_failures", "business_rule_failure", and "@rule". The "@rule" option is selected and highlighted. At the bottom are "OK" and "Cancel" buttons.

12. Drag and drop a new Message onto the Process Canvas.

Business Rules Activity

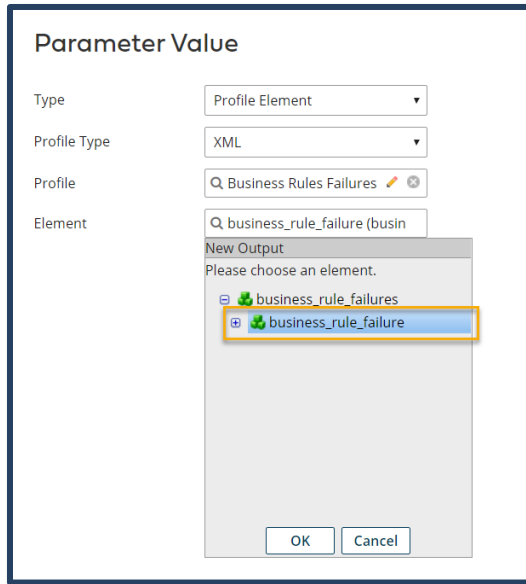


13. Configure the Message as shown below:

The screenshot shows the 'Message' configuration dialog box. The 'Display Name' is 'Format Results XML'. The 'Option' is 'Combine documents into a single message' (checked). The 'Message' field contains '{1}'. The 'Variables' section shows a list of variables: '{1} XML Profile - Business Rules Failures - business_rule_failure (business_rule_failures/business_rule_failure)'.

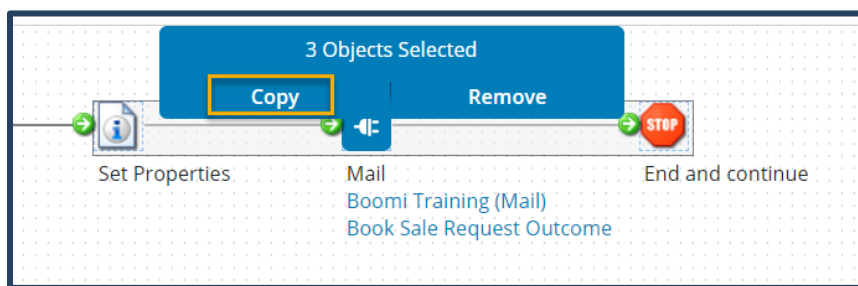
 You are not pulling the failure message directly from the Base Business Rules Result in the document property but from a profile element in your newly created Business Rule Failure XML profile.

Business Rules Activity

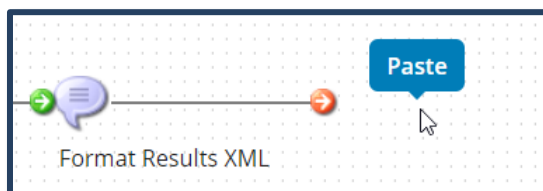


The 'Parameter Value' dialog box is shown. It has four fields: 'Type' set to 'Profile Element', 'Profile Type' set to 'XML', 'Profile' set to 'Business Rules Failures', and 'Element' set to 'business_rule_failure (busin'. A 'New Output' dropdown is open, showing 'business_rule_failures' and 'business_rule_failure' (the latter is highlighted with a blue border). 'OK' and 'Cancel' buttons are at the bottom.

14. Select the last three shapes in the Accept path and select **Copy**.



15. Paste the selected objects at the end of the Rejected path after the Format Results XML Message.



16. Update the Set Properties shape in the Rejected path. Set the Mail Subject to **The following books were not selected.**

Business Rules Activity

Display Name

Properties to Set

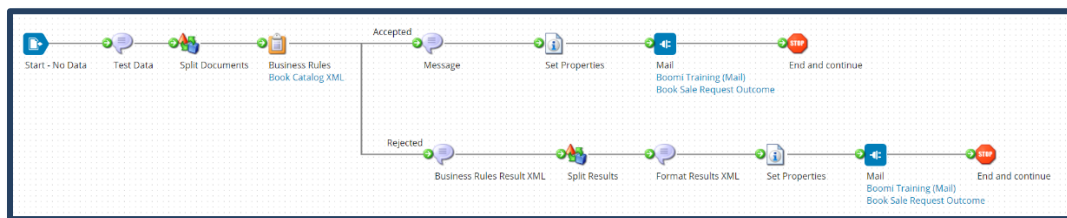
Mail - Subject

Parameters

Static value of 'The following books were not selected'

✓ Do not make any changes to the Mail Connection.

Your process should look like this:



17. **Save** and **Test** the process.

18. Review the output of both the Accepted and Rejected paths and compare this to the email(s) you received.

Rejected

Business Rules Result XML

Split Results

Format Results XML

Set Properties

Mail: Boomi Training (Mail) Book Sale Request Outcome

End and continue

Test Results

Logs

Shape Source Data

Connection Data

View

From Address

To Address

Subject

Message Id

Host

User

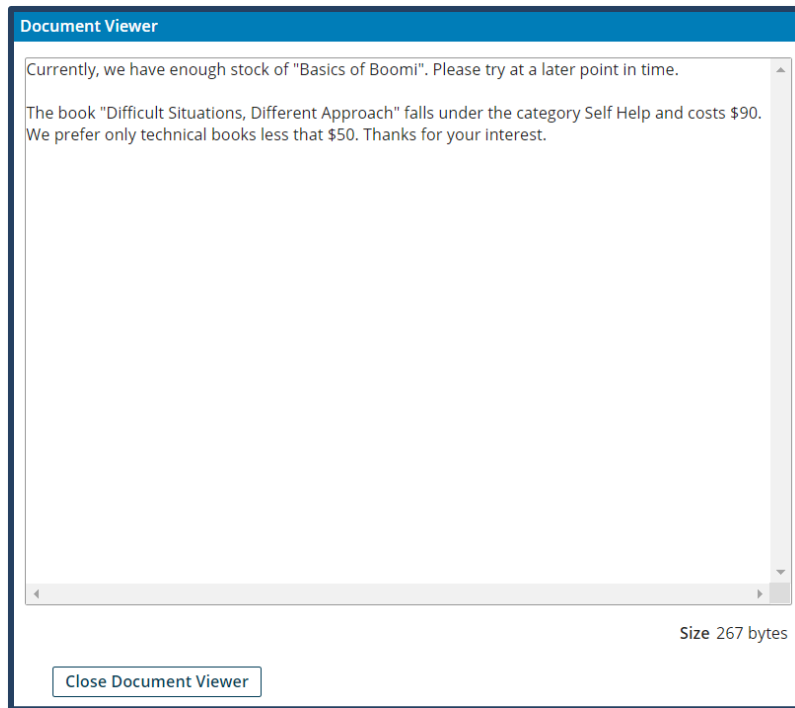
File Name

boomitrain@gmail.com

The following books were not

<1555521502.3.1581433325159.jax_email-smtp.us-west-2.am_AKIA0XMKUNQJDKH535H

Business Rules Activity



The process uses the same connection and operation components as the Mail connector shape in the Accepted path.

Using multiple connector shapes to reuse the same connection component does NOT affect connection licenses. It is more of a visual layout choice.

Document Cache Activity

With document caching, you can add documents to a cache or memory to reference later in a process or sub-process. You can index profile fields to use during the process execution. You can also activate multi-source profile mapping with cached documents to reference these stored multiple times when integrating with other systems.

Here are some document cache features:

- **Add to Cache** stores profile data at the front end of processing.
- **Load from Cache** gets the documents from a document cache.
- **Document Cache Component** is a reusable component for indexing and key definitions.
- **Add Cached Data** is a mapping feature with a source profile definition and a key linking the exposed corresponding cached profile fields.

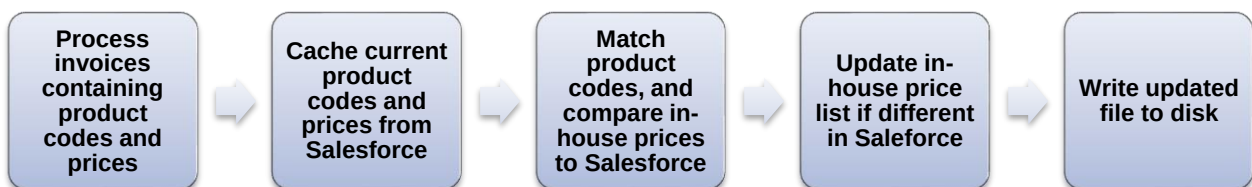
Scenario

Canyon Enterprises stores its product information and maintains the most up-to-date pricing in Salesforce. Campus Store, an independent reseller, carries many of Canyon Enterprise's products but keeps its in-house pricing database. Sometimes Campus's in-house prices can get out of sync with Canyon's prices.

Goal

Use document caching to keep Campus Store's prices in sync with Canyon Enterprises'.

The following graphic shows the high-level integration strategy and business workflow.



Create a subfolder within your account

1. Select the drop-down arrow next to the **Professional Developer** folder.
2. Choose New Folder and name it **Document Cache**.

Load the process from the Process Library

1. Open the Process Library by selecting **Browse Process Library** at the bottom of the Component Explorer.
2. In the right column on the search bar, enter **Document Cache** and select **Enter**.
3. Select **Install** next to **Document Cache Activity**.
4. Select **Browse** to Select Installation Location and navigate to the Document Cache folder.
5. Select **Install** in the lower-right corner.
6. The process is now installed in your account. Select **View Process**.

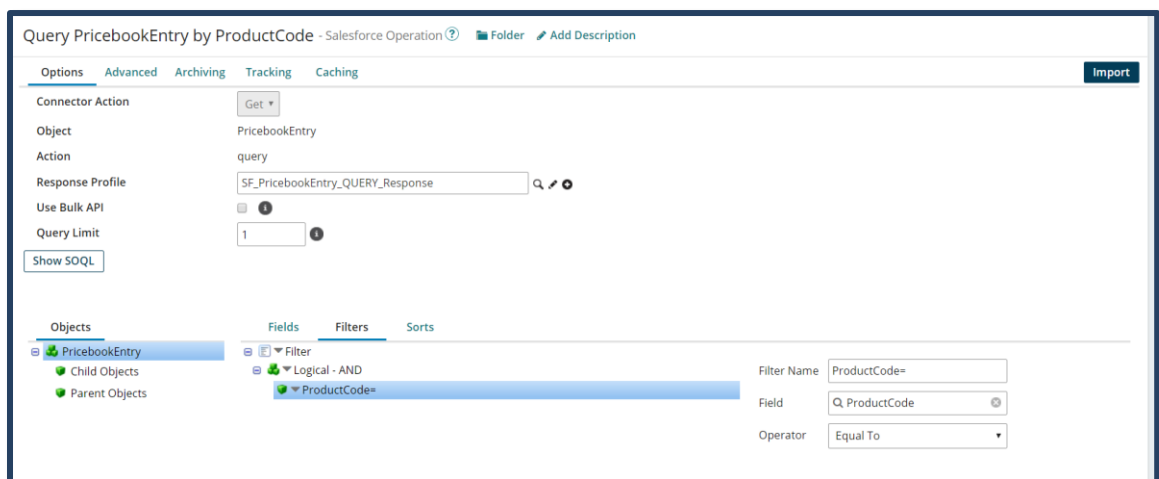
Examine the filter



To limit the number of returned Salesforce records, you created a query to return Salesforce products and prices only for products listed in the incoming in-house flat file document, *Product List FF*. A filter in the operation references the Salesforce *ProductCode* field. In the *Parameter* tab of the *Salesforce Connector*, this filter is applied and pulls records from Salesforce where the *ProductCode* matches the incoming flat file *ProductCode* field.

1. In the Salesforce Operation, under Objects, select **PricebookEntry**.
2. In the bottom-right pane, expand the **Filters** tab and select **ProductCode**.

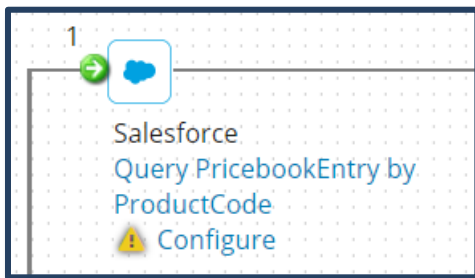
The Filter Name, Field, and Operator values are displayed.



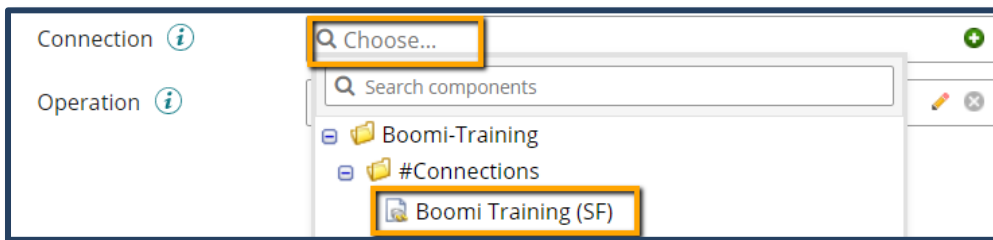
3. When finished examining the filter, select **Close**.

Load the Salesforce Connection and configure the action parameter

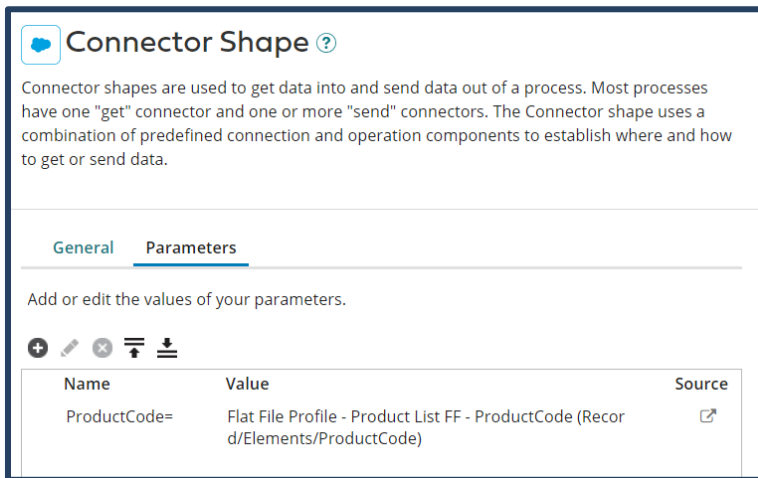
1. Open the Salesforce Connector.



2. Select the **Boomi Training (SF)** Connection.



3. Select the **Parameters** tab, and select the plus icon.
4. Reference Flat File Profile - Product List FF - ProductCode (Record/Elements/ProductCode).

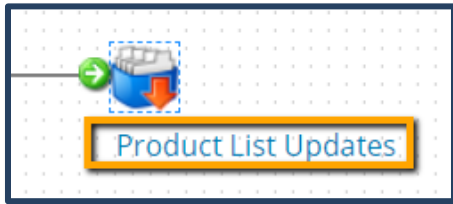


5. When finished, select **OK**.

You configured the Salesforce connector to query only records containing the same product code in the incoming flat file. To store the data in the cache, configure the Document Cache to index the Salesforce XML response data (SF_PricebookEntry_QUERY_Response) based on the ProductCode field.

Document Cache Activity

6. Select the Product List Updates hyperlink under the Add to Cache shape.



 This process supplies the *SF_PricebookEntry_QUERY_Response* XML profile. To build from scratch, run the process on the Build page to capture the response data from the Salesforce Connection before adding and configuring the Add to Cache shapes.

7. For Profile, select for SF_PricebookEntry_Query_Response and select **Add Index**.

A screenshot of the 'Product List Updates - Document Cache' configuration window. The 'Profile Type' is set to 'XML'. The 'Profile' field contains 'SF_PricebookEntry_QUE'. The 'Option' section has a checkbox for 'Enforce one index entry per document'. The 'Indexes' section has an 'Add Index' button highlighted with a yellow box.

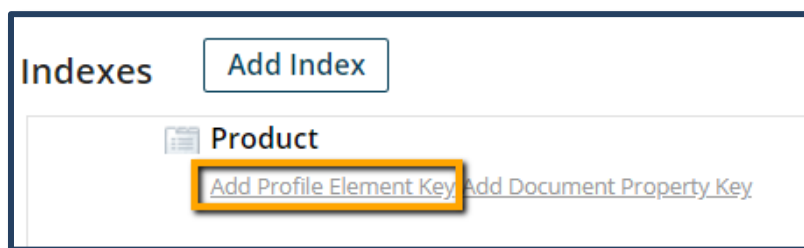
 The *ProductCode* field is added as an Index.

8. Name the new Index **Product** and select **OK**.

A screenshot of the 'Add Index' dialog box. The 'Index Name' field contains the text 'Product'. Below the field are 'OK' and 'Cancel' buttons. The 'OK' button is highlighted with a yellow box.

9. Select **Add Profile Element Key** hyperlink.

Document Cache Activity

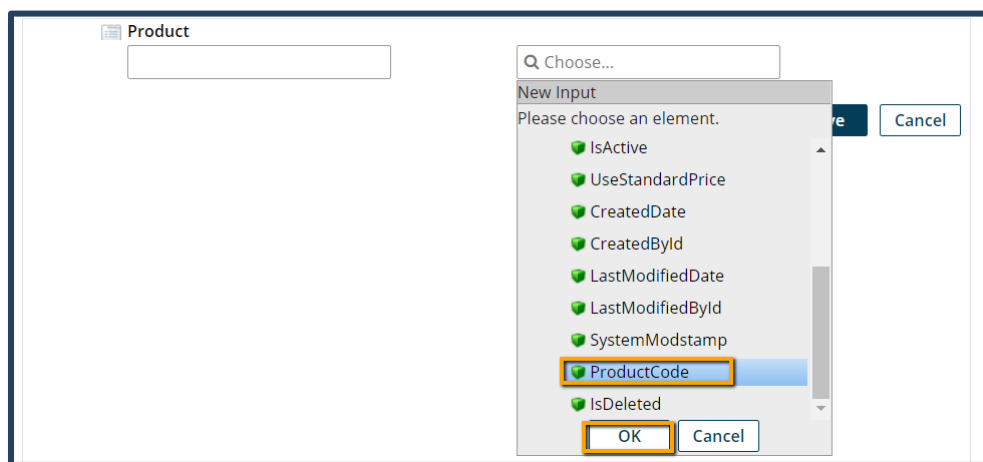


Indexes [Add Index](#)

Product

[Add Profile Element Key](#) [Add Document Property Key](#)

10. Select **ProductCode**.



Product

Choose...

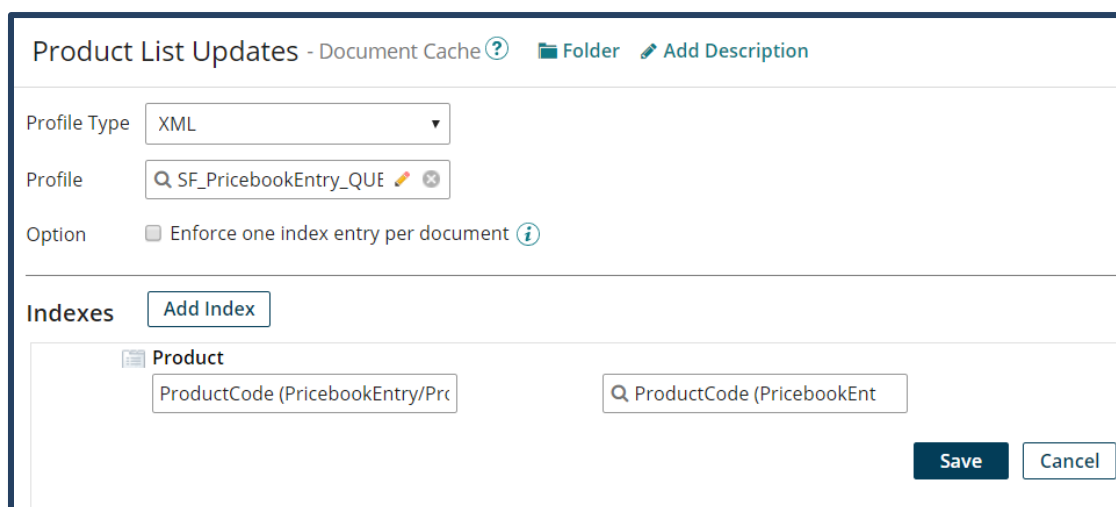
New Input

Please choose an element.

- IsActive
- UseStandardPrice
- CreatedDate
- CreatedByld
- LastModifiedDate
- LastModifiedByld
- SystemModstamp
- ProductCode**
- IsDeleted

[OK](#) [Cancel](#)

When complete, the index will look like the image below:



Product List Updates - Document Cache [?](#) [Folder](#) [Add Description](#)

Profile Type

Profile [✎](#) [✕](#)

Option ☒ Enforce one index entry per document [i](#)

Indexes [Add Index](#)

Product

ProductCode (PricebookEntry/Prc)

[Save](#) [Cancel](#)

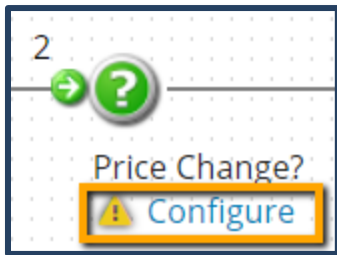
11. Select **Save** to return to the process. Select **Save** to save the process and finish the first branch.

Document Cache Activity

For each incoming flat file containing the product code and price, a Salesforce XML document is returned, and the Add to Cache shape stores the document in the cache. You can reference this shape anywhere in this process.

If the Salesforce product price does not match the in-house price in the incoming flat file document, update the in-house price in the second branch.

12. Select the **Configure** hyperlink under the Price Change? Decision.



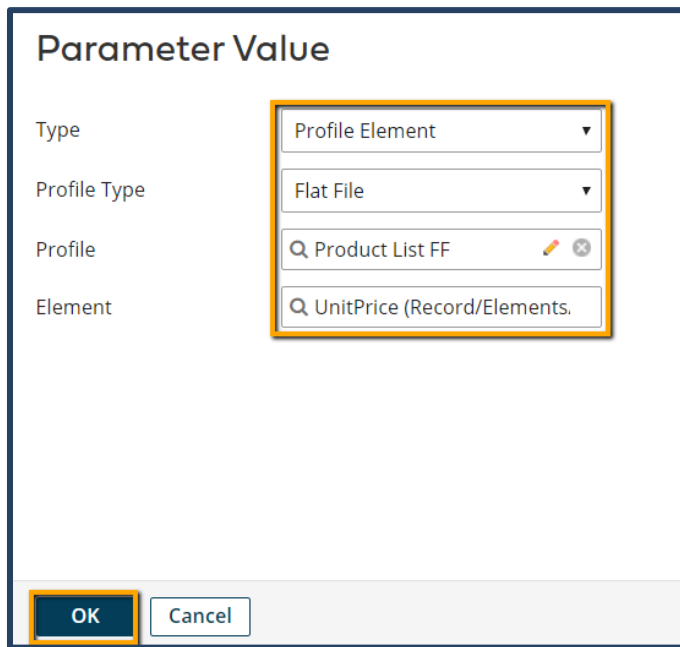
13. For First Value, select **Choose...**

A screenshot of the Boomi Decision Shape configuration window. The window has a title bar with a question mark icon and the text 'Decision Shape'. Below the title bar, there is a description: 'The Decision shape routes documents based on a true/false comparison of two values. Those values can be anything from field values in the current document (profile elements), static values, results of a database, an application query, and more.' The configuration area contains four fields: 'Display Name' with the value 'Price Change?', 'First Value' with a search icon and the text 'Choose...', 'Comparison' with a dropdown menu showing 'Not Equal To', and 'Second Value' with a search icon and the text 'Choose...'. The 'First Value' field is highlighted with an orange rectangular box.

14. In the Parameter Value window for Profile Element, set:

Type:	Profile Element
Profile Type:	Flat File
Profile:	ProductListFF
Element:	UnitPrice (Record/Elements/UnitPrice)

15. Select **OK**.

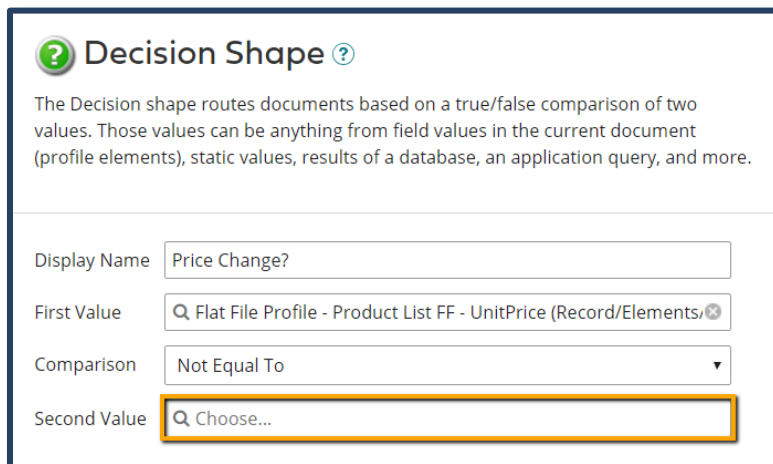


The 'Parameter Value' dialog box is shown with the following fields:

- Type: Profile Element
- Profile Type: Flat File
- Profile: Product List FF
- Element: UnitPrice (Record/Elements)

The 'OK' button is highlighted with an orange border.

16. For Second Value, select **Choose....**



The 'Decision Shape' configuration dialog is shown with the following fields:

- Display Name: Price Change?
- First Value: Flat File Profile - Product List FF - UnitPrice (Record/Elements)
- Comparison: Not Equal To
- Second Value: Choose...

The 'Second Value' field is highlighted with an orange border.

17. In the Parameter Value window for Document Cache Lookup, set:

Type:	Document Cache Lookup
Choose Document Cache:	Product List Updates
Cache Index	Product
Output Parameter:	UnitPrice PricebookEntry/UnitPrice)
Keys:	ProductCode (PricebookEntry/ProductCode

Document Cache Activity

Parameter Value

Type: Document Cache Lookup

Choose Document: Product List Updates

Cache:

Cache Index: Product

Output Parameter: UnitPrice (PricebookEntry/U

Keys: ProductCode (PricebookEntry/ProductCode)

Parameter value: Choose...

18. In the Parameter value window, select **Choose...** and configure the following:

Type:	Profile Element
Profile Type:	Flat File
Profile:	ProductListFF
Element:	ProductCode

19. Select **OK** when finished.

Parameter Value

Type: Profile Element

Profile Type: Flat File

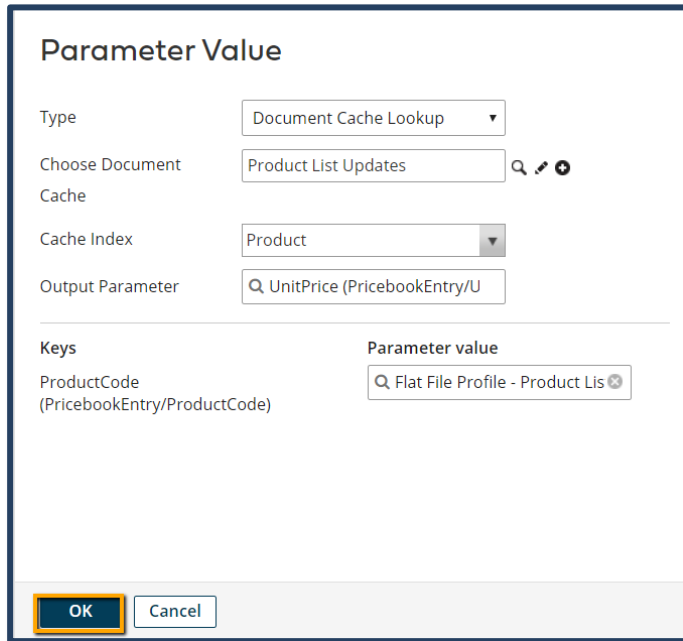
Profile: Product List FF

Element: ProductCode (Record/Elem

OK Cancel

Document Cache Activity

The result looks like the below:



The dialog box is titled "Parameter Value". It contains the following fields:

- Type: Document Cache Lookup (dropdown)
- Choose Document: Product List Updates (text input with search, edit, and delete icons)
- Cache: (empty text input)
- Cache Index: Product (dropdown)
- Output Parameter: UnitPrice (PricebookEntry/U (text input with search icon)

Below these fields is a table with two columns: "Keys" and "Parameter value".

Keys	Parameter value
ProductCode (PricebookEntry/ProductCode)	Flat File Profile - Product Lis (text input with search icon)

At the bottom are "OK" and "Cancel" buttons. The "OK" button is highlighted with a yellow border.

20. Select **OK**.



If the Keys and Parameter values match, the Output Parameter, UnitPrice, is returned from the cache and stored as the Second Value in the Decision.

The decision logic states: If the price changes, map the price field stored in the document cache from the Salesforce XML response to the updated Flat File. You can use Add Cached Data in the source profile, Product List FF, or the Document Cache Lookup function. You will use the second option, the Document Cache Lookup function.

Configure the Document Cache Lookup and map source and destination fields

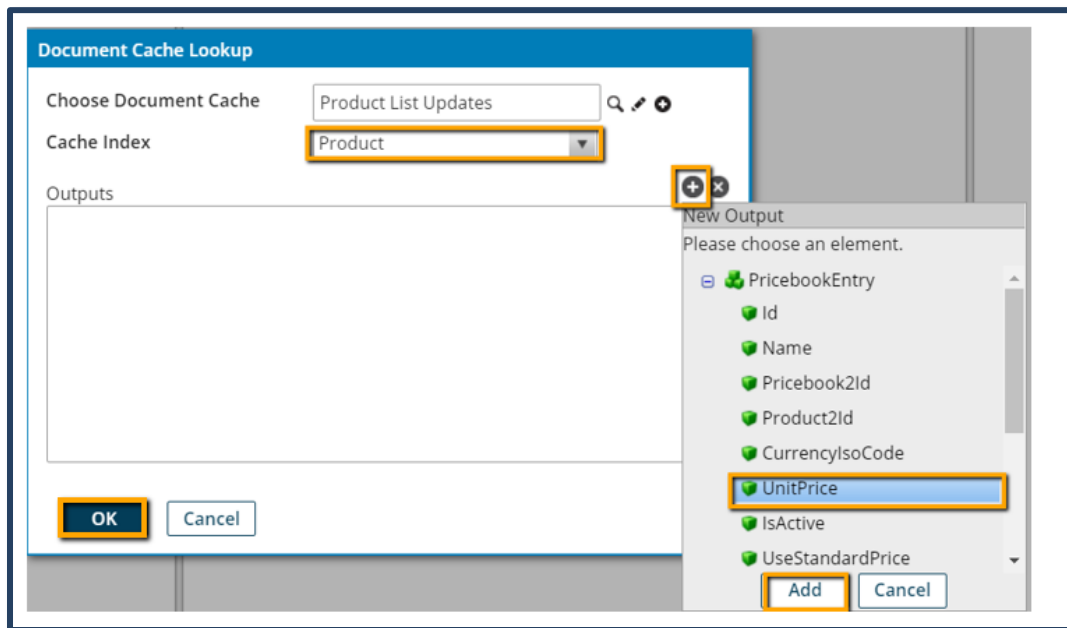
1. Open the **Price Updates** Map.
2. Select the Edit icon to open the **Document Cache Lookup**.



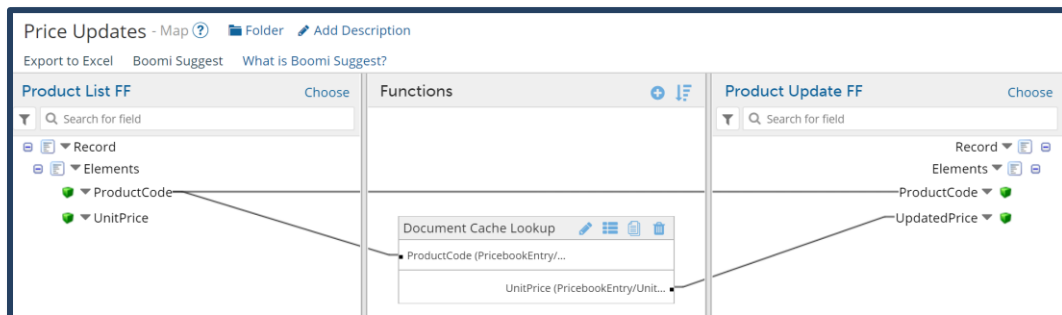
The "Functions" panel shows a list of functions. The "Document Cache Lookup" function is highlighted with a yellow border. To the right of the function name are icons for edit, add, and delete.

3. For the Cache Index, choose **Product**, and for Add an Output, select **UnitPrice**. Select **Add**, and select **OK** to return to the Map.

Document Cache Activity



4. Map **ProductCode** from the source profile into the Document Cache Lookup and map the **UnitPrice** retrieved from the Document Cache to the **UpdatedPrice** destination field of the Product Update FF.



The Document Cache Lookup function takes the incoming Flat File ProductCode and matches it to the Document Cache ProductCode, mapping the Document Cache UnitPrice to the UpdatedPrice field.

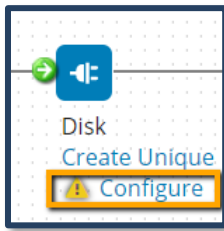
5. Select **Save and Close** when finished with the Map to return to the process.

Configure the Disk Connection and Run the process

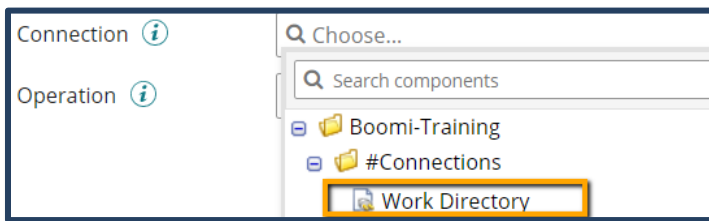
The Document Cache Activity uses a reusable Connection stored in your #Connections folder.

1. Select the **Configure** hyperlink under the Disk shape.

Document Cache Activity

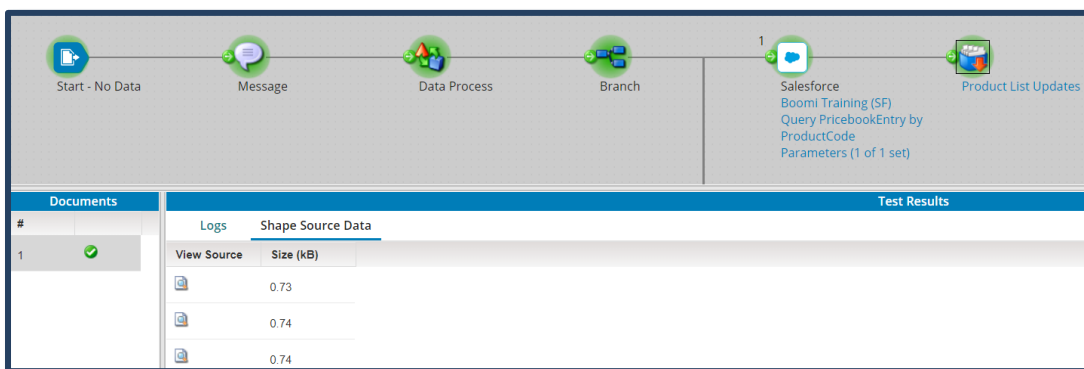


2. Select **Choose...** next to Connection, and select **Work Directory**.



✓ If you completed Boomi Essentials, the Work Directory will be in your #Connections folder. If not, you will find it in your Process Call folder and called Work.

3. Run the process and select **Product List Updates** to view the source. There should be three entries with updates.



Below is an example entry; notice the UnitPrice and ProductCode shown below and select **Close Document Viewer** when finished.

Document Cache Activity

```
1 <PricebookEntry>
2 <Id>01u0q000000LQjTAAW</Id>
3 <Name>GenWatt Diesel 10kW</Name>
4 <Pricebook2Id>01s0q0000008japAAA</Pricebook2Id>
5 <Product2Id>01t0q0000002oalAAA</Product2Id>
6 <CurrencyIsoCode>USD</CurrencyIsoCode>
7 <UnitPrice>5000.0</UnitPrice>
8 <IsActive>true</IsActive>
9 <UseStandardPrice>true</UseStandardPrice>
10 <CreateDate>2017-06-20T19:50:30.000Z</CreateDate>
11 <CreatedById>005400000003O16YAAS</CreatedById>
12 <LastModifiedDate>2017-06-20T19:50:30.000Z</LastModifiedDate>
13 <LastModifiedById>005400000003O16YAAS</LastModifiedById>
14 <SystemModstamp>2017-06-20T19:50:30.000Z</SystemModstamp>
15 <ProductCode>GC1020</ProductCode>
16 <IsDeleted>false</IsDeleted>
17 </PricebookEntry>
```

4. Select the **Disk Connector** in the True path and select the Shape Source Data tab to view the first entry - two products had their prices updated successfully.

The screenshot shows a Boomi process flow with steps: Price Change?, Price Updates, Data Process, Disk Work Directory, and End and continue. The 'Price Updates' step is selected. Below the flow, the 'Test Results' tab is active, showing a table with columns: #, View Source, Size (kB), and Connection Data. The first entry is visible with a green checkmark and a size of 0.03 kB.

The Document Viewer displays the following XML data:

```
GC1020*5000.0
GC1060*100000.0
```

Try/Catch Activity

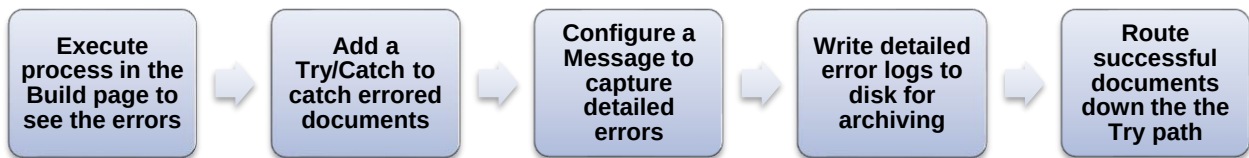
A Try/Catch is placed before the main processing shapes in a process to capture document-level errors for one or more documents that fail during execution. All successful Documents continue down the Try path for processing; failed Documents go down the Catch path.

Scenario

A process that oversees incoming purchase orders (PO) in batches has PO transaction instances not going through. In one example, a process batch containing three separate PO orders stops after the first document resulting in an error without writing to the data store. The first document failure prevents all remaining documents from flowing through the process.

Goal

Modify a prebuilt process using the Try/Catch to process all successful documents and capture and route out the main process flow of all error documents.



Create a subfolder within your account

1. Select the drop-down arrow next to the **Professional Developer** folder.
2. Choose New Folder and name it **Try/Catch**.

Load the Process from the Process Library

1. Open the Process Library by selecting **Browse Process Library** at the bottom of the Component Explorer.
2. In the right column, enter the search bar and enter the word Try.
3. Select Install next to **Try/Catch Activity**.
4. Select Browse Select Installation Location and navigate to the Try/Catch folder.
5. Select **Install**.
6. The Process is now installed in your account. Select **View Process**.

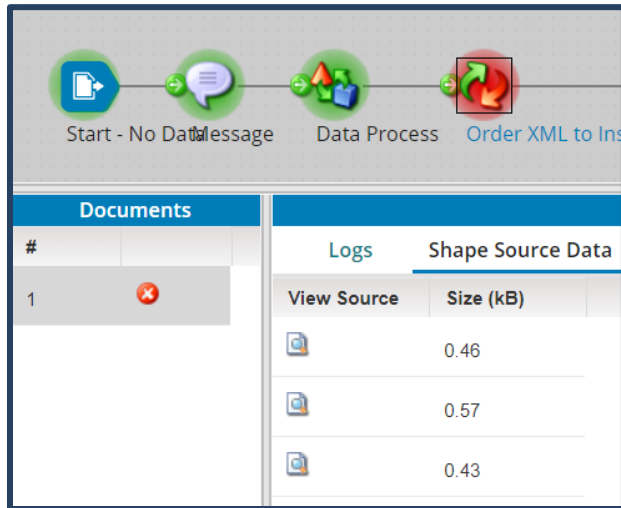
Configure the Disk Connections

Try/Catch Activity

1. The Try/Catch Activity leverages the same Disk Connection as the Document Cache activity. Open the **Disk Connector**, select Browse next to Connection, and select **Work Directory** from the #Connections folder.

Run a Test to see errors

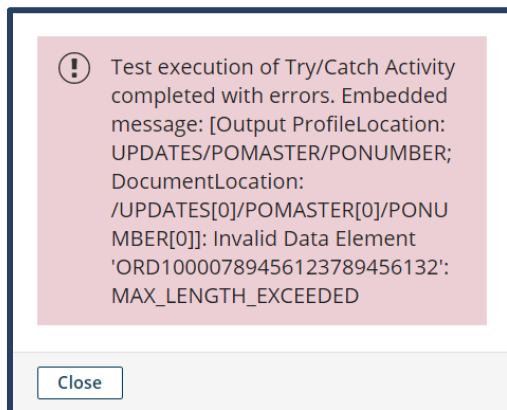
1. In the Build page, execute the process to see the errored document(s). One document fails, but it is hard to know which one.



The screenshot shows a process canvas with four activities: 'Start - No Data', 'Message', 'Data Process', and 'Order XML to Ins'. Below the canvas is a 'Documents' table with one row containing a red 'x' icon. To the right of the 'Documents' table is a 'Logs' table with two columns: 'View Source' and 'Size (kB)'. The 'Logs' table contains three rows of data.

Documents	
#	
1	

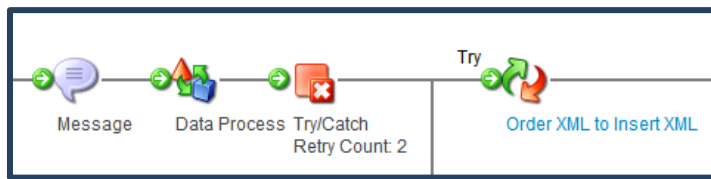
Logs	
View Source	Size (kB)
	0.46
	0.57
	0.43



Configure the Try/Catch

1. Drag Try/Catch onto the process canvas.

Try/Catch Activity



2. In the Try/Catch Properties, set the Retry Count to **2** and leave the Failure trigger at **Document Errors**.

Try/Catch Shape ⓘ

The Try/Catch shape captures process-level errors or document-level errors for one or more documents that fail during an execution. The Try/Catch shape must be placed before the main processing shapes in your process. The Try/Catch shape sends documents down the "Try" path to be processed. Documents that fail are caught and are sent down the "Catch" path.

* Required fields.

Display Name

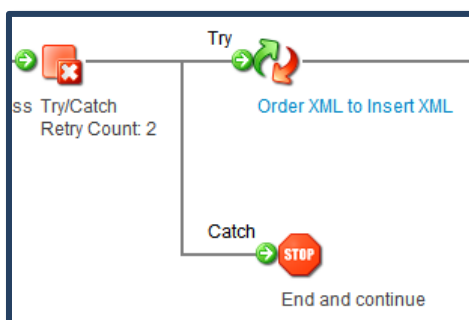
Retry Count* ⓘ

Failure Trigger

OK Cancel

- ✓ *The Retry Count tries to resend failed Documents down the Try path twice before going down the Catch path. This technique is helpful when resolving initial connection problems.*

3. Select **OK** when finished.
4. Drag and drop a **Stop** to the Catch path.



- ✓ *A Stop is added to the Catch path so successful documents flow through the Try path.*

5. Execute the process on the Build page.

Try/Catch Activity

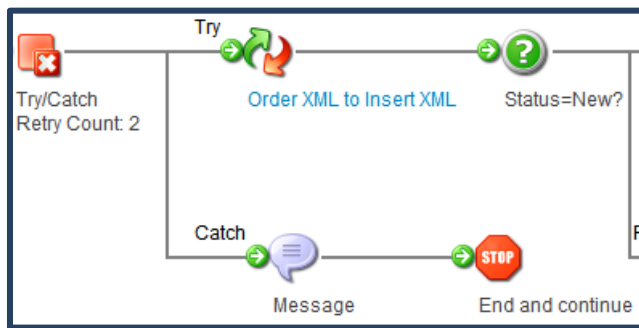


- ✓ Of the three documents, the process wrote one to disk, one failed on the Map, and one stopped on the FTP Connector shape.

Configure the Catch path

Your process now allows successful documents to route to completion. Now, configure the process to capture the remaining document errors and archive detailed error messages to the disk.

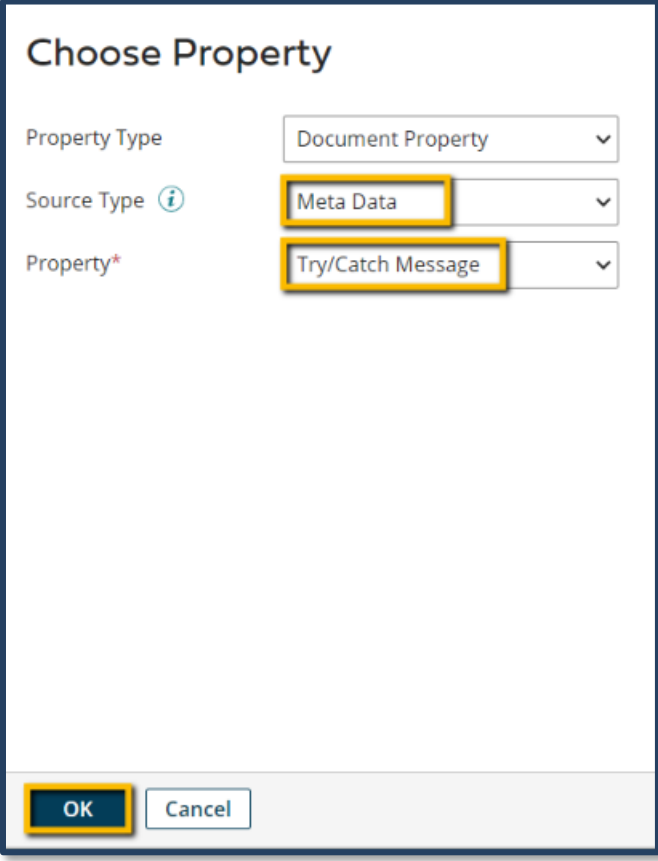
1. Place a **Message** before the **Stop** on the **Catch** path and open it.



2. In the Message Properties window under Parameters, select **Add** and select **Document Property**. Press the **magnifying glass** to choose a Document Property.

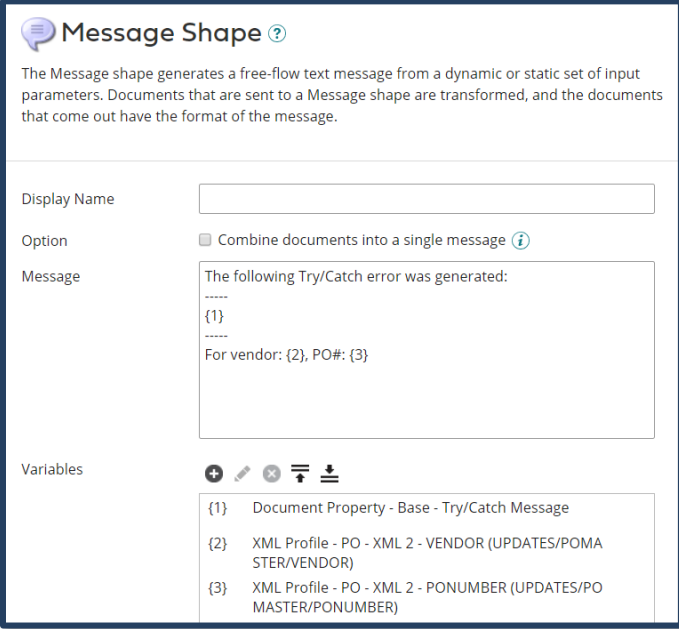
Try/Catch Activity

3. The Choose Property dialog displays. Select **Meta Data** for the Source type, **Try/Catch Message** for Property, and select **OK**.



The 'Choose Property' dialog box is shown. It has three dropdown menus: 'Property Type' set to 'Document Property', 'Source Type' set to 'Meta Data', and 'Property*' set to 'Try/Catch Message'. The 'OK' button is highlighted with a yellow border.

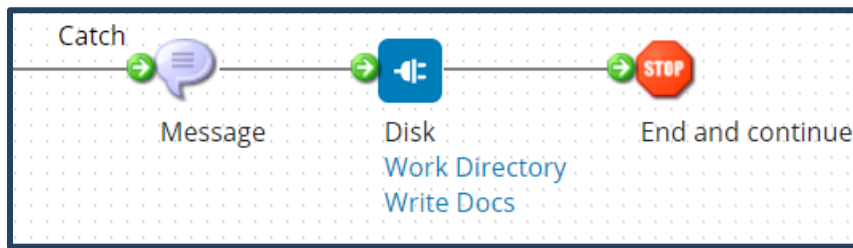
4. Complete the Message and select **OK** when finished.



The 'Message Shape' configuration window is shown. It includes a description: 'The Message shape generates a free-flow text message from a dynamic or static set of input parameters. Documents that are sent to a Message shape are transformed, and the documents that come out have the format of the message.' Below this, there are fields for 'Display Name', 'Option' (with a checked checkbox 'Combine documents into a single message'), and 'Message'. The 'Message' field contains the text: 'The following Try/Catch error was generated: {1} For vendor: {2}, PO#: {3}'. At the bottom, there is a 'Variables' section with a list of three variables: {1} Document Property - Base - Try/Catch Message, {2} XML Profile - PO - XML 2 - VENDOR (UPDATES/POMASTER/VENDOR), and {3} XML Profile - PO - XML 2 - PONUMBER (UPDATES/POMASTER/PONUMBER).

Try/Catch Activity

- Place a Disk Connector between the Message and Stops on the Catch path.



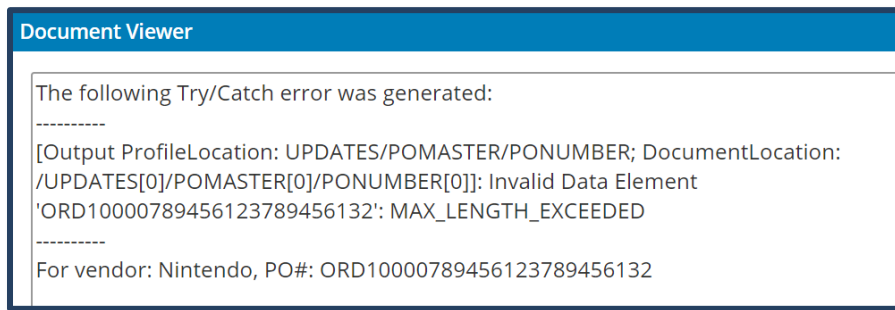
- Open the **Disk Connector** and change the action to **Send**. Browse the Connection and select **Work Directory** from the #Connections folder. Select Browse next to Operation and select **Write Docs** from the Try/Catch folder. Select **OK** and select **Save** to save the process.

- ✓ *This process reuses the same Connection and Operation components as the Disk Connector used in the Try path, writing all rejected documents in the same directory.*
 - ✓ *Connectors using the same Connection component DO NOT affect connection licenses.*
- Run the process on the Build page and examine the detailed error messages configured with the Message step. Select the Disk connector and view the connection data.

View	Directory	File Name
	/mnt/shared/Cloud_USA_East_Integration_Test_Cloud_01/ac	1582567888718.dat
	/mnt/shared/Cloud_USA_East_Integration_Test_Cloud_01/ac	1582567888731.dat

- The Document Viewer displays a detailed error message:

Try/Catch Activity

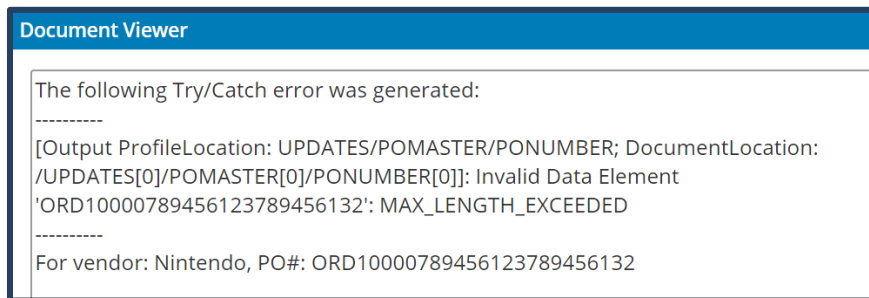


9. Select **Close Document Viewer** when finished.

This message makes it easier to archive and troubleshoot failed documents.

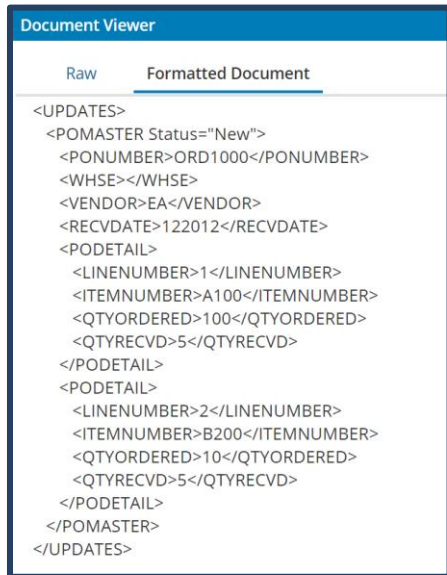
View the finished process

1. Examine the outputs beginning with the Catch branch. Recall the Message you configured earlier and how the output is displayed below:

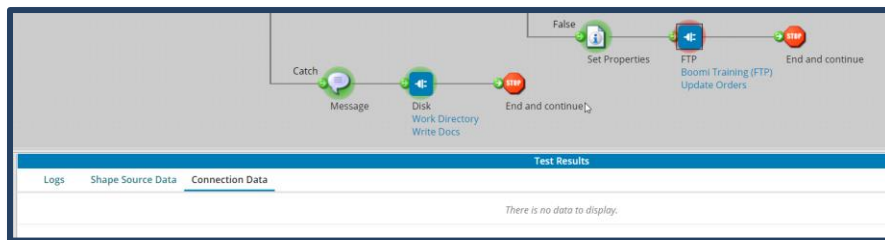


Try/Catch Activity

In the Try path, the output for the True branch is below:



2. The False branch contains no documents:



Error Handling - Activity 1 (Demo)

Scenario

In this process, the Start shape reads a database table, returns four documents, and maps them from a DB profile to an XML profile. The Route shape routes the documents by category, with each document flowing down a separate path.

Goal

To demonstrate frequent error situations and how they impact process execution.

Create a subfolder within your account

1. Select the drop-down arrow next to the **Professional Developer** folder.
2. Choose New Folder and name it **Error Handling**.

Load the process from the Process Library

1. Open the Process Library by selecting Browse Process Library at the bottom of the Component Explorer.
2. In the right column of the search bar, enter the **Dev Error Handling Activity 1** and select **Enter**.
3. Select Install next to Dev Error Handling Activity 1.
4. Browse next to Select Installation Location and navigate to the Error Handling folder.
5. Select Install in the lower-right corner.
6. The process is now installed in your account. Select View Process.

Instructor Demo - Review the Process

1. Review the process.

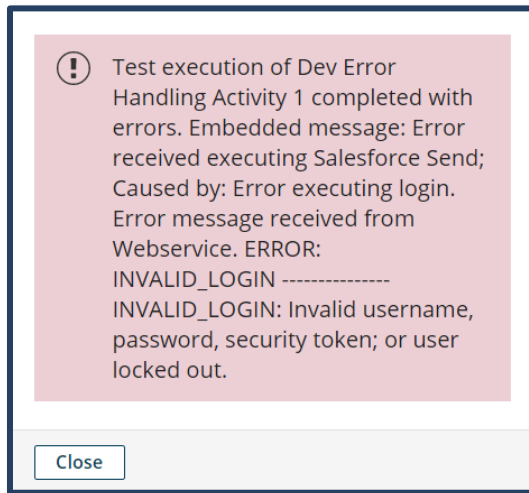


Three out of four documents contain errors.

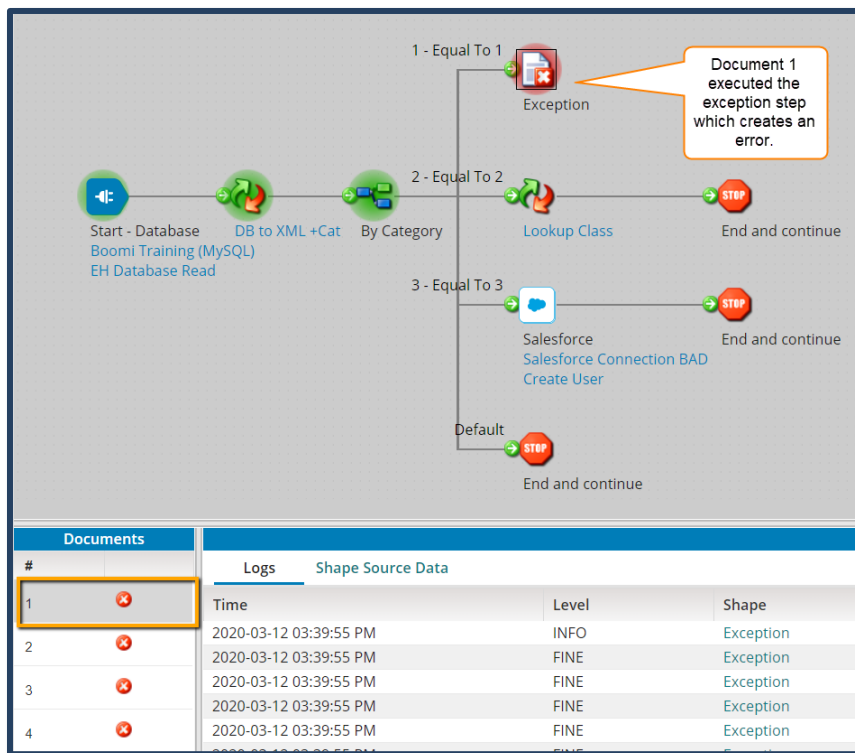
2. Configure the Start Shape Database Connector by choosing Boomi Training (MySQL) as the Connection component and Save the process.

Error Handling - Activity 1 (Demo)

- Run a test and observe the error generated below.

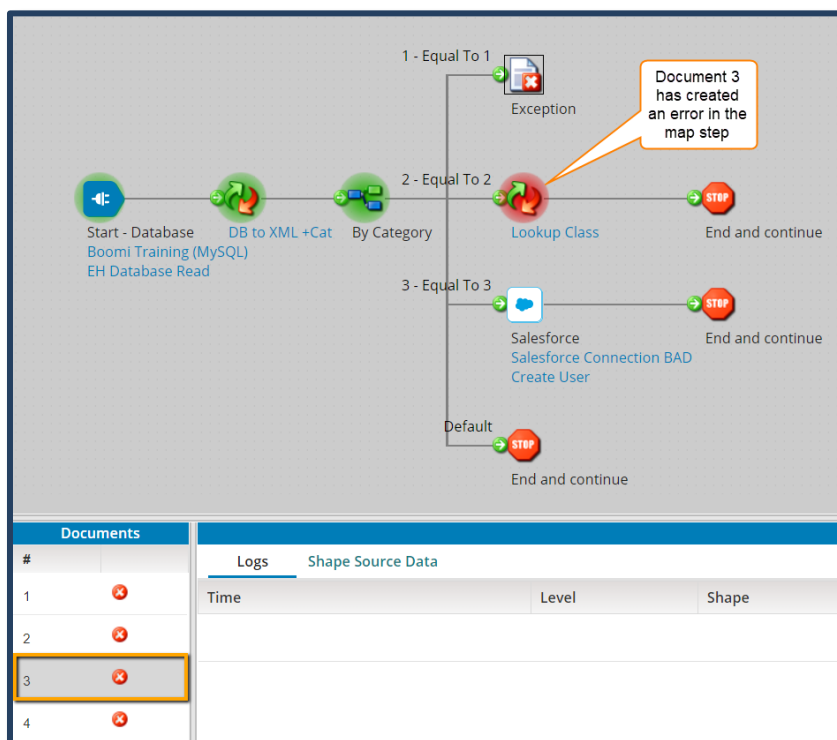
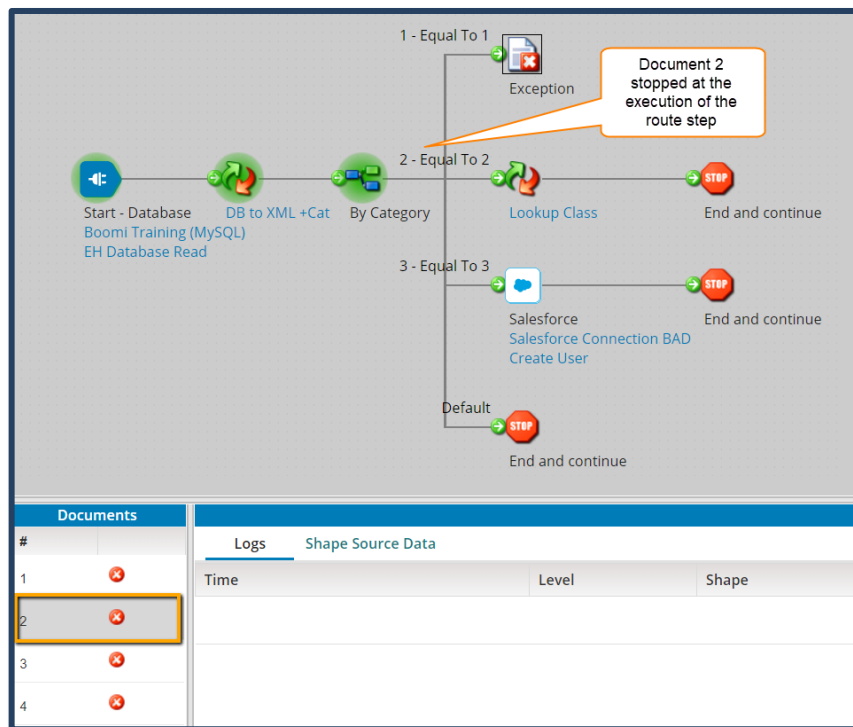


- Select **Close** to close the error message, and select each document to review its processing path.

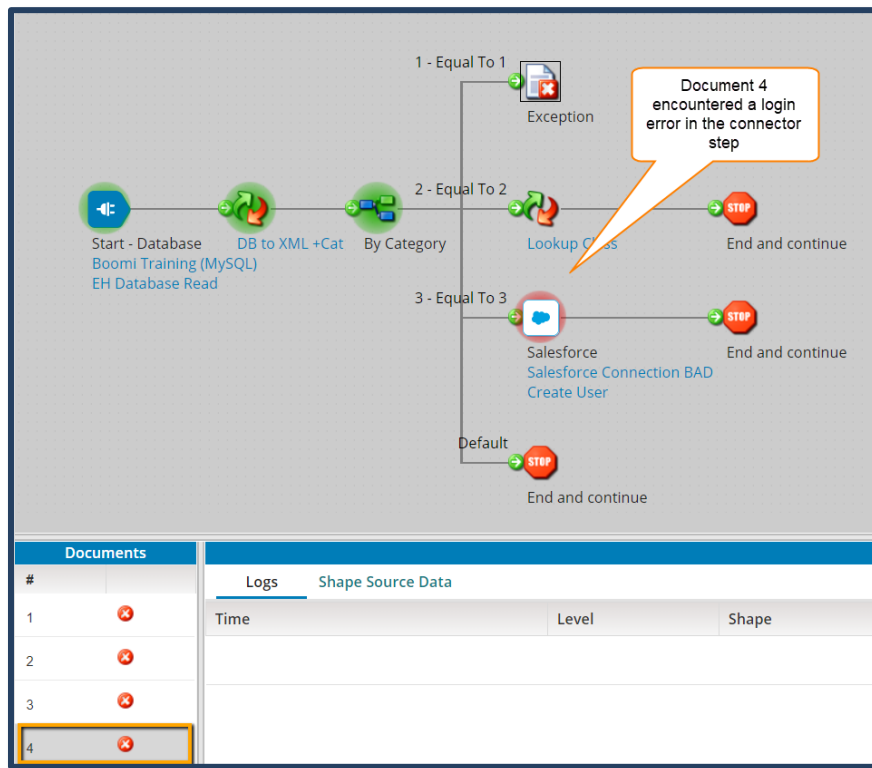


 The Process Route shape allows you to select an execution path dynamically at run time based on a value, such as a document property, data profile, extension value, or trading partner.

Error Handling - Activity 1 (Demo)



Error Handling - Activity 1 (Demo)

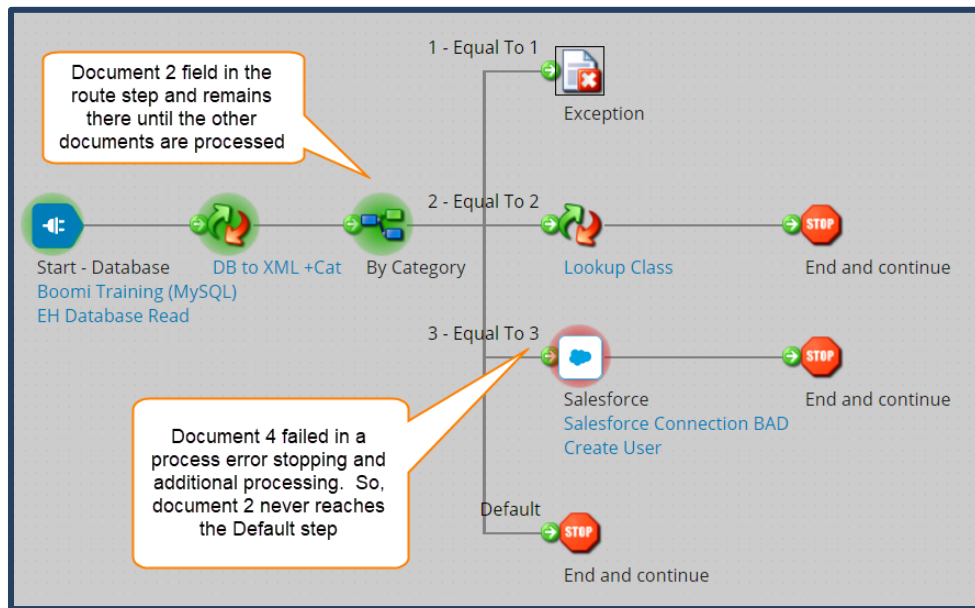


Questions:

1. The process generates three errors, one of the type process and two of the type document. Which was which? How do you know?
 - The Login error was a process error – it was the subject of the first pop-up error message.
2. What was the source of the errors?
 - An Exception
 - A mapping error (non-numeric value)
 - A bad connector login
3. How did each error impact document processing?
 - The document errors stopped the processing of the specific document. The process error stopped processing all documents.

4. Why did Document 2 stop at the Route shape?

- Due to the Category values, Document 2 was executed last through the Route shape. It had not progressed when Document 3 encountered a processing error and stopped all processing.



Next, you will review how errors impact document flow and discuss the different error-handling methods.

Error Handling - Activity 2

Scenario

Acme Engineering has implemented a referral program with its distributors. New contacts from distributors from sources (personal, website, events) and posted to a database without verification. The integration process reads these database records, maps them, and writes them to the Salesforce Account and Contact records.

Due to the varied sources, data quality is an issue. Currently, the integration process is consistently failing with error types. Your task is to design and implement an error-handling solution.

Goal

You begin with an existing integration process containing errors. You will analyze, design, and implement an error-handling approach using error-handling techniques.

Instal the process from the Process Library

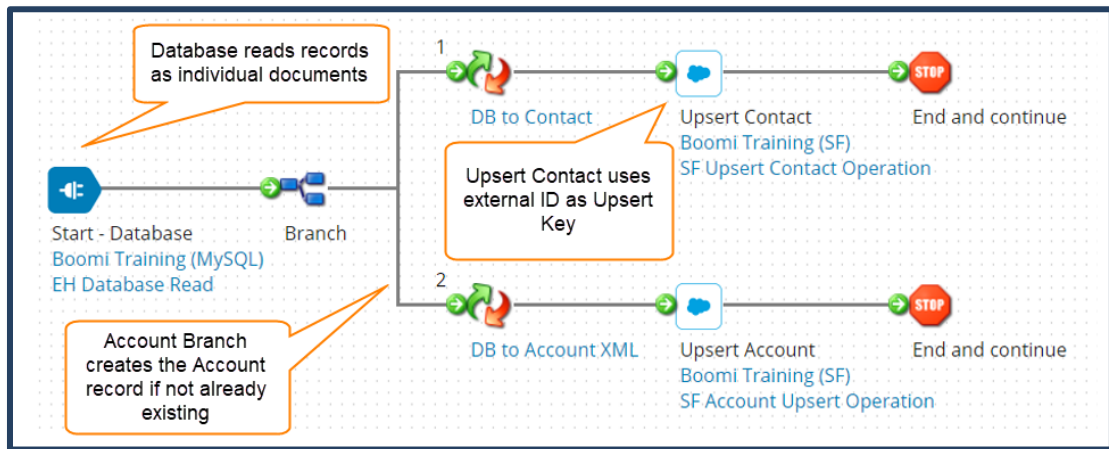
1. Open the Process Library by selecting Browse Process Library at the bottom of the Component Explorer.
2. In the right column in the search bar, enter the **Dev Error Handling Activity 2** and select **Enter**.
3. Select **Install** next to Dev Error Handling Activity 2
4. Select **Browse** next to Select Installation Location and navigate to the Error Handling folder.
5. **Install**.
6. The process is now installed. Select **View Process**.

Load the DB and Salesforce Connection Components and Review the Current Process

1. Open the Database Start shape and load the Boomi Training (MySQL) Connection from the #Connections folder.
2. Open each Salesforce Connector and load the Boomi Training (SF) Connection from the #Connections folder.
3. Select Save to save the process.

Error Handling - Activity 2

- Review the Error Handling Activity 2 process.



- Select Test and observe the results.

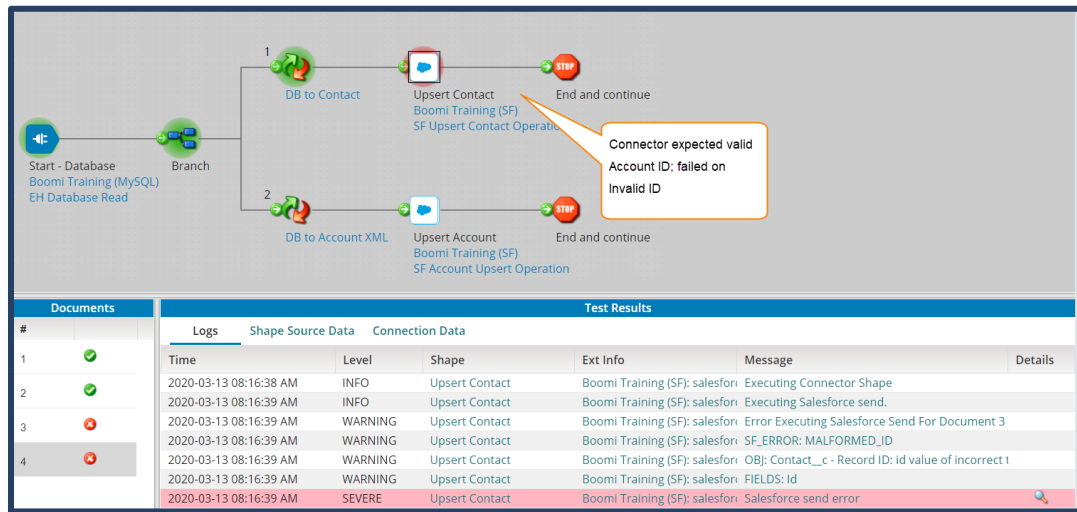
Due to the varied sources, data quality is an issue with the process. The process commonly throws two errors: Record 3 is missing a valid date, and it also encounters a mapping error in the DB to Account XML map.

The screenshot shows the Boomi Test Results interface. The top section displays the process flow with a red 'X' icon on the 'DB to Account XML' connector, indicating a failure. Below the flowchart, the 'Documents' table shows four records, with record 3 marked as failed. The 'Test Results' table provides detailed logs for each record, including the time, level, shape, and message.

Documents		Test Results					
#		Logs Shape Source Data					
		Time	Level	Shape	Ext Info	Message	Details
1	✓	2020-03-13 08:16:39 AM	INFO	Map	DB to Account XML	Executing Map Shape	
2	✓	2020-03-13 08:16:39 AM	FINEST	Map	DB to Account XML	Function: 2	
3	✗	2020-03-13 08:16:39 AM	FINEST	Map	DB to Account XML	Function: 2- Input[2:Key, G-45]	
4	✗	2020-03-13 08:16:39 AM	FINEST	Map	DB to Account XML	Function Step: 2:Simple Lookup	
		2020-03-13 08:16:39 AM	FINEST	Map	DB to Account XML	Function Execution: Simple Lookup- Output[3:Val	
		2020-03-13 08:16:39 AM	FINEST	Map	DB to Account XML	Function: 2- Output Element[SLA_Expiration_Date	
		2020-03-13 08:16:39 AM	SEVERE	Map	DB to Account XML	[Output ProfileLocation: Account...c/SLA_Expirati	

Error Handling - Activity 2

Document 4 fails in the Upsert Contact connector with a Bad External ID.



Student Design Goals:

- Create a User Alert with a WARNING severity containing the Contact ID and Error message for every error. This message is for business users who subscribe to User Alerts in AtomSphere.
- For each error, create an AtomSphere document error, which will alert IT operations to the error status.
- When possible, halt the processing of Contact records as soon as the error is detected.
- Create a separate User Warning for each connector error.
- OPTIONAL: Develop and Test in the Build page and run final tests in Deployed mode. Deploy to the Production Environment, using the Production 'Atom Cloud' atom.
- OPTIONAL: Subscribe to Errors and Warnings in Setup to confirm you are receiving errors.

Discussion about potential solutions. Questions:

- Who would use Try/Catch? Why?
- Who would use the Application error return? Why?
- Who would use Exception? Why?
- Who would use Notify? Why?

Load the process solution from the Process Library

1. Open the Process Library by selecting Browse Process Library at the bottom of the Component Explorer.
2. In the right column on the search bar, enter the **Dev Error Handling Activity 2 (Solution)** and select **Enter**.
3. Select **Install** next to Dev Error Handling Activity 2 (Solution).
4. Select the browse symbol Select Installation Location and navigate to the Error Handling folder.
5. **Select** Install in the lower right.
6. The process is now installed. Select **View Process**.

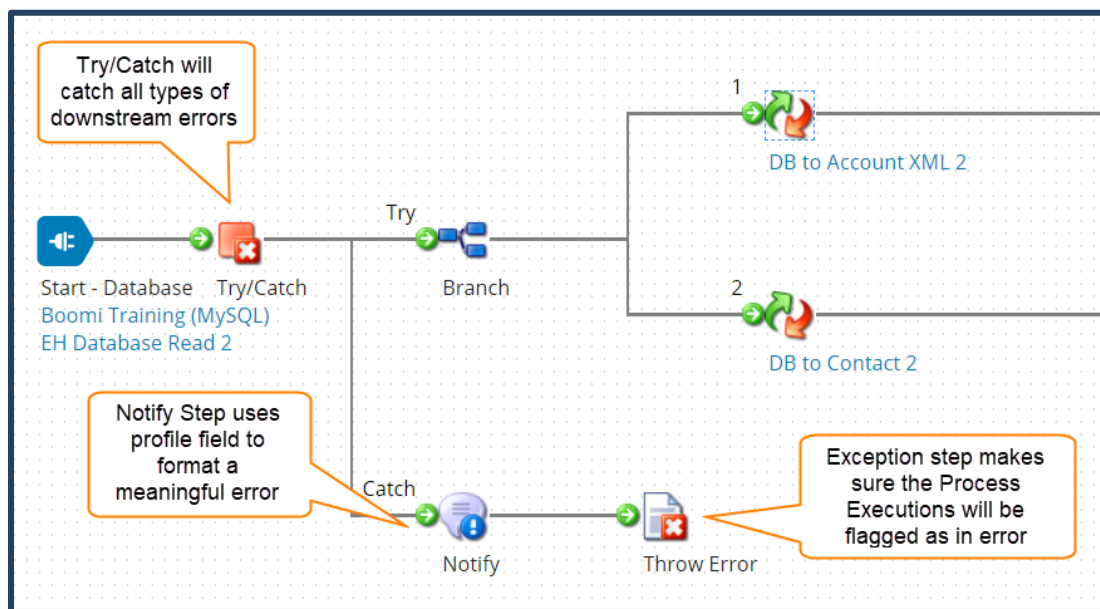
Configure Connections

1. Add Connections to the Database Start shape and Salesforce Connectors.

Review the Process

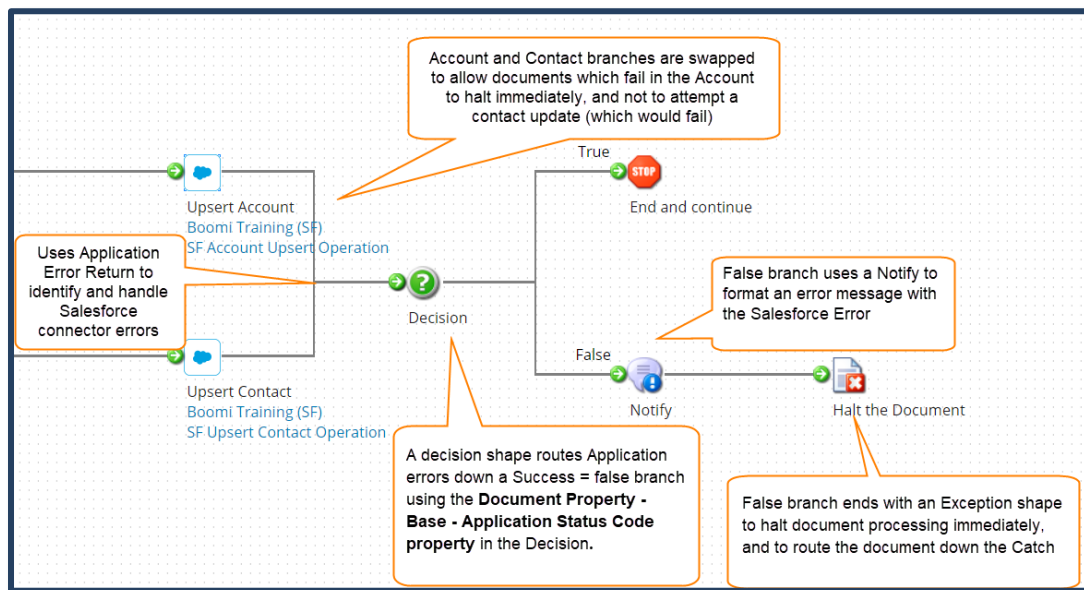
The solution:

1. Use Try/Catch to catch all errors in the Contact and Account branches.
2. Try/Catch catches all types of downstream errors.
3. Notify uses the profile field to format a meaningful error message.
4. Exception makes sure you flag the Process execution as “in error.”



Error Handling - Activity 2

5. Use Application Error Return to identify and handle Salesforce connector errors.
6. A Decision sends Application errors down a Success = false branch using the Document Property - Base - Application Status Code property in the Decision.
7. The false branch uses a Notify to format an error message with the Salesforce Error.
8. The false branch ends with an Exception to halt document processing immediately and route the document down the Catch path for consistent error processing.
9. You swap Account and Contact branches to allow documents to fail in the Account to stop and not attempt a contact update (which would fail).



Review the solution, particularly the shape configurations, including:

- Try/Catch
- Notifications
- Exceptions
- Salesforce connectors – 'Return Application Error Responses' is checked.
- Decision

Web Services Integration Activity

Scenario

A user enters contact information into an online form and sends it to the Boomi Web Server.

An Integration process validates the information and returns responses as either New Contacts or Invalid Emails. It emails validated contact information to the appropriate team member for logging and reporting.

Goal

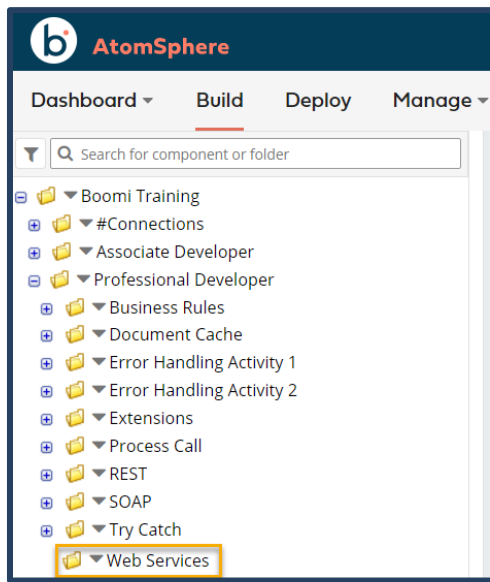
Change two processes to complete Web Services Integration.



Create a subfolder within your account

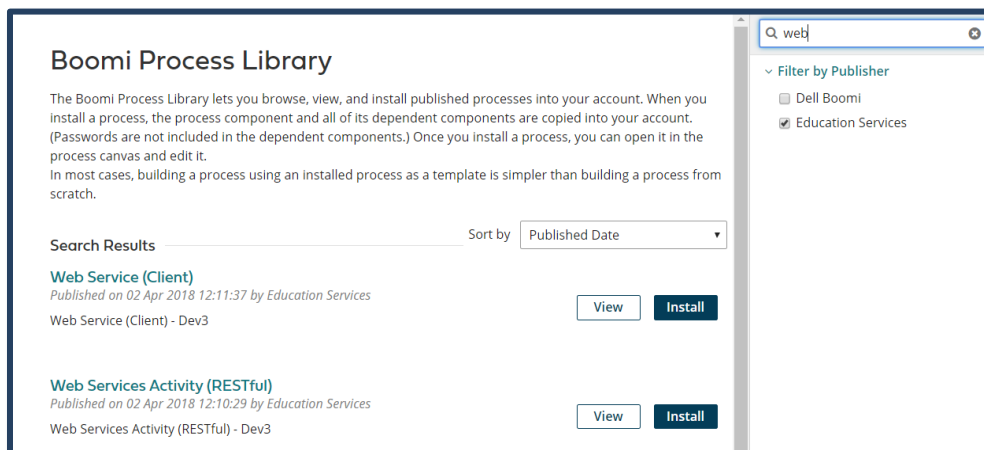
1. Select the drop-down arrow next to the **Professional Developer** folder.
2. Choose New Folder and name it **Web Services**.

Web Services Integration Activity



Install the Web Services Activity (RESTful) from the Process Library

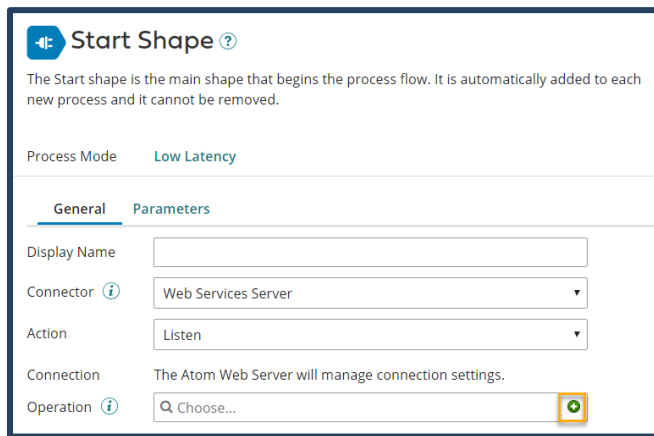
1. Open the Process Library by selecting **Browse Process Library** at the bottom of the Component Explorer.
2. Next to Web Services Activity (RESTful), select **Install**.



Configure the Web Services Activity (RESTful) Process

1. In the **Web Services Activity (RESTful)** process, select the **Start** shape, and select the plus icon to create a new Operation.

Web Services Integration Activity



Start Shape ?

The Start shape is the main shape that begins the process flow. It is automatically added to each new process and it cannot be removed.

Process Mode **Low Latency**

General Parameters

Display Name

Connector *i* Web Services Server ▼

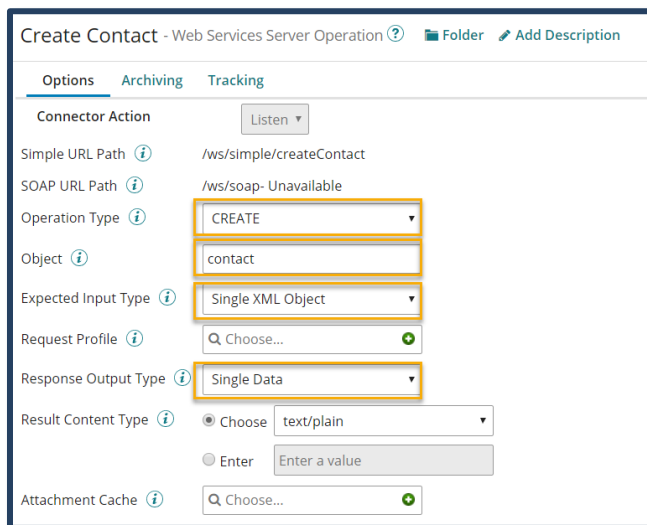
Action Listen ▼

Connection The Atom Web Server will manage connection settings.

Operation *i* 

2. Enter the following:

Title	Create Contact
Operation Type	Create
Object	Contact
Expected Input Type	Single XML Object
Expected Output Type	Single Data



Create Contact - Web Services Server Operation ? Folder Add Description

Options Archiving Tracking

Connector Action Listen ▼

Simple URL Path *i* /ws/simple/createContact

SOAP URL Path *i* /ws/soap- Unavailable

Operation Type *i* CREATE ▼

Object *i* contact

Expected Input Type *i* Single XML Object ▼

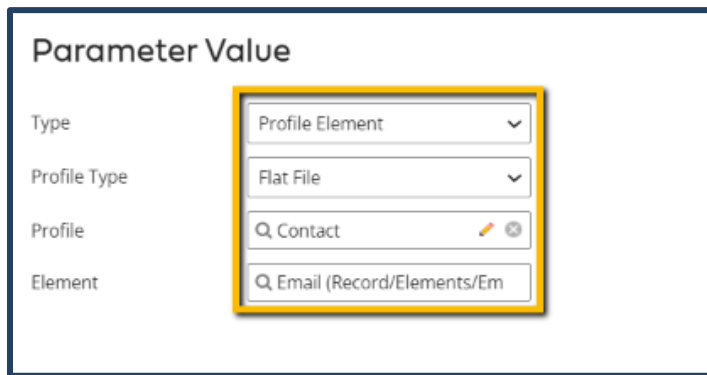
Request Profile *i* 

Response Output Type *i* Single Data ▼

Result Content Type *i* ☒ Choose text/plain ▼ ☐ Enter Enter a value

Attachment Cache *i* 

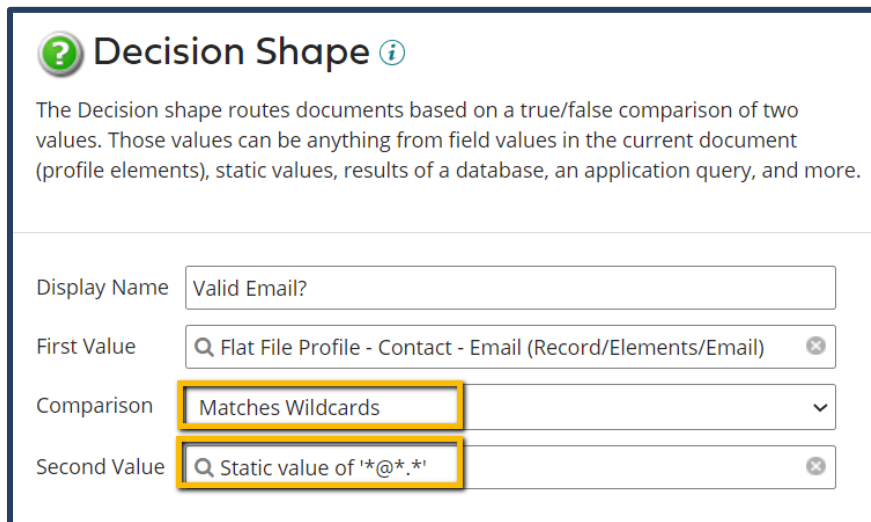
3. Select **Save and Close**.
4. Select **OK** in the Start Shape General tab to return to the Process Canvas.
5. Open the Decision to enter the decision logic. Choose the First Value. For Type Select, **Profile Element**, Profile Type Select **Flat File**, Profile Select **Contact**, and Element selects **Email**. Select **OK** to save your value.



The 'Parameter Value' window is shown with a yellow border. It contains four rows of configuration options:

Label	Value
Type	Profile Element
Profile Type	Flat File
Profile	Contact
Element	Email (Record/Elements/Em

- Set Comparison to **Matches Wildcards**, select **Static** for the Second Value, and enter ***@*.*** for the Static Value. Select **OK** to save the Decision.



The 'Decision Shape' window is shown with a yellow border. It contains the following configuration options:

Field	Value
Display Name	Valid Email?
First Value	Flat File Profile - Contact - Email (Record/Elements/Email)
Comparison	Matches Wildcards
Second Value	Static value of '*@*.*'

- Open the Mail Connector and select **Boomi Training (Mail)** from your #Connections folder.

Web Services Integration Activity

Connector: Mail

Action: Send

Connection: Choose...

Operation: Search components

- Boomi Training
- #Connections
- Boomi Training (Mail)

✓ If you do not have a #Connections folder or Boomi Training (Mail) is not in it, the connection will be in your Associate Developer -> Daily Customer Wins folder.

8. Create a new Operation by selecting the plus icon.

Connector Shape

Connector shapes are used to get data into and send data out of a process. Most processes have one "get" connector and one or more "send" connectors. The Connector shape uses a combination of predefined connection and operation components to establish where and how to get or send data.

General Parameters

Display Name:

Connector: Mail

Action: Send

Connection: Boomi Training (Mail)

Operation: Choose...

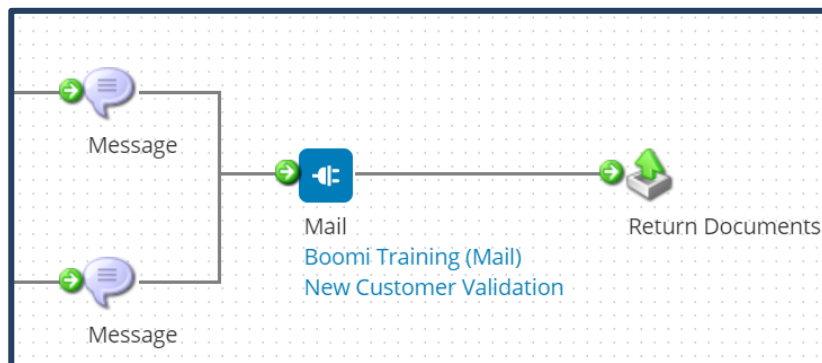
9. Name your new Operation **New Contact Validation** and configure it according to the following image:

Web Services Integration Activity

Be sure to enter your email address in the *To* field.

Leave the *Subject* blank. You can configure the *Set Properties* shapes to create subjects for documents in the *True* and *False* paths.

10. Select **Save and Close** to Save the Operation and **OK** to Save the Connector.
11. Add a **Return Documents** shape at the end of the process path. This step is necessary to return documents to the Web Service Client application.

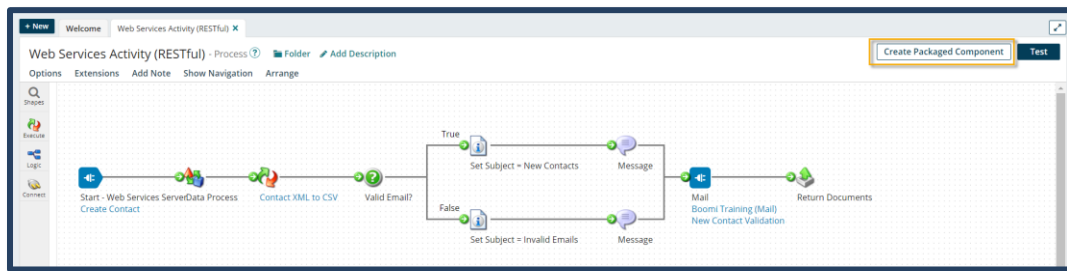


12. Select **Save**.

Create the Packaged Component and Deploy the process

1. Select **Create Packaged Component** in the upper right-hand corner of the Process Canvas.

Web Services Integration Activity



2. Accept the default settings, and select **Next**.

3. Add version information and package notes. Select **Create Packaged Component**, create the package, and select **Deploy**.

Field	Information to update
Version for all	PDEV-MMMYYYY-01
Package Notes for All	Initial package
Sharing	Unchecked

Web Services Integration Activity

4. Select **Production** as the Deployment Environment and enter **Initial Deployment** in the Deployment Notes. Proceed through the remaining deployment steps, accepting the defaults.

Deploy: Select Environment

Select the environment in which to deploy your packaged component(s), and optionally add notes about the deployment.

Deployment Environment:

Deployment Notes:

3981 characters remaining.

Deploy: Select Packaged Components

Environment: Production | **Deployment Notes:** Initial deployment.

Select one or more packaged components from the Component Explorer. By default, the latest version of each packaged component is selected. Packaged components can be created from the [Packaged Components](#) page. You can deploy a maximum of 250 packaged components at one time.

Actions	Name	Type	Selected Version	Deployed Version
	Web Services Activity (RESTful)	Process	PDEV-MAR2020-01 View details Select different package	N/A

Deploy: Review

You're almost done! Before deploying this version of your packaged component, confirm that the destination environment you have selected is correct.

Environment: Production
Deployment Notes: Initial deployment.

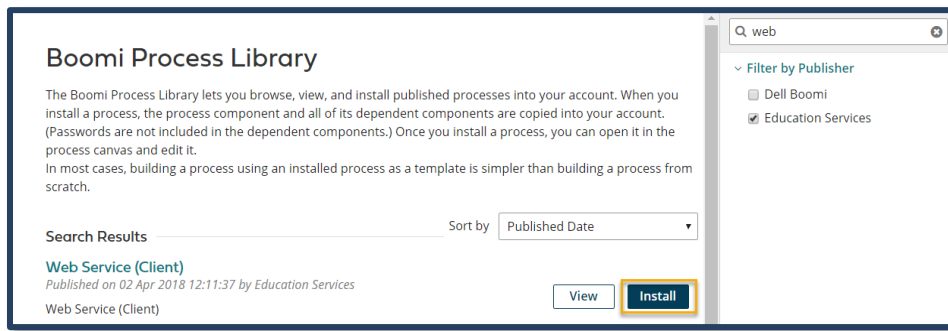
Name	Type	Selected Version	Deployed Version	Duplicate
Web Services Activity (RESTful)	Process	PDEV-MAR2020-01	N/A	—

Previous 1-1 of 1 Next

Install the Web Service Client from the Process Library

1. Open the Process Library by selecting **Browse Process Library** at the bottom of the Component Explorer.
2. Select **Install** next to Web Service Client and follow the instructions.

Web Services Integration Activity

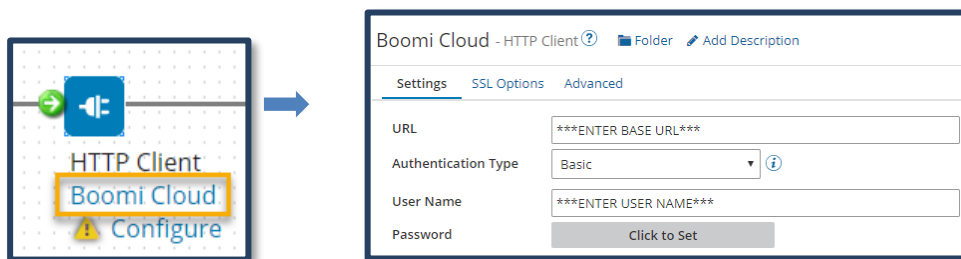


Configure the Web Service Client

Instead of using a third-party application to call the listener process, you can utilize the HTTP Client Connector to use Boomi AtomSphere as the Web Service Client.

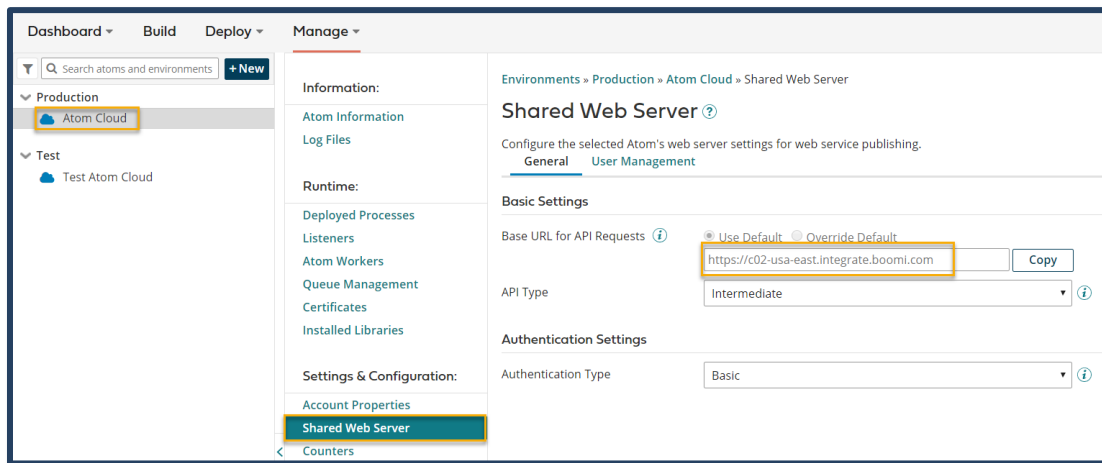
 You need to go between *Atom Management* and the *Build* page in this section. It is recommended to have a tab open for each.

1. On one tab, open the **Web Service (Client)** process and the HTTP Client Connection **Boomi Cloud**.

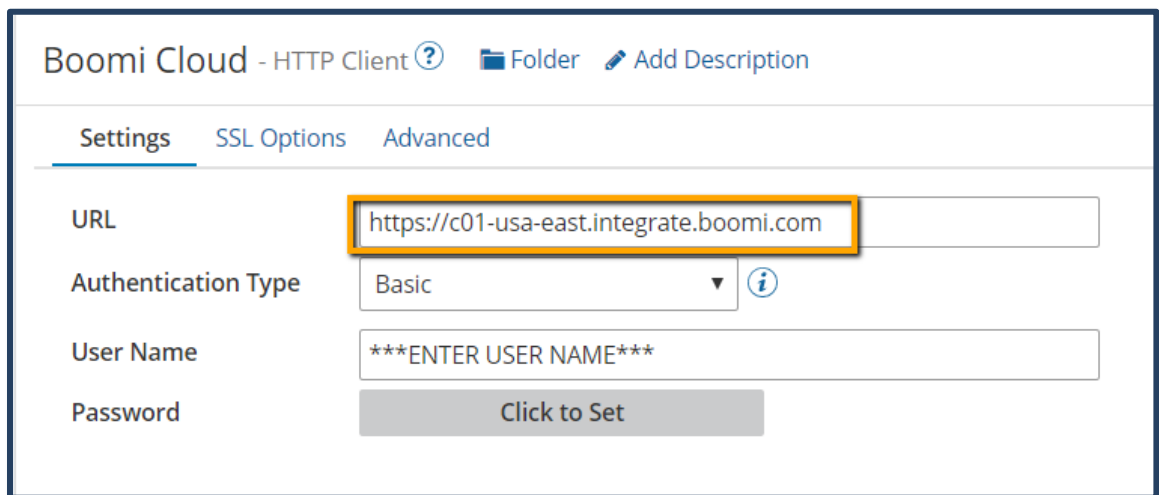


2. On another tab, open **Manage > Atom Management** and highlight the **Atom Cloud**. Select the **Shared Web Service** menu option. Use the **Copy** button to copy the Base URL for API Requests.

Web Services Integration Activity

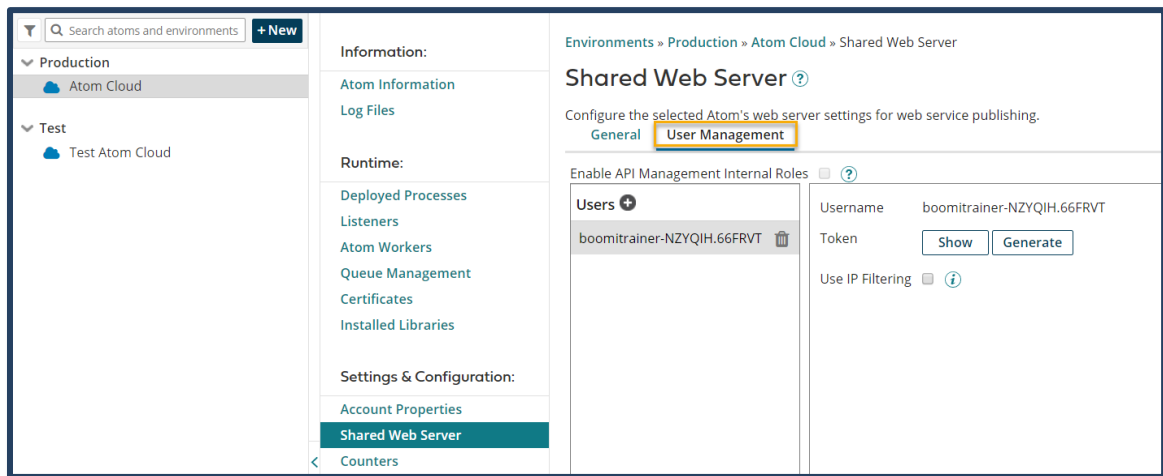


-  Your base URL may differ depending on your regional Production cloud.
 -  Confirm the API Type is Intermediate. If it is Advanced, you will receive a 404 error.
3. Go to the other tab containing the **HTTP Client Connection Boomi Cloud** and copy and paste the Base URL for your cloud.



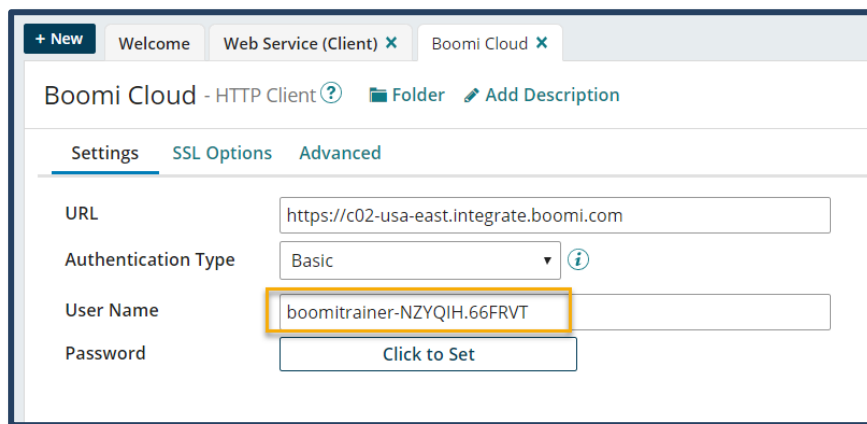
-  Your URL may differ depending on what regional Production cloud you are using.
4. Return to the **Atom Management** tab, select the **User Management** tab, and copy the Username field.

Web Services Integration Activity



- ✓ The default Username is the Atom's Instance ID, located under **Atom Information**. You may add more users, in which case the **full** username is: `username@AtomInstanceID`.

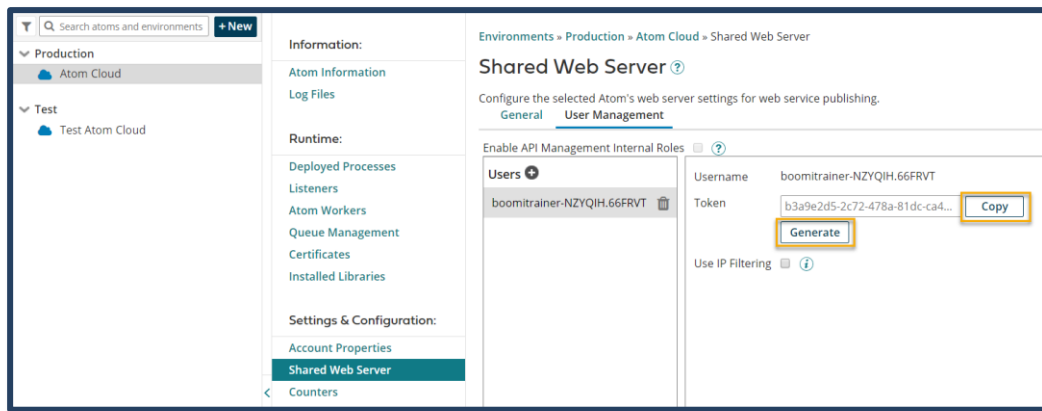
5. Return to the Build page containing the Web Service (Client) and paste it into the User Name field.



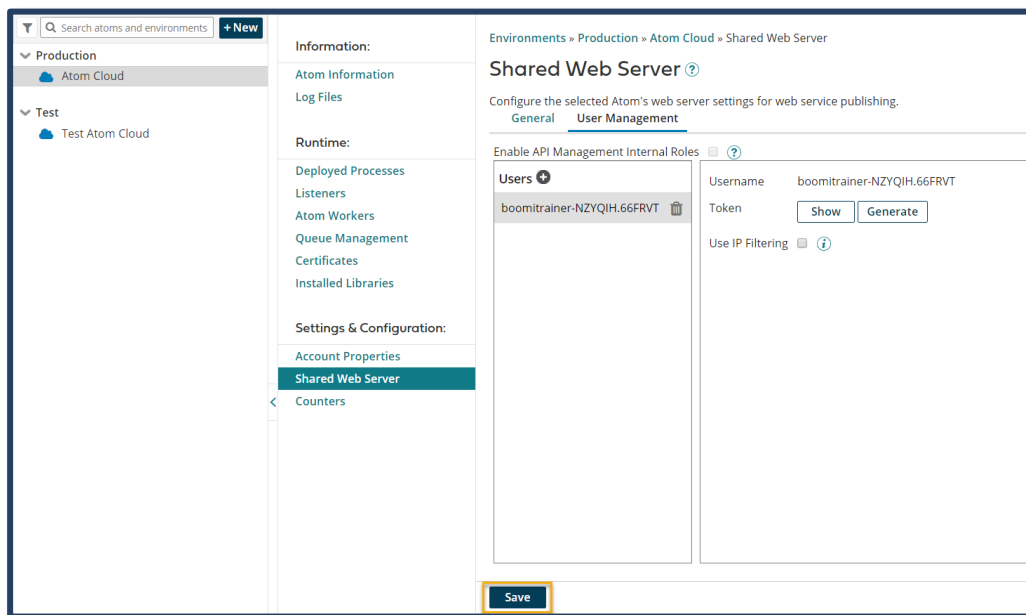
 Remove any white spaces.

6. Navigate to Atom Management, highlight the **Atom Cloud**, and select **Shared Web Server** to obtain the password. On the **User Management** tab, select **Generate** and **Copy**.

Web Services Integration Activity



7. Select **Save** to save the Shared Web Server settings.



 You **MUST** select **Save**, or you will get 400-level errors when testing the process.

8. Copy the token into the Password field on the HTTP Client Connection. Select **Save and Close**.

Web Services Integration Activity

Boomi Cloud - HTTP Client ? Folder Add Description

Settings SSL Options Advanced

URL

Authentication Type

User Name

Password

Password

Apply Cancel

Save Save and Close Close

9. On the HTTP Client Connector General tab, select the plus icon to create a new Operation.

Connector Shape ?

Connector shapes are used to get data into and send data out of a process. Most processes have one "get" connector and one or more "send" connectors. The Connector shape uses a combination of predefined connection and operation components to establish where and how to get or send data.

General Parameters

Display Name

Connector

Action

Connection

Operation

10. Name the connector Operation **Create New Contact**. The only change you must make is to check **Return HTTP Responses**.

Create New Contact - HTTP Client Operation ? Folder Add Description

Options Archiving Tracking Caching

Connector Action: Send ▼

Select Request Profile Type: NONE ▼

Select Response Profile Type: NONE ▼

Content Type: text/plain ▼

HTTP Method: POST ▼

Return HTTP Errors: ☐

Follow Redirects: ☐

Return HTTP Responses: ☒

You defined the base URL in step three by configuring the Connection component of the HTTP Client Connector. The remainder URL Path is the HTTP Client Connector Operation. This path may be copied and pasted from the Create Contact Operation in the Start Shape of the Web Services Activity (RESTful). To copy and paste, follow the instructions below:

11. In the **Web Services Activity (Restful)** process on the Start Shape, select the **Create Contact** operation.
12. Copy the Simple URL Path.

Create Contact - Web Services Server Operation ? Folder

Options Archiving Tracking

Connector Action: Listen ▼

Simple URL Path: ws/simple/createContact

✓ If copying and pasting, do not include the leading forward slash.

13. Select **Save and Close** to close the operation, **OK** to save the Connector, and **Save** to save the process.
14. In the **Web Service (Client)** process, open the HTTP Client Connector Operation and **Create New Client**.
15. Scroll to the **Resource Path**, select the plus icon, and paste **ws/simple/createContact**.

Create New Contact - HTTP Client Operation ? Folder Add Description

Options Archiving Tracking Caching

Resource Path

+ -

ws/simple/createContact

Name ws/simple/createContact

☐ Is replacement variable?

 Be sure to remove any leading spaces as well as the /.

16. Select **Save and Close** to save the new operation and select **Save** to save the process.

Call the listener process using the Web Service (Client) process and confirm the results

1. Run the **Web Service (Client)** test using the Test Atom Cloud.

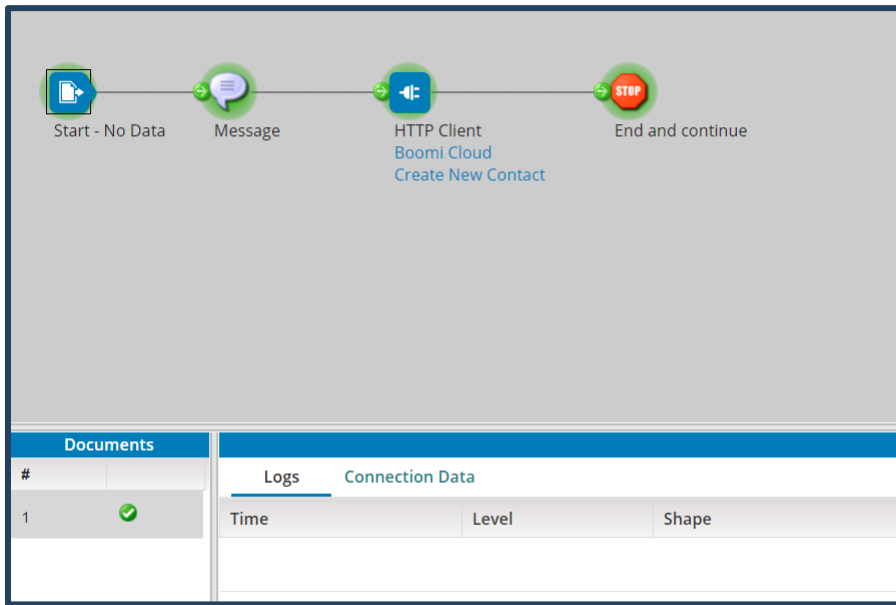
Test

Test Atom Cloud

Test Extensions

Run Test Cancel

Web Services Integration Activity



Both **New Contacts** and **Invalid Emails** will arrive in your Inbox.

New Contacts

✖ DELETE ← REPLY ⇐ REPLY ALL → FORWARD ⋮

Mark as unread

boomitrain@gmail.com

Conratulations! John Doe has been added as a new contact with a correctly formatted email: john@doe.com.

Congratulations! Jane Smith has been added as a new contact with a correctly formatted email: jane@smith.com.

Invalid Emails

✖ DELETE ← REPLY ⇐ REPLY ALL → FORWARD ⋮

Mark as unread

boomitrain@gmail.com

Email (john@smithcom) is not correctly formatted. Please resubmit the request.

Email (janedoe.com) is not correctly formatted. Please resubmit the request.
